# 01 PRIME FACTOR STUDIO

PAGE / 08

Multiply by Stacking: Tactile Prime Towers & 12-Socket Lid Tray

01 02 03 04

SKU: PRIME–01 • KEEP IN LID

SIDE A • COVER (TOP–LEFT)

**WHAT IS THIS BOX FOR? MEET THE BLOCKS THAT MULTIPLY WHEN YOU STACK THEM.**

Most counting blocks add (2 + 3 = 5). Ours use a secret logarithmic superpower: stack [ 2 ] on [ 3 ] and the tower height multiplies to 6!

## THE SECRET SUPERPOWER: MULTIPLYING BY HEIGHT

Every whole number is built from indestructible "prime atoms" (2, 3, 5, 7...). Because each block's height is scaled to the logarithm of its prime:

**Height(a × b) = Height(a) + Height(b)**

Whenever you snap blocks together into a tower, adding their physical heights automatically multiplies their numbers — no calculator required!

### 12-SOCKET COMPARISON LID TRAY

Place the 12-socket lid tray flat on your desk to hold prime towers upright side-by-side! Compare quantities and factorizations visually — whenever two towers stand at the exact same height on the lid tray, their products match.

### WHAT'S INSIDE YOUR BOX (45 PIECES)

- 43 Pure Prime Blocks: 24×[2], 9×[3], 6×[5], and 4×[7] (4–3–3–2 Eratosthenes Set)
- 1× 7U Stepped-Floor Storage Box + 1× 12-Socket Comparison Lid Tray
- Keep this folded guide inside your box lid for examples & part IDs!

★ YOUR FIRST 60 SECONDS WITH THIS BOX — START HERE!

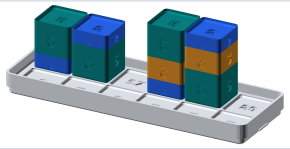
1. **POP THE 12-SOCKET LID TRAY:** Lift the snap-fit 12-socket Lid Tray off the 7U Stepped Box and set it flat on your desk as your stage.
2. **COMPARE TWO TOWERS ON THE LID TRAY:** Stand [2]+[3] (= 6) in one socket and [2]+[2]+[3] (= 12) next to it to see the height difference!
3. **PRESS THE BOX ICON KEY [g][FACT]:** Type 12 [g] [FACT] on your calculator—the factor tree icon on your box lid is this exact key!

**BOX ICON = CALCULATOR [PRIME] & [G][FACT] FACTOR TREE KEY**

The button icon embossed on your box lid (left) is the exact calculator key ([PRIME] / [g][FACT])! Build every lesson with your blocks, then press the key.

## COMMUTATIVE TOWERS: [2]×[3]×[5] = 30 ↔ BOX [PRIME] ICON

STEP A: HANDS-ON WITH BLOCKS ON LID TRAY — 1:1 CAD SCENE



Tower [2][5] = 10 — 23.25 mm | Tower [5][2] = 10 — 23.25 mm | Tower [2][3][5] — 34.34 mm (= 30)

▲ Stack prime blocks in any order on the Lid Tray: both 10-towers and both 30-towers stand dead-level!

STEP B: TRY IT ON YOUR CALCULATOR — BOX ICON: [PRIME] / [g][FACT

FACTOR TREE: 30 = 2 × 3 × 5 (34.34 MM)

30 (34.34 mm) → 6 (18.09 mm) × 5 → [2] Blue | [3] Gold | [5] White

7.00 + 11.09 + 16.25 = 34.34 mm

PRESS BOX ICON & RPN KEYS (SCG):

1. STACK 2×3×5: 2 ENTER 3 × 5 ×
2. BOX ICON KEY: g FACT 2×3×5 = 30

✓ DISPLAY (X) = 30

KEYSTROKES: 2 [ENTER] 3 [×] 5 [×] [g] [FACT]

Stack [2], [3], and [5] on the Lid Tray (= 30), then press [g][FACT] to see the prime tree!

WELCOME TO HANDS-ON MATH • KEEP THIS GUIDE IN YOUR BOX LID

STEP 1: UNFOLD RIGHT → PAGE 02

▲ FOLD 1: FOLD BOTTOM HALF BEHIND TOP ▲

# 02 HOW IT WORKS: THE CORE IDEA

PAGE / 08

Logarithmic Height Scaling: Stack to Multiply

01 02 03 04

SKU: PRIME–01 • KEEP IN LID

SIDE A • 1ST UNFOLD (TOP–RIGHT)

## THE LOGARITHMIC STACKING PRINCIPLE

Because logarithms turn multiplication into simple addition, every prime block has a visible height proportional to log10(p):

**h_vis(p) = K_log × log10(p)**

When you snap blocks together, the top socket completely swallows the bottom plug (1.85mm), so only the visible heights add up:

**H_tower(N) = 1.85 mm + K_log × log10(N)**

### HOW THE SCALE IS TUNED

- Base Unit Height: Prime [ 2 ] = 7.00 mm (matches our 1/12 Fraction block!)
- Logarithmic Scale Factor: K_log = 7.00 / log10(2) ≈ 23.25 mm per factor of 10
- Flush Snap Connector: 1.85 mm plug seats cleanly into 1.85 mm socket
- Unique Prime Fingerprint: Only one combination of primes reaches each height!

**COMMUTATIVE PROPERTY: A × B = B × A**

In math class, you're told "order doesn't matter when multiplying." Here, you can see why! Build [2]→[2]→[3]→[5] (= 60) and [5]→[3]→[2]→[2] (= 60). Seat them side-by-side on the lid tray: both towers are 41.34 mm tall. Rearranging the blocks never changes the total height of the tower!

3 CLASSIC MATH TRAPS THIS KIT MAKES IMPOSSIBLE — VISUAL PROOF

✗ TRAP 1 — Confusing 2² with 2 × 3:
✓ FIX: On the Lid Tray, [2][2][2] (= 8, 21mm) stands visibly taller than [2]+[3] (= 6, 18mm)!

✗ TRAP 2 — Mixing up GCD and LCM:
✓ FIX: On the Lid Tray, GCD is the shorter shared tower (6); LCM is the taller union tower (36).

✗ TRAP 3 — Thinking 1 is a prime:
✓ FIX: A '1' block would have height log(1) = 0 mm—it wouldn't change any tower's height at all!

CALCULATOR CHECK: 2 [ENTER] 3 [×] 5 [×] [g] [FACT] → X = 30

## PRIME BLOCK HEIGHT REFERENCE

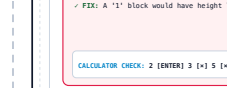
| PRIME | LOG10(P) | VISIBLE HEIGHT | UNITS (U) | COLOR |
|---|---|---|---|---|
| [ 2 ] | 0.3010 | 7.00 mm | 1.00 U | Blue (24x) |
| [ 3 ] | 0.4771 | 11.10 mm | 1.59 U | Red (9x) |
| [ 5 ] | 0.6990 | 16.25 mm | 2.32 U | White (6x) |
| [ 7 ] | 0.8451 | 19.65 mm | 2.81 U | Tan (4x) |
| 2×3 = 6 | 0.7782 | 18.10 mm | 2.59 U | Tower 6 |
| 2²×3=12 | 1.0792 | 25.10 mm | 3.59 U | Tower 12 |
| 2×3²=18 | 1.2553 | 29.20 mm | 4.17 U | Tower 18 |
| 2·3·5=30 | 1.4771 | 34.35 mm | 4.91 U | Tower 30 |
| 2⁶ = 64 | 1.8062 | 42.00 mm | 6.00 U | Waterline |
| Box Bay | Stepped | 56.40 mm | 7.00 U | Gray/Wht |
| Lid Tray | 12-Sock | 10.40 mm | 1.49 U | White |

**WHY YOUR BRAIN LOVES THIS**

Physical heights replace rote memorization — your hands and eyes spot common factors and multiples before you even write an equation!

1:1 CAD REFERENCE VIEW — BOX ICON & FORMULA QUICK-REFERENCE

1:1 BLOCK CAD ANATOMY




✓ FLUSH PLUG-AND-SOCKET JOINT
Socket swallows plug completely: Only 7.00mm true heights add!

Box Button Icon Key: [PRIME] / [g][FACT] Prime Factorize

Tower Height Formula: H_vis(N) = 23.25 mm × log₁₀(N)

Multiply Two Towers: Stack A on B → A [ENTER] B [×]

Divide Tower A by B: Pop B off A → A [ENTER] B [÷]

GCD & LCM Keystrokes: A [ENTER] B [GCF] or [f] [LCM]

FUNDAMENTAL THEOREM OF ARITHMETIC • EVERY NUMBER HAS ONE PRIME DNA

STEP 2: UNFOLD DOWN ▼ PAGES 03–04

▼ UNFOLD STEP 2: OPEN DOWN FOR PAGES 03 & 04 ▼

# 03 EXAMPLE 1: HANDS-ON WALKTHROUGH

PAGE / 08

Build Prime Towers for 12 & 18 Side-by-Side on the Lid Tray

01 02 03 04

SKU: PRIME–01 • KEEP IN LID

SIDE A • 2ND UNFOLD (BOT–LEFT)

## 1 PRIME DECOMPOSITION & TOWER CONSTRUCTION

**GOAL: Find the Least Common Multiple of 12 and 18: LCM(12, 18)**

1. DECOMPOSE NUMBER 12 INTO PRIMES:
12 = 2 × 6 = 2 × 2 × 3 = 2² × 3
→ Gather two [ 2 ] blocks and one [ 3 ] block.

2. DECOMPOSE NUMBER 18 INTO PRIMES:
18 = 2 × 9 = 2 × 3 × 3 = 2 × 3²
→ Gather one [ 2 ] block and two [ 3 ] blocks.

3. BUILD & SEAT BOTH TOWERS ON THE LID TRAY:
Snap the blocks into Tower 12 and Tower 18 and seat them on the Lid Tray.

### LID TRAY SIDE-BY-SIDE HEIGHT COMPARISON

Seat both towers into adjacent sockets on the 12-socket Lid Tray:
- **Tower 12 Height: Visible height = 7.00 + 7.00 + 11.09 = 25.09 mm.** Built from two Blue [ 2 ] blocks and one Gold [ 3 ] block (2² × 3 = 12).
- **Tower 18 Height: Visible height = 7.00 + 11.09 + 11.09 = 29.18 mm.** Stands 4.09 mm taller on the lid tray because a [ 3 ] replaces a [ 2 ]!
- ✓ **Physical confirmation: Shared base [2]×[3] = 6 is visible across both towers.**

STEP B: TRY EXAMPLE #1 ON YOUR CALCULATOR — [PRIME] / [g][FACT

STEP 1 (FACTOR 12): Press 12 [g] [FACT] → Shows 2² × 3, matching two Blue [2]s + one Gold [3] (25.09 mm).

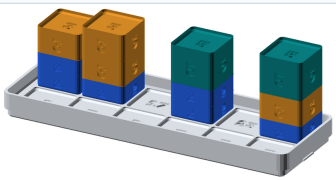
STEP 2 (FACTOR 18): Press 18 [g] [FACT] → Shows 2 × 3², matching one Blue [2] + two Gold [3]s (29.18 mm).

STEP 3 (FIND GCF & LCM): Press 12 [ENTER] 18 [GCF] (→ 6) and 12 [ENTER] 18 [f] [LCM] (→ 36) to verify both towers!

LIVE 4-REGISTER RPN STACK READOUT (AFTER KEYSTROKES):
T: 0 | Z: 0 | Y: 12 | X: 18 → DISPLAY (X) = 18

## LESSON 1: BLOCKS ON LID TRAY ↔ [g][FACT] KEYSTROKES

STEP A: HANDS-ON WITH BLOCKS ON LID TRAY — 1:1 CAD SCENE



Tower 12 = [2][2][3] — 25.09 mm | Tower 18 = [2][3][3] — 29.18 mm | Tower 30 = [2][3][5] — 34.34 mm

▲ Seat prime towers in the 12-socket Lid Tray to compare quantities and heights side-by-side!

STEP B: TRY IT ON YOUR CALCULATOR — BOX ICON: [PRIME] / [g][FACT

FACTOR TREES: 12 = 2²×3 vs 18 = 2×3²

12 vs 18 ON LID TRAY

12 = 4 × 3 → [2][2] [3]
18 = 2 × 9 → [2] [3][3]

12 (25.09 mm) vs 18 (29.18 mm)

PRESS BOX ICON & RPN KEYS (SCG):

1. FACTOR 12: 1 2 g FACT 2×3
2. FACTOR 18: 1 8 g FACT 2×3²

✓ DISPLAY (X) = 18

KEYSTROKES: 12 [g] [FACT] 18 [g] [FACT]

Build the factor towers on the Lid Tray, then press the Box Icon operation [g][FACT] to verify!

✓ 4-REGISTER STACK: T: 0 | Z: 0 | Y: 12 | X: 18 → X = 18

HANDS-ON EXPLORATION • MULTIPLY BY STACKING BLOCKS

CONTINUE RIGHT → PAGE 04

# 04 EXAMPLE 2: CORE MATHEMATICAL PRINCIPLE

PAGE / 08

Tower Leveling: Why GCD × LCM Always Equals A × B

01 02 03 04

SKU: PRIME–01 • KEEP IN LID

SIDE A • 2ND UNFOLD (BOT–RIGHT)

## 2 PRIME UNION & THE CONSERVATION THEOREM

1. **FIND SHARED PRIMES (INTERSECTION):**
Tower 12 {2, 2, 3} and Tower 18 {2, 3, 3} share: { [ 2 ], [ 3 ] }
→ Tower GCD = [ 2 ] + [ 3 ] = 6 (GCD(12, 18) = 6)

2. **FIND PRIME UNION (HIGHEST POWERS):**
Take maximum of each prime: two [2]s and two [3]s
→ Tower LCM = [ 2 ] + [ 2 ] + [ 3 ] + [ 3 ] = 36 (LCM(12, 18) = 36)

✓ **THEY ARE THE EXACT SAME HEIGHT!**

**GCD(A, B) × LCM(A, B) = A × B**

**6 × 36 = 12 × 18 = 216**

Stack Tower 12 on top of Tower 18: [2]+[2]+[3] + [2]+[3]+[3] = 216.
Stack Tower GCD (6) on top of Tower LCM (36): [2]+[3] + [2]+[2]+[3]+[3] = 216.
**Both stacks contain exactly three [2]s and three [3]s!**

### LAW OF CONSERVATION OF PRIME MATTER

Primes are indestructible arithmetic matter. In finding GCD and LCM, no blocks are created or destroyed — only partitioned and rearranged!

★ 2-PLAYER GAME: 'LID TRAY TOWER MATCH' (2 MINS)

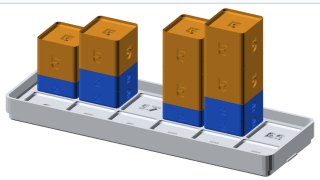
Player A builds a 3-block prime tower on the Lid Tray (like [2][3][5] = 30). Player B must build a tower with the exact same height using a different block order ([5][3][2]) **and press [g][FACT] on the calculator to prove both equal 30!**

⊙ WHERE THIS SHOWS UP IN REAL-WORLD ENGINEERING & CRYPTOGRAPHY

Logarithmic addition (log(A×B) = log(A)+log(B)) is how engineers multiplied numbers for centuries! Modern internet security (RSA encryption) relies on the fact that stacking two giant prime **blocks is easy, while figuring out which primes built a giant tower is supercomputer-hard.**

## LESSON 2: GCD & LCM TOWERS ↔ [GCF] & [f][LCM] KEYS

STEP A: HANDS-ON WITH BLOCKS ON LID TRAY — 1:1 CAD SCENE



GCD(12,18) = 6 — 18.09 mm | LCM(12,18) = 36 — 36.18 mm | 6 × 36 = 12 × 18 — 216 (54.27 mm)

▲ On the Lid Tray, GCD = [2][3] (6) is the shared core and LCM = [2][2][3][3] (36) is the full union!

STEP B: TRY IT ON YOUR CALCULATOR — BOX ICON: [PRIME] / [g][FACT

FACTOR TREE: LCM(12,18) = 36 & GCD = 6

LCM = 36 (36.18 mm) → GCD = 6 [2] Blue [2] Blue | × 6 = 36 [3] Gold [3] Gold

GCD=6 (18.09mm) • LCM=36 (36.18mm)

PRESS BOX ICON & RPN KEYS (SCG):

1. FIND GCF: 12 ENTER 18 GCF = 6
2. FIND LCM: 12 ENTER 18 f LCM = 36

✓ DISPLAY (X) = 36

KEYSTROKES: 12 [ENTER] 18 [GCF] 12 [ENTER] 18 [f] [LCM]

Press [GCF] for the shared prime tower (6) and [f][LCM] for the full union tower (36).

✓ 4-REGISTER STACK: T: 0 | Z: 0 | Y: 6 | X: 36 → X = 36

THE PHYSICAL PROOF • CONSERVATION OF PRIME BUILDING BLOCKS

STEP 3: FLIP SHEET OVER ↻ PAGE 05

# 05 — EXAMPLE 3 & DESK CHALLENGES

PAGE / 08

Commutative Towers: 2 × 3 × 5 = 30 + 3 Desk Challenges

05 06 07 08

SKU: PRIME-01 • KEEP IN LID

SIDE B • FLIP SIDE (TOP-LEFT)

## 3 COMMUTATIVE TOWERS: 2 × 3 × 5 = 30 IN ANY ORDER

**THE GOAL: Prove that block order never changes tower height or product!**

**1. BUILD TOWER [2][3][5]:**

Stack Blue [2] (7.0mm) + Red [3] (11.1mm) + White [5] (16.3mm) = 34.34 mm.

**2. BUILD REVERSED TOWER [5][3][2]:**

Stack White [5] + Red [3] + Blue [2] on the Lid Tray in reverse order.

**3. COMPARE ON THE LID TRAY:**

Seat both towers in adjacent sockets—their summits stand dead-level at 34.34 mm!

**✓ THE COMMUTATIVE LAW IN 3D: 2 × 3 × 5 = 30**

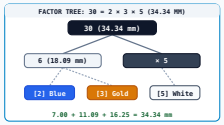
**Every block's height is proportional to its logarithm: h(p) = K_log × log10(p).**
Since heights add linearly: h(2) + h(3) + h(5) = h(5) + h(3) + h(2) = 34.34 mm.
Order never changes the sum of heights or the product of factors (2 × 3 × 5 = 30).
Verify on calculator: 2 [ENTER] 3 [×] 5 [×] [g][FACT] proves both equal 30!

**PRO TIP: CLOSE YOUR EYES & FEEL THE ANSWER**

Once you build an example, run your fingers across the pieces. Your hands remember physical shapes and heights way longer than memorized rules!

EXAMPLE 3: VISUAL DECOMPOSITION TREE

FACTOR TREE: 30 = 2 × 3 × 5 (34.34 MM)

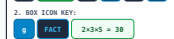



STEP B: PRESS BOX ICON & RPN KEYS (SC6)

TACTILE KEYSTROKE WALKTHROUGH:

1. STACK 2×3×5:

2. BOX ICON KEY:




✓ CALCULATOR DISPLAY (X) = 30

FULL KEYSTROKES: 2 [ENTER] 3 [×] 5 [×] [g] [FACT]

✓ 4-REGISTER STACK: T: 0 | Z: 0 | Y: 0 | X: 30 → X = 30

## 3 DESK CHALLENGES — TRY THESE NOW!

**WARM-UP #1** — **Build 8 and 12 on the Lid Tray. Find GCD & LCM!**

Step A: Build with blocks on your Lid Tray • Step B: Verify on SC-32 Calculator
✓ Build first, then flip booklet upside-down to check Answer #1 below!

**LEVEL-UP #2** — **Find the only 3-block tower that equals 30.**

Step A: Build with blocks on your Lid Tray • Step B: Verify on SC-32 Calculator
✓ Build first, then flip booklet upside-down to check Answer #2 below!

**BRAIN-MELTER #3** — **Build 20 and 50. Verify GCD × LCM = 20 × 50.**

Step A: Build with blocks on your Lid Tray • Step B: Verify on SC-32 Calculator
✓ Build first, then flip booklet upside-down to check Answer #3 below!

SELF-CHECK ANSWER KEY (ROTATE BOOKLET 180° TO READ)

#3: GCD = 10, LCM = 100 (10 × 100 = 1,000)

#2: GCD = [2][3][5] = 4, LCM = [2][2][3][5] = 2x | 2x [2] × [3] × [5] = 30 (spheric number!)

**FOR PARENTS & TEACHERS (Grades 4–9 (Number Theory, Factors, GCD & LCM))**

- CCSS.MATH.CONTENT.6.NS.B.4: Find greatest common factor & least common multiple.
- CCSS.MATH.CONTENT.4.OA.B.4: Find factor pairs & identify prime vs. composite numbers.
- TEKS 111.26.b.6.7.A: Generate equivalent expressions using prime factorization.
- **CCSS.MATH.PRACTICE.MP2, MP4, MP5, MP7: Reason, Model, Use Tools & Structure.**

Next up on Page 06 → How to stack everything back in the box in 10 seconds!

**BOSS CHALLENGE #4** — **BLOCKS → BOX ICON KEY SPEED-RUN!**

1. Build Examples #1, #2, and #3 hands-on with your physical pieces.
2. Press the Box Button Icon ([PRIME] / [g][FACT]) and keystrokes on your calculator:
   • 2 [ENTER] 3 [×] 5 [×] [g] [FACT] → X = 30

**CHECK OFF YOUR MASTERY MILESTONES:**

- ☐ 1. Demonstrated Lessons #1, #2 & #3 hands-on with the physical blocks
- ☐ 2. Solved all 3 Desk Challenges above (check upside-down key if stuck!)
- ☐ 3. Demonstrated each lesson on the calculator using the Box Button Icon
- ☐ 4. Packed all pieces back into the box in under 10 seconds (see Page 06 →)

TEST YOURSELF • ANSWERS INCLUDED SO YOU CAN CHECK RIGHT AWAY

NEXT: HOW TO PACK UP → PAGE 06

SIDE B (PAGES 05–08) • READ TOP ROW FIRST

# 06 — HOW TO STACK IT ALL BACK IN

PAGE / 08

The 10-Second Self-Checking Pack-Up Puzzle

05 06 07 08

SKU: PRIME-01 • KEEP IN LID

SIDE B • PACK-UP GUIDE (TOP-RIGHT)

## TOP-DOWN BOX MAP: WHERE EVERY PIECE LIVES

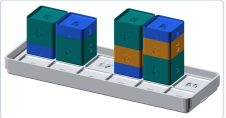
Putting the blocks away is a 10-second puzzle. Match each piece to its labeled home bay below so everything nests flush and the lid snaps shut:

| | | | | | |
|---|---|---|---|---|---|
| 6x [2] "2" (0mm) | 6x [2] "2" (0mm) | 2x [5] "5" (+9.5) | 2x [5] "5" (+9.5) | 2x [5] "5" (+9.5) | 2x [7] "7" (+2.7) |
| 6x [2] "2" (0mm) | 6x [2] "2" (0mm) | 3x [3] "3" (+8.7) | 3x [3] "3" (+8.7) | 3x [3] "3" (+8.7) | 2x [7] "7" (+2.7) |

**✓ THE 42.0 MM STORAGE BOX WATERLINE CHECK (▼)**

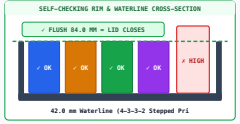
Each of the 12 storage box columns (4-3-3-2 layout) has a stepped floor pedestal calibrated to its
When all 43 prime blocks are stacked in their home columns inside the 7U Storage Box,
all 12 towers rise to the exact same 42.00 mm (2* = 64) waterline so the lid snaps shut.
If one column dips low, you know immediately which prime block is still on the desk!

1. STORAGE CASE & COMPARTMENTS



2. WHY THE LID SELF-CHECKS EVERY PIECE

SELF-CHECKING RIM & WATERLINE CROSS-SECTION




▲ SELF-INDEXING BAYS & EMBOSSED BOX BUTTON ICON:
Calibrated bays guide every piece home, and the lid icon links directly to your calculator.

## THE 10-SECOND PACK-UP ROUTINE

**HOW TO STACK EVERYTHING BACK IN:**

1. **SORT BY COLOR & NUMBER:**
Group matching blocks or tiles together on your desk first.
2. **STACK & DROP INTO LABELED BAYS:**
Follow the Top-Down Box Map on the left. If a tower looks too tall or too short, a block is in the wrong bay—swap it until everything levels!
3. **TUCK THIS CARD IN THE LID & SNAP SHUT:**
Fold this guide back to [01] and rest it right on top before closing.

**12-SOCKET LID TRAY & COMPACT 7U BOX**

- Twelve recessed sockets on the Lid Tray hold towers upright while you compare heights.
- Seat the 12-socket Lid Tray directly onto the 7U Stepped Box for secure backpack storage.
- Easy care: wipe clean or rinse in lukewarm soapy water (never put in dishwasher).

**SHORT A PIECE AT CLEAN-UP TIME?**

First check under notebooks and chair legs! If a piece truly vanished, **see PAGE 08 (bottom-right) for your complete Part ID Reference Table.**

✓ PRIME-01 4-POINT CLEAN-UP CHECKLIST

| | |
|---|---|
| 1. 42MM WATERLINE: | All 12 prime columns (4-3-3-2 stepped floor) stand dead-level at 42.0 mm. |
| 2. CLEAR LID TRAY: | Remove any comparison towers still seated in the 12 Lid Tray sockets. |
| 3. FOLD CARD TO [01]: | Fold this reference booklet to Panel [01] and lay inside the lid. |
| 4. SNAP LID FLUSH: | Snap the 12-socket Lid Tray onto the 7U box rim for rattle-free carry! |

**MODULAR CLASSROOM & BOOKSHELF NESTING (42.0 MM PITCH)**

Every StackCalc box is sized in exact multiples of 42.0 mm (1×1, 3×1, 3×2). All 9 kit boxes interlock side-by-side in a classroom cart or drawer with zero wasted space, and every embossed box icon matches its calculator operation key!

SELF-CHECKING WATERLINE • SPOT ANY MISSING BLOCK IN 2 SECONDS

NEXT: CONNECTED LAB ▼ PAGE 07

THEN READ BOTTOM ROW (PAGE 07 → PAGE 08)

# 07 — THE STACKCALC CONNECTED LAB

PAGE / 08

How Your Kits & RPN Calculator Stack Work Together

05 06 07 08

SKU: PRIME-01 • KEEP IN LID

SIDE B • ECOSYSTEM (BOT-LEFT)

**ONE CONNECTED SYSTEM FROM KINDERGARTEN TO CALCULUS**

Every StackCalc box uses the same modular grid and unit heights—so pieces from different kits actually fit and work together.

## WHERE YOUR BOX FITS IN THE 9-KIT LAB

| | | |
|---|---|---|
| BASE-10 **Ten-Frames Lab** Early Counting & Regrouping | DOMI-01 **Math Domino Studio** Doubles & 12-Frame Part-Whole | FRAC-01 **Fraction Studio** Proportional Towers & 4-Way Faces |
| PRIME-01 ◂ YOUR BOX **Prime Factor Studio** Multiply by Stacking Prime Towers | ALGB-01 **Algebra Studio** Quadratic Area Arrays & Factoring | TIME-01 **Planetary Clock** 12:1 Geared Hands & Time Wedges |
| PROB-01 **Probability Dice** Fair Polyhedrals & 10-Sector Spinner | VOLV-01 **Unit Circle Volvelle** Spin-to-Read Exact Trig Radicals | SC-32-HW **SC-32 RPN Calculator** 4-Register Stack Scientific Engine |

**WHY THIS MATTERS: ZERO "THROW-AWAY" MATH**

In most classrooms, every grade uses a totally unrelated plastic toy.
In StackCalc, the 7mm unit block you count with in 1st grade is the exact same height as the 1/12th fraction block and the [2] prime block later on!

1. PRIME-01 BOX LID ICON

2. SC-6 ELEMENTARY RPN

3. SC-32 SCIENTIFIC RPN

▲ BOX ICON ↔ CALCULATOR KEY CONTINUITY:
Every physical box lid carries the exact icon of the calculator key that computes that lesson!

## HOW BLOCKS CONNECT TO THE CALCULATOR

**SHARED MODULAR FOOTPRINT & HEIGHTS**

- Universal 7mm base unit: Prime Block [2] (7.0mm) matches the 1/12th Fraction block & ±05 Clock block.
- Recessed sockets swallow the bottom peg so only true heights add.
- All storage boxes nest neatly side-by-side on any desk or shelf.

**WHY WE CALL IT "STACKCALC" (THE RPN SECRET)**

- Stacking blocks mirrors the SC-32 RPN calculator: [2] [ENTER] [3] [×] builds the tower digitally.
- Placing a block on the table = pressing [ENTER] to push onto the stack.
- Snapping two towers together = pressing [+] or [×] with zero parentheses!
- Mirrors the 4 stack levels on the SC-32 calculator: X, Y, Z, and T.

**FREE ACTIVITY DECKS & INTERACTIVE SIMULATORS**

Download extra printable puzzle cards, teacher lesson guides, and interactive browser simulations anytime—no login needed.

Explore All 9 Kits & Free Guides at stackcalc.io

LIVE 4-REGISTER RPN STACK

T: 0 (Top Memory)
Z: 0 (3rd Level)
Y: 6 (GCF) (1st Block)
X: 36 (LCM) (Active Display)

HOW BLOCKS MAP TO STACK KEYS

1. **PLACE 1ST PIECE ON STAGE:**
Type number + [ENTER] → lifts to Y
2. **ADD / STACK 2ND PIECE:**
Type 2nd number in X, then [+]/[×]
3. **PRESS BOX ICON KEY:**
Press [PRIME] / [g][FACT]
✓ **Zero Parentheses • Always Visible!**

BOX ICON = [PRIME] / [g][FACT] • VERIFIED LESSON SEQUENCES:
• Lesson 1: 12 [g] [FACT] 18 [g] [FACT] → X = 18
• Lesson 2: 12 [ENTER] 18 [GCF] 12 [ENTER] 18 [f] [LCM] → X = 36

MODULAR 9-KIT CONTINUUM • FROM EARLY COUNTING TO AP CALCULUS

NEXT: PART ID TABLE → PAGE 08

# 08 — KEEP THIS CARD! PART IDs & VIDEO

PAGE / 08

Complete 45-Piece Reference & 60-Second 3D Video Guide

05 06 07 08

SKU: PRIME-01 • KEEP IN LID

SIDE B • PART REFERENCE (BOT-RIGHT)

## PART ID REFERENCE TABLE (45 TOTAL PIECES)

**✓ KEEP THIS CARD INSIDE YOUR BOX LID!**

Backpacks eat blocks. Dogs chew dice. Vacuum cleaners take no prisoners. Keep this card in your lid so you always have the exact Part IDs below for single-piece refills!

| PART NAME | QTY | COLOR / FINISH | PART ID |
|---|---|---|---|
| Prime Block [ 2 ] (7.00mm) | 24x | Matte Blue | PRIM-BLCK-02 |
| Prime Block [ 3 ] (11.10mm) | 9x | Muted Red | PRIM-BLCK-03 |
| Prime Block [ 5 ] (16.25mm) | 6x | White PLA | PRIM-BLCK-05 |
| Prime Block [ 7 ] (19.65mm) | 4x | Tan PLA | PRIM-BLCK-07 |
| 7U Stepped-Floor Box Body | 1x | Gray / White | PRIM-BOX-07U |
| 12-Socket Comparison Lid | 1x | White PLA | PRIM-LID-12S |

**NEED A REPLACEMENT PIECE? NO NEED TO BUY A WHOLE NEW BOX.**

Enter any Part ID above at stackcalc.io/parts to order individual replacement pieces.

COLOR & PIECE IDENTIFICATION KEY — MATCH ANY LOOSE PART

UNIVERSAL STACKCALC HEIGHT PITCH (1U = 7.00 MM • 12U = 84.00 MM ONE WHOLE)

Measure any loose piece in 7mm units (U) against this scale bar to identify it:

✓ DISTRICT PO & SAFETY: 10-Pack SKU PRIME-01-CLASS10 • CPSIA / ASTM F963 Non-Toxic Washable PLA/ABS

## WATCH THE 60-SECOND 3D VIDEO GUIDE

POINT CAMERA TO WATCH DEMO

stackcalc.io/products/prime-studio/

See animated 3D examples & pack-up walkthrough

**QUESTIONS, CLASSROOM SETS & SUPPORT**

- Email us anytime: support@stackcalc.io
- School & District POs: school-orders@stackcalc.io

**Designed in Mountain View, CA • Non-Toxic & Washable**

COMPANION MANUAL CHAPTERS, STANDARDS & KEYSTROKES

**BOOK CHAPTERS:** SC-6 Guide Ch. 3 (Prime Towers) • SC-32 Manual Ch. 2
**STANDARDS:** CCSS 4.OA.B.4, 6.NS.B.4 • TEKS 5.4A, 6.7A (Primes, GCF & LCM)
**WATERLINE SPEC:** 42.0 mm Waterline (4-3-3-2 Stepped Prime Lanes)

ALL 3 VERIFIED CALCULATOR LESSON SEQUENCES (SCG):
Ex 1: 12 [g] [FACT] 18 [g] [FACT] → X=18
Ex 2: 12 [ENTER] 18 [GCF] 12 [ENTER] 18 [f] [LCM] → X=36
Ex 3: 2 [ENTER] 3 [×] 5 [×] [g] [FACT] → X=30

BOX ICON KEY: [PRIME] / [g][FACT]
✓ Cross-Referenced to Companion Book & Calculator Engine
Fold back to [01] & keep this reference card inside your box lid!

LOST A PIECE? GRAB SINGLE REPLACEMENTS ANYTIME AT stackcalc.io/parts

✓ KEEP THIS GUIDE IN BOX LID!

# 01 PAGE / 08 — FRACTION STUDIO PROPORTIONAL TOWERS

See & Feel Fractions: 4-Sided Blocks from 1 Whole to 1/12

01 02 03 04 — SKU: FRAC-01 • KEEP IN LID — SIDE A • COVER (TOP-LEFT)

**WHAT IS THIS BOX FOR? FRACTIONS WITHOUT THE TEARS (OR THE ROOT CANAL).**

Every block in this box is scaled to a 1.0 Whole tower (84mm). Stack two towers side-by-side: if their tops are level, the fractions are equal!

## 4-SIDED BLOCKS: SPIN TO TRANSLATE

Why are 1/4, 0.25, and 25% taught in separate months of school?

- Face 1: Fraction (e.g. 1/4)
- Face 2: Decimal (e.g. 0.25)
- Face 3: Percentage (e.g. 25%)
- Face 4: Calculator Reciprocal Key ([1/x])

Just spin a block in your fingers to see all four languages at once!

### THE 84MM "ONE WHOLE" RULE

Why 84 millimeters for 1 Whole? Because 84 divides cleanly by 2, 3, 4, 6, and 12! Every fraction column levels off at the exact same 84.00 mm ceiling.

### WHAT'S INSIDE YOUR BOX (65 PIECES)

- 63 Fraction Blocks across 12 columns (10 denominators: 1/1 through 1/12)
- 1× 13U Socketed Storage Box + 1× Comparison Stage Lid
- Keep this folded guide inside your box lid for examples & part IDs!

★ YOUR FIRST 60 SECONDS WITH THIS BOX — START HERE!

1. **SPIN THE PURPLE [1/4] BLOCK:** Rotate it 90° at a time in your fingers: read 1/4 → 0.25 → 25% → [4][1/x] on its four sides!
2. **COMPARE TOWERS ON THE LID TRAY:** Stand [1/2], two [1/4]s, three [1/6]s, and six [1/12]s in the Lid Tray sockets—every tower is 42.0 mm!
3. **PRESS THE BOX ICON KEY [FDISP]:** Add 1/3 + 1/4 on your calculator and press [FDISP] (your box lid icon!) to flip 0.5833 → 7/12.

FDISP — **BOX ICON = CALCULATOR [FDISP] FRACTION↔DECIMAL TOGGLE KEY**

The button icon embossed on your box lid (left) is the exact calculator key ([FDISP] & [g][SIMP])! Build every lesson with your blocks, then press the key.

## 4-SIDED BLOCKS: FRACTION ↔ DECIMAL ↔ [FDISP] BOX ICON

STEP A: HANDS-ON WITH BLOCKS ON LID TRAY — 1:1 CAD SCENE

Face 1: Fraction — 1/2 + 1/4 + 1/4 = 1 | Face 2: Decimal — 0.50 + 0.25 + 0.25 | Face 4: Box Icon — [FDISP] & [f][1/x]

▲ Spin any block 90° at a time to read Fraction, Decimal, Percentage, and RPN Reciprocal keys!

STEP B: TRY IT ON YOUR CALCULATOR — BOX ICON: [FDISP] & [g][SIMP]

DECOMPOSITION TREE: 1 WHOLE = 1/2 + 1/4 + 1/4

1 WHOLE (84.0 mm) → 1/2 (42 mm) + 2/4 = 1/2 (42 mm) → 0.50 (50%) + 1/4 (0.25) + 1/4 (0.25)

42 + 21 + 21 = 84.0 mm (100%)

PRESS BOX ICON & RPN KEYS (SCG):

1. ADD PARTS: 1 ENTER 2 + → + 1/4 + 1/4
2. BOX ICON KEY: FDISP → = 1 (Whole)

✓ DISPLAY (X) = 1

KEYSTROKES: 1 [ENTER] 2 [÷] 1 [ENTER] 4 [÷] [÷] 1 [ENTER] 4 [÷] [÷] [F

Build 1/2 + 1/4 + 1/4 on the Lid Tray (= 84mm Whole), then verify on the calculator with [FDISP]!

WELCOME TO HANDS-ON MATH • KEEP THIS GUIDE IN YOUR BOX LID

STEP 1: UNFOLD RIGHT → PAGE 02

▲ FOLD 1: FOLD BOTTOM HALF BEHIND TOP ▲

# 02 PAGE / 08 — HOW IT WORKS: THE CORE IDEA

True Proportional Heights & 4-Way Notation Faces

01 02 03 04 — SKU: FRAC-01 • KEEP IN LID — SIDE A • 1ST UNFOLD (TOP-RIGHT)

## INVERSE PROPORTIONALITY: BIGGER DENOMINATOR = SHORTER BLOCK

A fraction 1/n slices 1 whole unit into n equal pieces.
The visible height of every block is 84.00 mm divided by n:

h_vis(1/n) = H_whole / n = 84.00 mm / n

When you snap blocks into a tower, the bottom plug disappears inside the socket below it, so the heights add cleanly:

H_tower = 1.85 mm + Σ (84.00 mm / n_i)

### HOW THE FRACTION SYSTEM FITS TOGETHER

- Whole Tower Height: H_whole = 84.00 mm (12U Unit Stacking Pitch)
- Base Unit Block: 1/12th block = 7.00 mm = 1U unit height
- Cross-Kit Harmony: 1/12 block matches Prime Block [2] (7.00mm)!
- Instant Check: Run your finger across two towers to feel if they are equal

### FOUR LANGUAGES ON EVERY BLOCK

Switching between fractions, decimals, and percentages trips up almost everyone at first. Because all four faces are engraved around the same block, you see immediately that 1/4, 0.25, and 25% are the exact same physical amount wearing different outfits!

3 CLASSIC MATH TRAPS THIS KIT MAKES IMPOSSIBLE — VISUAL PROOF

✗ TRAP 1 — Adding denominators (1/2 + 1/3 ≠ 2/5):
✓ FIX: On the Lid Tray, [1/2] + [1/3] is 70mm (5/6)-over twice as tall as 2/5 (33.6mm)!

✗ TRAP 2 — Thinking bigger denominator = bigger piece:
✓ FIX: Seat [1/12] (7mm) next to Green [1/2] (42mm) on the Lid Tray: more slices = shorter blocks.

✗ TRAP 3 — Treating decimals as separate math:
✓ FIX: Spinning any block or pressing [FDISP] proves 0.25 and 1/4 are the exact same 21mm height.

CALCULATOR CHECK: 1 [ENTER] 2 [÷] 1 [ENTER] 4 [÷] [÷] 1 [ENTER] 4 [÷] [÷] → X = 1

## 10 CURRICULUM DENOMINATOR TIERS REFERENCE

| TIER | DECIMAL | PERCENT | HEIGHT | UNITS (U) |
|---|---|---|---|---|
| 1 / 1 | 1.000 | 100.0% | 84.00 mm | 12.00 U |
| 1 / 2 | 0.500 | 50.0% | 42.00 mm | 6.00 U |
| 1 / 3 | 0.333 | 33.3% | 28.00 mm | 4.00 U |
| 1 / 4 | 0.250 | 25.0% | 21.00 mm | 3.00 U |
| 1 / 5 | 0.200 | 20.0% | 16.80 mm | 2.40 U |
| 1 / 6 | 0.167 | 16.7% | 14.00 mm | 2.00 U |
| 1 / 8 | 0.125 | 12.5% | 10.50 mm | 1.50 U |
| 1 / 9 | 0.111 | 11.1% | 9.33 mm | 1.33 U |
| 1 / 10 | 0.100 | 10.0% | 8.40 mm | 1.20 U |
| 1 / 12 | 0.083 | 8.33% | 7.00 mm | 1.00 U |
| Box & Lid | 13U Case | 12-Lane | 98.40 mm | 13.00 U |

### WHY YOUR BRAIN LOVES THIS

When two fraction towers are level to your fingertips, you don't need to guess common denominators — you can physically see and feel why they match!

1:1 CAD REFERENCE VIEW — BOX ICON & FORMULA QUICK-REFERENCE

1:1 BLOCK CAD ANATOMY

✓ FLUSH PLUG-AND-SOCKET JOINT
Socket swallows plug completely: Only 7.00mm true heights add!

Box Button Icon Key: [FDISP] Fraction ↔ Decimal Toggle
Block Height Law: H(1/n) = 84.00 mm ÷ n
Face 1 → Face 2 → Face 3: 1/n → Decimal → Percent (%)
Face 4 (Reciprocal): Press [f] [1/x] (SC-6) or [1/x] (SC-32)
Simplify Fractions: [g] [SIMP] reduces to lowest terms

PROPORTIONAL HEIGHT SYSTEM • H(1/N) = 84.0 MM ÷ N

STEP 2: UNFOLD DOWN ▼ PAGES 03–04

▼ UNFOLD STEP 2: OPEN DOWN FOR PAGES 03 & 04 ▼

# 03 PAGE / 08 — EXAMPLE 1: HANDS-ON WALKTHROUGH

Equal-Height Proof: Why 1/2 + 1/4 + 1/4 Equals 1 Whole

01 02 03 04 — SKU: FRAC-01 • KEEP IN LID — SIDE A • 2ND UNFOLD (BOT-LEFT)

## 1 ESTABLISH 1.0 DATUM & EQUIVALENCE

1. **SEAT THE 1.0 WHOLE REFERENCE TOWER:**
Place the gold 84.00mm Whole block into the comparison socket.
Its summit establishes the benchmark ceiling for 1 whole unit.

2. **BUILD EQUIVALENT TOWERS SIDE-BY-SIDE:**
- Take two 1/2 blocks (42mm each): stack them → exactly 84.00 mm!
- Take one 1/2 block and two 1/4 blocks: 42 + 21 + 21 = 84.00 mm!
- Take four 1/4 blocks: 21 × 4 = 84.00 mm!

3. **ROTATE MULTI-NOTATION FACES:**
Turn blocks to verify decimal and percent sums match 1.0 and 100%:
0.50 + 0.25 + 0.25 = 1.00 | 50% + 25% + 25% = 100%

### BENCHMARK LEVELING PROOF

Slide a straight-edge across the top of both towers. They meet seamlessly with zero step, zero gap, and zero wobble. The physical height proves mathematical equivalence directly to the student's fingertips.

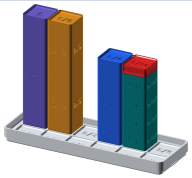
STEP B: TRY EXAMPLE #1 ON YOUR CALCULATOR — [FDISP] & [g][SIMP]

STEP 1 (ENTER 4/6): Press 4 [ENTER] 6 [÷] → Pushes 0.6667 (four Blue [1/6] blocks = 56.0 mm) onto the stack.
STEP 2 (ADD 1/12): Press 1 [ENTER] 12 [÷] [+] → Adds 0.0833 (7.0 mm) to reach 0.7500 (63.0 mm total height).
STEP 3 (PRESS BOX ICON [FDISP]): Press [FDISP] → Converts 0.75 into exact reduced fraction 3/4 (matching three [1/4] blocks)!

LIVE 4-REGISTER RPN STACK READOUT (AFTER KEYSTROKES):
T: 0 | Z: 0 | Y: 0 | X: 3/4 → DISPLAY (X) = 3/4

## LESSON 1: LID TRAY EQUIVALENCE ↔ [FDISP] KEYSTROKES

STEP A: HANDS-ON WITH BLOCKS ON LID TRAY — 1:1 CAD SCENE



1.0 Whole Tower — 84.00 mm (12U) | 3/4 Tower — 63.00 mm (9U) | 4/6 + 1/12 Tower — 63.00 mm (9U)

▲ Seat fraction towers on the 12-socket Lid Tray: when two towers stand level, the fractions are equal!

STEP B: TRY IT ON YOUR CALCULATOR — BOX ICON: [FDISP] & [g][SIMP]

EQUIVALENCE TREE: 4/6 + 1/12 = 3/4 (63.0 MM)

9/12 = 3/4 (63.0 mm) → 4 × [1/6] (56mm) + 1×[1/12] (7mm) → 4/6 = 8/12 + 1/12 = 3/4 (63mm)

63.0 mm = Three [1/4] Blocks on Lid!

PRESS BOX ICON & RPN KEYS (SCG):

1. 4/6 + 1/12: 4 ENTER 6 ÷ 1 ENTER 12 ÷
2. BOX ICON KEY: ÷ FDISP = 3/4

✓ DISPLAY (X) = 3/4

KEYSTROKES: 4 [ENTER] 6 [÷] 1 [ENTER] 12 [÷] [÷] [FDISP]
Compare fraction towers on the Lid Tray, then press the Box Icon key [FDISP] for exact reduced fractions!
✓ 4-REGISTER STACK: T: 0 | Z: 0 | Y: 0 | X: 3/4 → X = 3/4

4 NOTATIONS ON EVERY BLOCK • FRACTION, DECIMAL, PERCENT & RPN

CONTINUE RIGHT → PAGE 04

# 04 PAGE / 08 — EXAMPLE 2: CORE MATHEMATICAL PRINCIPLE

Adding Unlike Denominators: 1/3 + 1/4 = 7/12 by Leveling

01 02 03 04 — SKU: FRAC-01 • KEEP IN LID — SIDE A • 2ND UNFOLD (BOT-RIGHT)

## 2 UNLIKE DENOMINATOR ADDITION: 1/3 + 1/4

1. **THE CLASSIC FRACTION DILEMMA: ADD 1/3 + 1/4**
Students frequently make the error of adding across: 1/3 + 1/4 ≠ 2/7.
Stack one 1/3 block (28.0mm) and one 1/4 block (21.0mm) together:
Combined Tower Height = 28.00 + 21.00 = 49.00 mm

2. **FIND THE COMMON LEVELING MEASURING ROD:**
- Can 1/6 blocks measure it? 14 × 3 = 42mm (gap), 14 × 4 = 56mm (overshoots!).
- Can 1/8 blocks measure it? 10.5 × 4 = 42mm, 10.5 × 5 = 52.5mm (overshoots!).
- **Test the 1/12 blocks (7.00 mm):**

Stack 1/12 blocks one by one alongside the tower...

✓ EXACTLY 7 TWELFTH BLOCKS LEVEL PERFECTLY!

7 × (1/12) = 7 × 7.00 mm = 49.00 mm
1/3 + 1/4 = 4/12 + 3/12 = 7/12

Four 1/12 blocks level with 1/3 (28mm). Three 1/12 blocks level with 1/4 (21mm). Together, 4 + 3 = 7 blocks level with the summit with zero remainder!

### PHYSICAL COMMON DENOMINATOR PROOF

Students never have to memorize finding the LCD blindly again. The physical leveling proves why 12 is the common denominator and why the answer must be 7/12.

★ 2-PLAYER GAME: 'MYSTERY TOWER MATCH' (2 MINS)

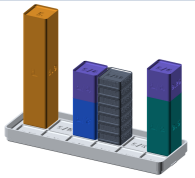
Player A builds a 2-block tower on the Lid Tray using unlike denominators (e.g., [1/2] + [1/6] = 56 mm). Player B must find a single-color tower of matching blocks that stands at the exact same height **(two [1/3] blocks = 2/3!) and verify with [FDISP] on the calculator.**

⚙ WHERE THIS SHOWS UP IN ENGINEERING, OPTICS & ELECTRONICS

Parallel resistors and camera lenses add by reciprocals: 1/R_total = 1/R₁ + 1/R₂!
When you stack [1/3] + [1/4] = 7/12 on the Lid Tray and verify with [FDISP] on your calculator, **you are using the exact harmonic fraction math electrical engineers use every day.**

## LESSON 2: 1/3 + 1/4 = 7/12 ON LID TRAY ↔ [FDISP] KEY

STEP A: HANDS-ON WITH BLOCKS ON LID TRAY — 1:1 CAD SCENE



1/3 Block (Blue) — 28.00 mm | 1/4 Block (Purple) — 21.00 mm | Seven 1/12 (Gray) — 7 × 7mm = 49.00 mm

▲ On the Lid Tray, [1/3] + [1/4] (28 + 21 = 49 mm) stands dead-level with seven [1/12] blocks!

STEP B: TRY IT ON YOUR CALCULATOR — BOX ICON: [FDISP] & [g][SIMP]

COMMON DENOMINATOR TREE: 1/3 + 1/4 = 7/12

1/3 + 1/4 = 7/12 (49 mm) → 1/3 = 4/12 (28mm) + 1/4 = 3/12 (21mm) → 4×[1/12] + 3×[1/12] = 7/12

LCD(3,4) = 12 • 28 + 21 = 49.0 mm

PRESS BOX ICON & RPN KEYS (SCG):

1. 1/3 + 1/4: 1 ENTER 3 ÷ 1 ENTER 4 ÷
2. BOX ICON KEY: ÷ FDISP = 7/12

✓ DISPLAY (X) = 7/12

KEYSTROKES: 1 [ENTER] 3 [÷] 1 [ENTER] 4 [÷] [÷] [FDISP]
Pressing [FDISP] (the exact icon embossed on your Fraction Studio box!) toggles 0.5833 ↔ 7/12!
✓ 4-REGISTER STACK: T: 0 | Z: 0 | Y: 0 | X: 7/12 → X = 7/12

COMMON DENOMINATORS MADE VISIBLE • MATCH HEIGHTS TO ADD

STEP 3: FLIP SHEET OVER ↻ PAGE 05

# 05 — EXAMPLE 3 & DESK CHALLENGES

PAGE / 08

Rotate Any Block to Translate Fractions, Decimals & Percents

05 06 07 08

SKU: FRAC-01 • KEEP IN LID

SIDE B • FLIP SIDE (TOP-LEFT)

## 3 INSTANT FRACTION ↔ DECIMAL ↔ PERCENT TRANSLATION

**THE GOAL: Prove that 3/4, 0.75, and 75% are the exact same quantity.**

**1. STACK THREE 1/4 BLOCKS:**

Snap three Matte Blue [1/4] blocks together on the Lid Tray (height = 63 mm).

**2. SPIN THE TOWER 90° AT A TIME:**

Face 1 reads 1/4 + 1/4 + 1/4 = 3/4. Spin 90°: Face 2 reads 0.25 × 3 = 0.75.

**3. COMPARE AGAINST 3/4 OF A WHOLE:**

Spin again: Face 3 reads 25% × 3 = 75%. Face 4 shows the RPN key [4][1/x]!

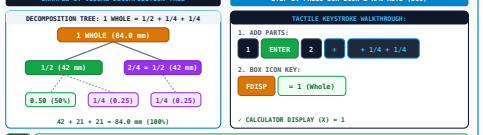
**✓ FOUR LANGUAGES FOR ONE PHYSICAL REALITY**

**Fractions, decimals, and percents aren't three different kinds of math.**
They are just four ways of naming the exact same 21 mm block height!
When students rotate a block in their hand, 1/5 = 0.20 = 20% becomes obvious.
Use Face 4 ([n][1/x]) with your SC-32 calculator to see live decimal reciprocals.

**PRO TIP: CLOSE YOUR EYES & FEEL THE ANSWER**

Once you build an example, run your fingers across the pieces. Your hands remember physical shapes and heights way longer than memorized rules!

EXAMPLE 3: VISUAL DECOMPOSITION TREE




STEP B: PRESS BOX ICON & RPN KEYS (SC6)




FDISP

FULL KEYSTROKES: 1 [ENTER] 2 [+] 1 [ENTER] 4 [+] [+] 1 [ENTER] 4 [+] [+] [FDISP]

✓ 4-REGISTER STACK: T: 0 | Z: 0 | Y: 0 | X: 1 → X = 1

## 3 DESK CHALLENGES — TRY THESE NOW!

**WARM-UP #1** — **Build 4 different single-color towers that match the [1/2] block!**

Step A: Build with blocks on your Lid Tray • Step B: Verify on SC-32 Calculator

✓ Build first, then flip booklet upside-down to check Answer #1 below!

**LEVEL-UP #2** — **Stack [1/3] + [1/6]. Which single block is the exact same height?**

Step A: Build with blocks on your Lid Tray • Step B: Verify on SC-32 Calculator

✓ Build first, then flip booklet upside-down to check Answer #2 below!

**BRAIN-MELTER #3** — **Find the height gap between [3/4] and [1/3] using [1/12] blocks!**

Step A: Build with blocks on your Lid Tray • Step B: Verify on SC-32 Calculator

✓ Build first, then flip booklet upside-down to check Answer #3 below!

SELF-CHECK ANSWER KEY (ROTATE BOOKLET 180° TO READ)

#3: Five [1/12] blocks (5/12)

#2: 2/4, 3/6, 4/8, 6/12 (2⨯3 = 1⨯6 = 4⨯2=8)

#1: 2/4 [1/2] blocks (2⨯3 = 1⨯6 = 4⨯2=8)

**FOR PARENTS & TEACHERS (Grades 3–8 (Fractions, Decimals, Percents & Ratios))**

- CCSS.MATH.CONTENT.3.NF.A.3: Explain equivalence of fractions and compare by size.
- CCSS.MATH.CONTENT.4.NF.C.6: Use decimal notation for fractions and compare quantities.
- CCSS.MATH.CONTENT.5.NF.A.1: Add and subtract fractions with unlike denominators.
- **CCSS.MATH.PRACTICE.MP2, MP4, MP5, MP7: Reason, Model, Use Tools & Structure.**

Next up on Page 06 → How to stack everything back in the box in 10 seconds!

**BOSS CHALLENGE #4** — **BLOCKS → BOX ICON KEY SPEED-RUN!**

1. Build Examples #1, #2, and #3 hands-on with your physical pieces.
2. Press the Box Button Icon ([FDISP] & [g][SIMP]) and keystrokes on your calculator:

• 1 [ENTER] 2 [+] 1 [ENTER] 4 [+] [+] 1 [ENTER] 4 [+] [+] [FDISP] → X = 1

**CHECK OFF YOUR MASTERY MILESTONES:**

- ☐ 1. Demonstrated Lessons #1, #2 & #3 hands-on with the physical blocks
- ☐ 2. Solved all 3 Desk Challenges above (check upside-down key if stuck!)
- ☐ 3. Demonstrated each lesson on the calculator using the Box Button Icon
- ☐ 4. Packed all pieces back into the box in under 10 seconds (see Page 06 →)

TEST YOURSELF • ANSWERS INCLUDED SO YOU CAN CHECK RIGHT AWAY

NEXT: HOW TO PACK UP → PAGE 06

# 06 — HOW TO STACK IT ALL BACK IN

PAGE / 08

The 10-Second Self-Checking Pack-Up Puzzle

05 06 07 08

SKU: FRAC-01 • KEEP IN LID

SIDE B • PACK-UP GUIDE (TOP-RIGHT)

## TOP-DOWN BOX MAP: WHERE EVERY PIECE LIVES

Putting the blocks away is a 10-second puzzle. Match each piece to its labeled home bay below so everything nests flush and the lid snaps shut:

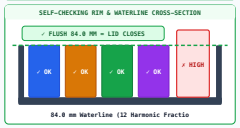
| | | | | | |
|---|---|---|---|---|---|
| 1x Whole "1" | 1x Whole "1" | 2x Half "1/2" | 2x Half "1/2" | 3x Third "1/3" | 4x Fourth "1/4" |
| 5x Fifth "1/5" | 6x Sixth "1/6" | 8x Eighth "1/8" | 9x Ninth "1/9" | 10x Tenth "1/10" | 12x Twelfth "1/12" |

**✓ THE 84.0 MM STORAGE BOX WATERLINE CHECK (▼)**

The 13U Storage Box has 12 columns across 10 denominators (2×[1], 2×[1/2], plus [1/3] to [1/12]). Because N blocks of size 1/N always add up to 1 Whole (84.0 mm), every single column levels off at the exact same 84.0 mm line when all 63 blocks are home. If any column looks short, you can tell in 2 seconds which fraction is still on the table!

1. STORAGE CASE & COMPARTMENTS

2. WHY THE LID SELF-CHECKS EVERY PIECE




▲ **SELF-INDEXING BAYS & EMBOSSED BOX BUTTON ICON:**
Calibrated bays guide every piece home, and the lid icon links directly to your calculator.

## THE 10-SECOND PACK-UP ROUTINE

**HOW TO STACK EVERYTHING BACK IN:**

1. **SORT BY COLOR & NUMBER:**
Group matching blocks or tiles together on your desk first.
2. **STACK & DROP INTO LABELED BAYS:**
Follow the Top-Down Box Map on the left. If a tower looks too tall or too short, a block is in the wrong bay—swap it until everything levels!
3. **TUCK THIS CARD IN THE LID & SNAP SHUT:**
Fold this guide back to [01] and rest it right on top before closing.

**12-SOCKET COMPARISON STAGE LID**

- Twelve comparison sockets on the Lid Tray let you test fraction towers side-by-side.
- Snap the Stage Lid directly onto the 13U Box Body to keep all 63 blocks secure.
- Easy to clean: wipe with a damp cloth or rinse in lukewarm soapy water.

**SHORT A PIECE AT CLEAN-UP TIME?**

First check under notebooks and chair legs! If a piece truly vanished, **see PAGE 08 (bottom-right) for your complete Part ID Reference Table.**

**✓ FRAC-01 4-POINT CLEAN-UP CHECKLIST**

| | |
|---|---|
| 1. 84MM WATERLINE: | All 12 fraction columns (2×1, 2×1/2, 1/3..1/12) level off at 84.0 mm. |
| 2. CLEAR LID TRAY: | Remove any fraction blocks still seated in the 10 Lid Tray sockets. |
| 3. FOLD CARD TO [01]: | Fold this reference booklet to Panel [01] and lay inside the lid. |
| 4. SNAP LID FLUSH: | Snap the Stage Lid flush onto the 13U box rim for silent carry! |

**MODULAR CLASSROOM & BOOKSHELF NESTING (42.0 MM PITCH)**

Every StackCalc box is sized in exact multiples of 42.0 mm (1×1, 3×1, 3×2). All 9 kit boxes interlock side-by-side in a classroom cart or drawer with zero wasted space, and every embossed box icon matches its calculator operation key!

84MM WATERLINE CHECK • EVERY COLUMN LEVELS AT THE TOP

NEXT: CONNECTED LAB ▼ PAGE 07

SIDE B (PAGES 05–08) • READ TOP ROW FIRST

THEN READ BOTTOM ROW (PAGE 07 → PAGE 08)

# 07 — THE STACKCALC CONNECTED LAB

PAGE / 08

How Your Kits & RPN Calculator Stack Work Together

05 06 07 08

SKU: FRAC-01 • KEEP IN LID

SIDE B • ECOSYSTEM (BOT-LEFT)

**ONE CONNECTED SYSTEM FROM KINDERGARTEN TO CALCULUS**

Every StackCalc box uses the same modular grid and unit heights—so pieces from different kits actually fit and work together.

## WHERE YOUR BOX FITS IN THE 9-KIT LAB

| | | |
|---|---|---|
| BASE-10 **Ten-Frames Lab** Early Counting & Regrouping | DOMI-01 **Math Domino Studio** Doubles & 12-Frame Part-Whole | FRAC-01 ◂ YOUR BOX **Fraction Studio** Proportional Towers & 4-Way Faces |
| PRIME-01 **Prime Factor Studio** Multiply by Stacking Prime Towers | ALGB-01 **Algebra Studio** Quadratic Area Arrays & Factoring | TIME-01 **Planetary Clock** 12:1 Geared Hands & Time Wedges |
| PROB-01 **Probability Dice** Fair Polyhedrals & 10-Sector Spinner | VOLV-01 **Unit Circle Volvelle** Spin-to-Read Exact Trig Radicals | SC-32-HW **SC-32 RPN Calculator** 4-Register Stack Scientific Engine |

**WHY THIS MATTERS: ZERO "THROW-AWAY" MATH**

In most classrooms, every grade uses a totally unrelated plastic toy. In StackCalc, the 7mm unit block you count with in 1st grade is the exact same height as the 1/12th fraction block and the [2] prime block later on!

1. FRAC-01 BOX LID ICON

[FDISP] & [g][SIMP]
EMBOSSED ON BOX LID
Build First → Press Key!

2. SC-6 ELEMENTARY RPN

3. SC-32 SCIENTIFIC RPN

▲ **BOX ICON ↔ CALCULATOR KEY CONTINUITY:**
Every physical box lid carries the exact icon of the calculator key that computes that lesson!

## HOW BLOCKS CONNECT TO THE CALCULATOR

**SHARED MODULAR FOOTPRINT & HEIGHTS**

- Shared 7mm unit pitch: the 1/12th block (7.0mm) matches Prime Block [2] and Time Block :05.
- Recessed sockets swallow the bottom peg so only true heights add.
- All storage boxes nest neatly side-by-side on any desk or shelf.

**WHY WE CALL IT "STACKCALC" (THE RPN SECRET)**

- Face 4 shows the SC-32 RPN reciprocal key: press [4] [1/x] on the calculator to get 0.25.
- Placing a block on the table = pressing [ENTER] to push onto the stack.
- Snapping two towers together = pressing [+] or [×] with zero parentheses!
- Mirrors the 4 stack levels on the SC-32 calculator: X, Y, Z, and T.

**FREE ACTIVITY DECKS & INTERACTIVE SIMULATORS**

Download extra printable puzzle cards, teacher lesson guides, and interactive browser simulations anytime—no login needed.

Explore All 9 Kits & Free Guides at stackcalc.io

LIVE 4-REGISTER RPN STACK

T: 0 (Top Memory)
Z: 0 (3rd Level)
Y: 0 (1st Block)
X: 3/4 (63.0 mm) (Active Display)

HOW BLOCKS MAP TO STACK KEYS

1. **PLACE 1ST PIECE ON STAGE:**
Type number + [ENTER] → lifts to Y
2. **ADD / STACK 2ND PIECE:**
Type 2nd number in X, then [+]/[×]
3. **PRESS BOX ICON KEY:**
Press [FDISP] & [g][SIMP]
✓ **Zero Parentheses • Always Visible!**

FDISP

BOX ICON = [FDISP] & [g][SIMP] • VERIFIED LESSON SEQUENCES:

- Lesson 1: 4 [ENTER] 6 [+] 1 [ENTER] 12 [÷] [=] [FDIS → X = 3/4
- Lesson 2: 1 [ENTER] 3 [+] 1 [ENTER] 4 [+] [FDISP → X = 7/12

MODULAR 9-KIT CONTINUUM • FROM EARLY COUNTING TO AP CALCULUS

NEXT: PART ID TABLE → PAGE 08

# 08 — KEEP THIS CARD! PART IDs & VIDEO

PAGE / 08

Complete 65-Piece Reference & 60-Second 3D Video Guide

05 06 07 08

SKU: FRAC-01 • KEEP IN LID

SIDE B • PART REFERENCE (BOT-RIGHT)

## PART ID REFERENCE TABLE (65 TOTAL PIECES)

**✓ KEEP THIS CARD INSIDE YOUR BOX LID!**

Backpacks eat blocks. Dogs chew dice. Vacuum cleaners take no prisoners. Keep this card in your lid so you always have the exact Part IDs below for single-piece refills!

| PART NAME | QTY | COLOR / FINISH | PART ID |
|---|---|---|---|
| 1.0 Whole Blocks (84mm) | 2x | Tan PLA | FRAC-FRAC-01W |
| 1/2 Half Blocks (42mm) | 4x | Tan PLA | FRAC-FRAC-02H |
| 1/3 Third Blocks (28mm) | 3x | Muted Red | FRAC-FRAC-03T |
| 1/4 Fourth Blocks (21mm) | 4x | Matte Blue | FRAC-FRAC-04F |
| 1/5 Fifth Blocks (16.8mm) | 5x | White PLA | FRAC-FRAC-05F |
| 1/6 Sixth Blocks (14mm) | 6x | Muted Red | FRAC-FRAC-06S |
| 1/8 Eighth Blocks (10.5mm) | 8x | Matte Blue | FRAC-FRAC-08E |
| 1/9 Ninth Blocks (9.33mm) | 9x | Muted Red | FRAC-FRAC-09N |
| 1/10 Tenth Blocks (8.4mm) | 10x | White PLA | FRAC-FRAC-10T |
| 1/12 Twelfth Blocks (7mm) | 12x | Matte Blue | FRAC-FRAC-12T |

**NEED A REPLACEMENT PIECE? NO NEED TO BUY A WHOLE NEW BOX.**

Enter any Part ID above at stackcalc.io/parts to order individual replacement pieces.

COLOR & PIECE IDENTIFICATION KEY — MATCH ANY LOOSE PART

| Tan PLA | Muted Red | Matte Blue | White PLA |
|---|---|---|---|
| Base-2: [1] (2x), [1/2] (4 | Base-3: [1/3], [1/6], [1/9 | Base-4: [1/4], [1/8], [1/1 | Base-5: [1/5], [1/10], Lid |

**UNIVERSAL STACKCALC HEIGHT PITCH (1U = 7.00 MM • 12U = 84.00 MM ONE WHOLE)**

Measure any loose piece in 7mm units (U) against this scale bar to identify it:

0U (0mm) 1U (7mm) 2U (14mm) 3U (21mm) 4U (28mm) 6U (42mm) 9U (63mm) 12U (84mm)

✓ DISTRICT PO & SAFETY: 10-Pack SKU FRAC-01-CLASS10 • CPSIA / ASTM F963 Non-Toxic Washable PLA/ABS

## WATCH THE 60-SECOND 3D VIDEO GUIDE

**POINT CAMERA TO WATCH DEMO**

stackcalc.io/products/fraction-studio/

See animated 3D examples & pack-up walkthrough

**QUESTIONS, CLASSROOM SETS & SUPPORT**

- Email us anytime: support@stackcalc.io
- School & District POs: school-orders@stackcalc.io

**Designed in Mountain View, CA • Non-Toxic & Washable**

COMPANION MANUAL CHAPTERS, STANDARDS & KEYSTROKES

**BOOK CHAPTERS:** SC-6 Guide Ch. 4 (Fraction Towers) • SC-32 Manual Ch. 3
**STANDARDS:** CCSS 3.NF.A.3, 4.NF.A.1, 5.NF.A.1 • TEKS 4.3C, 5.3H (Equivalence & LCD)
**WATERLINE SPEC:** 84.0 mm Waterline (12 Harmonic Fraction Lanes)

**ALL 3 VERIFIED CALCULATOR LESSON SEQUENCES (SCG):**

Ex 1: 4 [ENTER] 6 [÷] 1 [ENTER] 12 [÷] [=] [FDISP] → X=3/4
Ex 2: 1 [ENTER] 3 [+] 1 [ENTER] 4 [+] [=] [FDISP] → X=7/12
Ex 3: 1 [ENTER] 2 [÷] 1 [ENTER] 4 [÷] [=] 1 [ENTER] → X=1

FDISP

**BOX ICON KEY:** [FDISP] & [g][SIMP]
✓ Cross-Referenced to Companion Book & Calculator Engine
Fold back to [01] & keep this reference card inside your box lid!

LOST A PIECE? GRAB SINGLE REPLACEMENTS ANYTIME AT stackcalc.io/parts

✓ KEEP THIS GUIDE IN BOX LID!

# 01 — ALGEBRA FACTORING STUDIO & POLYNOMIAL TILES

PAGE / 08

Turn Quadratics Into Rectangles: +x², +x, and +1 Area Tiles

01 02 03 04

SKU: ALGB-01 • KEEP IN LID

SIDE A • COVER (TOP-LEFT)

**WHAT IS THIS BOX FOR? WHY "X-SQUARED" IS LITERALLY A SQUARE.**

Snap the corner rails together, slide in your tiles, and turn scary quadratic equations into a satisfying tabletop jigsaw puzzle!

## THE AREA MODEL: MULTIPLYING BY BUILDING RECTANGLES

The area of any rectangle is Width × Height. Our tiles turn algebra expressions into physical shapes you can arrange on your desk:

- **Quadratic Tile (x²): A big x-by-x square (38.6 × 38.6 mm)**
- **Linear Rod (x): An x-by-1 rectangle (38.6 × 19.0 mm)**
- **Unit Square (1): A 1-by-1 little square (19.0 × 19.0 mm)**

Green = Positive (+x², +x) | Purple = Negative (−x², −x) | Yellow/Red = (+1, −1).

### CORNER L-RAIL FACTORING FRAME

Snap the two L-rails together on your desk. Arrange your tiles inside the corner into a solid rectangle — then read the factors along the outer edges!

### WHAT'S INSIDE YOUR BOX (43 PIECES)

- 40 Algebra Tiles: 4× x² squares, 12× x rods, 24× unit squares (+ and −)
- 2× Interlocking Corner L-Rails + 1× 7U Storage Box
- Keep this folded guide inside your box lid for examples & part IDs!

★ YOUR FIRST 60 SECONDS WITH THIS BOX — START HERE!

1. **TEST WHY [+x] IS INCOMMENSURABLE:**
   Line up Yellow [+1] tiles (19.0mm) along a Green [+x] rod (38.6mm): x is a true variable length!
2. **BUILD YOUR FIRST RECTANGLE:**
   Snap the two Gray L-Rails together and pack 1×[+x²], 5×[+x], and 6×[+1] into a (x+2)(x+3) rectangle.
3. **PRESS THE BOX ICON KEY [r][POLY]:**
   Press [r] [POLY] on your SC-32 and enter 1 [A] 5 [B] 6 [C] 0 [D] [EXEC] to find roots −2 and −3!

EQN **BOX ICON = CALCULATOR [r][EQN] & [r][POLY] QUADRATIC SOLVER**

The button icon embossed on your box lid (left) is the exact calculator key ([r][EQN] & [r][POLY])! Build every lesson with your blocks, then press the key.

## L-RAIL FACTORING FRAME ↔ BOX [r][POLY] SOLVER ICON

STEP A: HANDS-ON WITH TILES IN L-RAILS — 1:1 CAD SCENE

Top Rail — Reads Width (x + 3)

Left Rail — Reads Height (x + 2)

Box Icon Key — [r][POLY] Roots −2, −3

▲ Snap the two Gray L-Rails together at 90° and pack your tiles inside—then verify roots with [r][POLY]!

EQN STEP B: TRY IT ON YOUR CALCULATOR — BOX ICON: [r][EQN] & [r][POL

FACTOR TREE: x² + 5x + 6 = (x + 2)(x + 3)

x² + 5x + 6 = 0

Factor (x+2)(x+3) | [r][POLY] Roots

[+x²] Green | 5×[+x] Green | r=−2, −3

Edges (x+2)(x+3) → Roots X=−2, Y=−3

PRESS BOX ICON & RPN KEYS (SC32):

1. [r][POLY]: r | POLY | 1 | A | 5 | B | 6 | C

2. SOLVE ROOTS: 0 | D | EXEC | X=−2, Y=−3

✓ DISPLAY (X) = −2

KEYSTROKES: [r] [POLY] 1 [A] 5 [B] 6 [C] 0 [D] [EXEC] [RCL] 1 [RCL] 3

Press [r][POLY] (the equation icon on your box!) to verify roots −2 and −3 of (x+2)(x+3)!

WELCOME TO HANDS-ON MATH • KEEP THIS GUIDE IN YOUR BOX LID

STEP 1: UNFOLD RIGHT → PAGE 02

▲ FOLD 1: FOLD BOTTOM HALF BEHIND TOP ▲

# 02 — HOW IT WORKS: THE CORE IDEA

PAGE / 08

Area = Length × Width: Factoring Is Just Building a Rectangle

01 02 03 04

SKU: ALGB-01 • KEEP IN LID

SIDE A • 1ST UNFOLD (TOP-RIGHT)

## POLYNOMIALS AS GEOMETRIC PUZZLES

When you multiply two binomials on paper (FOIL), you get four parts:

**(x + a)(x + b) = x² + ax + bx + ab**

Our Algebra Tiles turn those four parts into a literal rectangle: every piece has a real physical area:

- **Area of x² square = x × x**
- **Area of x rod = x × 1**
- **Area of unit tile = 1 × 1**

Factoring is just playing in reverse: grab the tiles for an expression, pack them into a solid rectangle, and read the side lengths!

### WHY THE 'X' ROD HAS SMOOTH UNMARKED SIDES

Because x is a variable that can stand for any number, the x rod is smooth with no unit tick marks on its face — reminding your brain that x is an unknown variable length, while the 1-tiles are fixed constants.

### THE ZERO-PAIR SUPERPOWER (+x AND −x = 0)

Place any positive tile next to its negative twin: (+x) + (−x) = 0, and (+1) + (−1) = 0! Adding a zero-pair doesn't change the total value, but it gives you the missing pieces needed to complete a rectangle.

3 CLASSIC MATH TRAPS THIS KIT MAKES IMPOSSIBLE — VISUAL PROOF

✗ **TRAP 1 — Writing (x + 3)² = x² + 9:**
✓ **FIX:** Build an (x+3)×(x+3) square in the rails—you immediately see the six Green [+x] wing rods you forgot!

✗ **TRAP 2 — Combining x² and x into '2x²':**
✓ **FIX:** A 38.6×38.6mm [+x²] square and a 38.6×19.0mm [+x] rod have different dimensions and cannot merge.

✗ **TRAP 3 — Guessing signs when factoring:**
✓ **FIX:** Purple/Red negative tiles (−x, −1) and Green/Yellow positive tiles (+x, +1) make zero-pairs visible.

CALCULATOR CHECK: [r] [POLY] 1 [A] 5 [B] 6 [C] 0 [D] [EXEC] [RCL] 1 [RCL] → X = −2

## TILE DIMENSIONS & COLOR GUIDE

| COMPONENT | DIMENSIONS | VALUE | COLOR |
|---|---|---|---|
| Quadratic (+x²) | 38.6 × 38.6 mm | +x² | Green |
| Quadratic (−x²) | 38.6 × 38.6 mm | −x² | Purple |
| Linear Rod (+x) | 38.6 × 19.0 mm | +x | Green |
| Linear Rod (−x) | 38.6 × 19.0 mm | −x | Purple |
| Unit Block (+1) | 19.0 × 19.0 mm | +1 | Yellow |
| Unit Block (−1) | 19.0 × 19.0 mm | −1 | Red |
| Corner L-Rails | 123.0 × 19.9 mm | Guides | Gray |
| 3x1 Storage Bin | 126 × 42 × 49 mm | 3x1x7U | Black |

### WHY YOUR BRAIN LOVES THIS

- Self-Checking Geometry: If your tiles form a solid rectangle with no missing corners or leftover pieces, your factoring is 100% correct!
- Custom Recessed Bays: Every tile has a home so cleanup takes 15 seconds.

1:1 CAD REFERENCE VIEW — BOX ICON & FORMULA QUICK-REFERENCE

1:1 BLOCK CAD ANATOMY

SOCKET (1.85mm) ▲ 1.85mm

1U = 7.00 mm VISIBLE DATUM

7mm

BASE PLUG (1.85mm) ▼ 1.85mm

✓ FLUSH PLUG-AND-SOCKET JOINT

Socket swallows plug completely: Only 7.00mm true heights add!

**Box Button Icon Key:**
[r][EQN] & [r][POLY] Polynomial Solver

**Rectangle Area Law:**
Area = Width × Height = ax² + bx + c

**Zero-Pair Identity:**
[+x] + [−x] = 0 and [+1] + [−1] = 0

**Complete the Square:**
Split b rods in half → add (b/2)² units

**SC-32 Quadratic Roots:**
[r][POLY] a[A] b[B] c[C] 0[D] [EXEC]

GEOMETRIC ALGEBRA • EVERY TRINOMIAL IS A RECTANGLE PUZZLE

STEP 2: UNFOLD DOWN ▼ PAGES 03–04

▼ UNFOLD STEP 2: OPEN DOWN FOR PAGES 03 & 04 ▼

# 03 — EXAMPLE 1: HANDS-ON WALKTHROUGH

PAGE / 08

Build (x + 2)(x + 3) = x² + 5x + 6 Inside the Corner Rails

01 02 03 04

SKU: ALGB-01 • KEEP IN LID

SIDE A • 2ND UNFOLD (BOT-LEFT)

## 1 DEPLOY RAILS & MODEL TRINOMIALS

**GOAL: Factor the quadratic expression x² + 5x + 6**

1. DEPLOY THE FACTORING CORNER RAILS:
Lock the two L-rails together on your desk. The inside corner establishes the Cartesian origin (0, 0).

2. GATHER THE POLYNOMIAL TILES:
Take one +x² tile, five +x linear rods, and six +1 unit squares.

3. ARRANGE INTO A SOLID RECTANGLE:
- Seat the +x² tile flush against the corner origin.
- Place three +x rods horizontally along the bottom rail.
- Place two +x rods vertically along the side rail.
- Fill the remaining corner void with the six +1 unit tiles!

### READ FACTORS DIRECTLY FROM OUTER EDGES

Look at the outer dimensions of your completed rectangle:

- **Horizontal Width: One x length + three unit widths = (x + 3)**
- **Vertical Height: One x length + two unit heights = (x + 2)**
- ✓ **Physical Factoring: x² + 5x + 6 = (x + 2)(x + 3)**

EQN STEP B: TRY EXAMPLE #1 ON YOUR CALCULATOR — [r][EQN] & [r][POL

**STEP 1 (BUILD RECTANGLE):**
Pack 1×[+x²], 5×[+x], and 6×[+1] inside the Gray L-Rails to form edges (x + 2) and (x + 3).

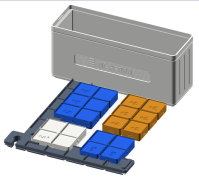
**STEP 2 (EVALUATE AT x=4):**
Press 4 [ENTER] 2 [+] (= 6) then 4 [ENTER] 3 [+] (= 7) and [×] → X = 42 (matches 4²+5(4)+6!).

**STEP 3 (SOLVE WITH [r][POLY]):**
Press [r][POLY] 1 [A] 5 [B] 6 [C] 0 [D] [EXEC] [RCL] 1 [RCL] 3 → Y = −3, X = −2!

LIVE 4-REGISTER RPN STACK READOUT (AFTER KEYSTROKES):
T: 0 | Z: 0 | Y: 0 | X: 42 → DISPLAY (X) = 42

## LESSON 1: (x+2)(x+3) RECTANGLE ↔ RPN AREA EVALUATION

STEP A: HANDS-ON WITH TILES IN L-RAILS — 1:1 CAD SCENE



Corner [+x²] Green — 38.6 × 38.6 mm

Five [+x] Green — 38.6 × 19.0 mm

Six [+1] Yellow — 19.0 × 19.0 mm

▲ Slide +x², five +x rods, and six +1 units tightly into the L-Rails: the outer edges read (x + 2) by (x + 3)!

EQN STEP B: TRY IT ON YOUR CALCULATOR — BOX ICON: [r][EQN] & [r][POL

FOIL AREA TREE: (x + 2)(x + 3) AT x = 4

(x + 2)(x + 3) = 42

Width: 4+2 = 6 | Height: 4+3 = 7

x²=16 | +5x=20 | +6=6

6 × 7 = 16 + 20 + 6 = 42

PRESS BOX ICON & RPN KEYS (SC32):

1. WIDTH x=2: 4 | ENTER | 2 | + | = 6

2. × HEIGHT: 4 | ENTER | 3 | + | × | = 42

✓ DISPLAY (X) = 42

KEYSTROKES: 4 [ENTER] 2 [+] 4 [ENTER] 3 [+] [×]

Build the (x+2)(x+3) rectangle in the L-Rails, then test x=4 on the RPN stack: 6 × 7 = 42!

✓ 4-REGISTER STACK: T: 0 | Z: 0 | Y: 0 | X: 42 → X = 42

READ FACTORS ON THE RAILS • OUTER EDGES REVEAL (x + a)(x + b)

CONTINUE RIGHT → PAGE 04

# 04 — EXAMPLE 2: CORE MATHEMATICAL PRINCIPLE

PAGE / 08

Zero-Pair Magic: Factoring x² − 4 (Difference of Squares)

01 02 03 04

SKU: ALGB-01 • KEEP IN LID

SIDE A • 2ND UNFOLD (BOT-RIGHT)

## 2 ZERO-PAIR ANNIHILATION: FACTOR x² − 4

**THE PUZZLE: How do you build a rectangle with x² and −4?**
If you place +x² and four −1 tiles, there is a giant corner gap. You cannot form a rectangle without adding linear rods!

**THE ZERO-PAIR BREAKTHROUGH:**
- Add two positive +x rods AND two negative −x rods.

**(+2x) + (−2x) = 0 (The total value of the expression is unchanged!)**

- Place the two +x rods along the horizontal base.
- Place the two −x rods along the vertical side.
- Fill the remaining corner with four red −1 unit tiles!

**(+2) × (−2) = −4 (Matches the constant term perfectly!)**

### ✓ DIFFERENCE OF SQUARES PHYSICAL THEOREM

**x² − a² = (x + a)(x − a)**

**x² − 4 = (x + 2)(x − 2)**

The outer rail lengths now read (x + 2) and (x − 2) directly. The positive and negative linear terms cancel out in the middle!

### AHA! DISCOVERY MOMENT

Students physically witness zero-pairs in action. Annihilation is not just an abstract symbol manipulation; it is a physical symmetry that completes geometric shapes!

★ 2-PLAYER GAME: 'SECRET RECTANGLE PUZZLE' (2 MINS)

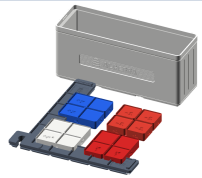
Player A picks a factor pair (like (x+3)(x−1)), expands it, and hands the simplified pile of tiles (1×[+x²], 2×[+x], 3×[−1]) to Player B. Player B must add zero-pairs ([+x]+[−x]) until all tiles **pack into a tight rectangle inside the L-Rails, then verify roots with [r][POLY]!**

⊙ WHERE THIS SHOWS UP IN ARCHITECTURE, GRAPHICS & AI

Every time al-Khwarizmi solved an equation in 820 AD, he literally 'completed the square' by moving rectangular strips around a square! Today, parabolic trajectories in physics and polynomial **splines in 3D animation use this exact root-and-area geometry.**

## LESSON 2: DIFFERENCE OF SQUARES (x²−4) ↔ [r][POLY]

STEP A: HANDS-ON WITH TILES IN L-RAILS — 1:1 CAD SCENE



Top-Left: [+x²] — 38.6 × 38.6 mm

Wings: +2x & −2x — Zero-Pair Cancel

Corner: 4×[−1] — x² − 4 = (x+2)(x−2)

▲ Adding two (+x, −x) zero-pairs completes the (x+2)(x−2) rectangle and reveals roots x = +2 and x = −2!

EQN STEP B: TRY IT ON YOUR CALCULATOR — BOX ICON: [r][EQN] & [r][POL

DIFFERENCE OF SQUARES: x² − 4 = (x+2)(x−2)

x² + 0x − 4 = 0

Add 2 Zero-Pairs | +2x − 2x = 0

[+x²] | 2×[+x] | 2×[−x] | 4×[−1]

(x+2)(x−2) → [r][POLY] Roots X=2, Y=−2

PRESS BOX ICON & RPN KEYS (SC32):

1. x²−4 POLY: r | POLY | 1 | A | 0 | B | 4 +/− | C

2. READ ROOTS: 0 | D | EXEC | X=2, Y=−2

✓ DISPLAY (X) = 2

KEYSTROKES: [r] [POLY] 1 [A] 0 [B] 4 [+/−] [C] 0 [D] [EXEC] [RCL] 1 [R

Adding two (+x)/(−x) zero-pairs completes (x+2)(x−2) and matches [r][POLY] roots X=2, Y=−2!

✓ 4-REGISTER STACK: T: −2 | Z: 0 | Y: −2 | X: 2 → X = 2

ZERO-PAIR SYMMETRY • WHY (+x) AND (−x) CANCEL IN MID-AIR

STEP 3: FLIP SHEET OVER ↻ PAGE 05

# 05 — EXAMPLE 3 & DESK CHALLENGES

PAGE / 08

Factoring Trinomials: x² + 5x + 6 + 3 Desk Challenges

05 06 07 08

SKU: ALGB-01 • KEEP IN LID

SIDE B • FLIP SIDE (TOP-LEFT)

## 3 FACTORING TRINOMIALS: x² + 5x + 6 = (x + 2)(x + 3)

**THE GOAL: Factor x² + 5x + 6 into a solid rectangle using the corner rails!**

**1. SEAT [+x²] AT THE ORIGIN:**
Place the large quadratic square tile at the intersection of the two L-rails.

**2. SPLIT FIVE [+x] RODS (2 & 3):**
Place two linear rods vertically and three rods horizontally along rails.

**3. FILL SIX [+1] UNITS:**
Drop six constant squares into the 2 × 3 corner to complete the rectangle.

### ✓ READ BINOMIAL FACTORS DIRECTLY OFF THE RAILS

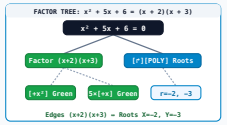
**The outer horizontal rail length is (x + 3); the vertical rail length is (x + 2).**
Area = Length × Width: x² + 5x + 6 = (x + 2)(x + 3) with zero pencil guessing!
Roots are where factors equal zero: x + 2 = 0 → x = −2, and x + 3 = 0 → x = −3.
Verify on calculator: [↱][POLY] 1 [A] 5 [B] 6 [C] [EXEC] finds roots x = −2, −3!

### PRO TIP: CLOSE YOUR EYES & FEEL THE ANSWER

Once you build an example, run your fingers across the pieces. Your hands remember physical shapes and heights way longer than memorized rules!

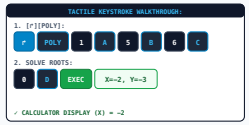
EXAMPLE 3: VISUAL DECOMPOSITION TREE

FACTOR TREE: x² + 5x + 6 → (x + 2)(x + 3)




STEP B: PRESS BOX ICON & RPN KEYS (SC32)

TACTILE KEYSTROKE WALKTHROUGH:




EQN

FULL KEYSTROKES: [↱] [POLY] 1 [A] 5 [B] 6 [C] 0 [D] [EXEC] [RCL] 1 [RCL] 3

✓ 4-REGISTER STACK: T: −3 | Z: 0 | Y: −3 | X: −2 ← X = −2

## 3 DESK CHALLENGES — TRY THESE NOW!

**WARM-UP #1** — Arrange 1x [+x²], 4x [+x], and 3x [+1] into a rectangle.
Step A: Build with blocks on your Lid Tray • Step B: Verify on SC-32 Calculator
✓ Build first, then flip booklet upside-down to check Answer #1 below!

**LEVEL-UP #2** — Use [+x] and [−x] zero-pairs to factor x² − x − 2.
Step A: Build with blocks on your Lid Tray • Step B: Verify on SC-32 Calculator
✓ Build first, then flip booklet upside-down to check Answer #2 below!

**BRAIN-MELTER #3** — How many [+1] tiles complete the square for x² + 4x?
Step A: Build with blocks on your Lid Tray • Step B: Verify on SC-32 Calculator
✓ Build first, then flip booklet upside-down to check Answer #3 below!

SELF-CHECK ANSWER KEY (ROTATE BOOKLET 180° TO READ)

ƨɹǝʍsuⱯ ʇɔǝʜƆ-ꟻƎꟻ ʇuǝʌǝ ƨ,ʇI ⊃ ɹ,(Z/Þ) ↑Z#
(ɪ + X)(Z − X) ↑Z# | Ɛ + Xþ + ,x = (Ɛ + X)(ɪ + X) ↑I#

### FOR PARENTS & TEACHERS (Grades 7–11 (Pre-Algebra, Algebra I & Algebra II))

- CCSS.MATH.CONTENT.HSA.SSE.A.2: Use the structure of an expression to rewrite it.
- CCSS.MATH.CONTENT.HSA.REI.B.4.A: Use the method of completing the square.
- TEKS 111.39.c.10.E: Factor trinomials and difference of two squares.
- **CCSS.MATH.PRACTICE.MP2, MP4, MP5, MP7: Reason, Model, Use Tools & Structure.**

Next up on Page 06 → How to stack everything back in the box in 10 seconds!

**BOSS CHALLENGE #4** — BLOCKS → BOX ICON KEY SPEED-RUN!

1. Build Examples #1, #2, and #3 hands-on with your physical pieces.
2. Press the Box Button Icon ([↱][EQN] & [↱][POLY]) and keystrokes on your calculator:
• [↱] [POLY] 1 [A] 5 [B] 6 [C] 0 [D] [EXEC] [RCL] 1 [RCL] 3 → X = −2

**CHECK OFF YOUR MASTERY MILESTONES:**

- ☐ 1. Demonstrated Lessons #1, #2 & #3 hands-on with the physical blocks
- ☐ 2. Solved all 3 Desk Challenges above (check upside-down key if stuck!)
- ☐ 3. Demonstrated each lesson on the calculator using the Box Button Icon
- ☐ 4. Packed all pieces back into the box in under 10 seconds (see Page 06 →)

TEST YOURSELF • ANSWERS INCLUDED SO YOU CAN CHECK RIGHT AWAY

NEXT: HOW TO PACK UP → PAGE 06

SIDE B (PAGES 05–08) • READ TOP ROW FIRST

# 06 — HOW TO STACK IT ALL BACK IN

PAGE / 08

The 10-Second Self-Checking Pack-Up Puzzle

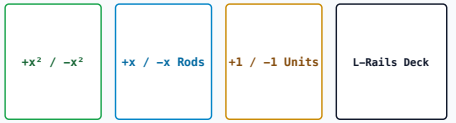
05 06 07 08

SKU: ALGB-01 • KEEP IN LID

SIDE B • PACK-UP GUIDE (TOP-RIGHT)

## TOP-DOWN BOX MAP: WHERE EVERY PIECE LIVES

Putting the blocks away is a 10-second puzzle. Match each piece to its labeled home bay below so everything nests flush and the lid snaps shut:




### ✓ CUSTOM-SHAPED BAYS FOR EVERY TILE TYPE

The box is divided into dedicated bays for Quadratic Squares (+x²/−x²), Linear Rods (+x/−x), Unit Constants (+1/−1), and the two Corner L-Rails. When all 40 tiles and 2 rails are stacked in their matching bays, the top surface sits completely flat with zero rattle.

1. STORAGE CASE & COMPARTMENTS

2. WHY THE LID SELF-CHECKS EVERY PIECE

SELF-CHECKING RIM & WATERLINE CROSS-SECTION

▲ SELF-INDEXING BAYS & EMBOSSED BOX BUTTON ICON:
Calibrated bays guide every piece home, and the lid icon links directly to your calculator.

## THE 10-SECOND PACK-UP ROUTINE

### HOW TO STACK EVERYTHING BACK IN:

**1. SORT BY COLOR & NUMBER:**
Group matching blocks or tiles together on your desk first.

**2. STACK & DROP INTO LABELED BAYS:**
Follow the Top-Down Box Map on the left. If a tower looks too tall or too short, a block is in the wrong bay—swap it until everything levels!

**3. TUCK THIS CARD IN THE LID & SNAP SHUT:**
Fold this guide back to [01] and rest it right on top before closing.

### CORNER L-RAILS & FACTOR AXES

- Place the two L-Rails on your desk to form a 90° factoring frame for any rectangle.
- Why doesn't [+x] equal a whole number of [+1] tiles? So you never confuse x with a fixed constant!
- Easy care: wipe clean with a damp cloth or lukewarm soapy water.

### SHORT A PIECE AT CLEAN-UP TIME?

First check under notebooks and chair legs! If a piece truly vanished, **see PAGE 08 (bottom-right) for your complete Part ID Reference Table.**

### ✓ ALGB-01 4-POINT CLEAN-UP CHECKLIST

| | |
|---|---|
| 1. SORT 4 TILE BAYS: | Stack +x²/−x² (38.6mm), +x/−x (38.6×19mm), and +1/−1 (19mm) in their bays |
| 2. SEAT L-RAILS: | Nest both Gray Corner L-Rails (123.0 mm) flat along the side rail deck. |
| 3. FOLD CARD TO [01]: | Fold this reference booklet to Panel [01] and lay inside the lid. |
| 4. SNAP LID FLUSH: | Close the 3×1 box lid—all 43 pieces lock with zero rattle! |

### MODULAR CLASSROOM & BOOKSHELF NESTING (42.0 MM PITCH)

Every StackCalc box is sized in exact multiples of 42.0 mm (1×1, 3×1, 3×2). All 9 kit boxes interlock side-by-side in a classroom cart or drawer with zero wasted space, and every embossed box icon matches its calculator operation key!

CUSTOM TILE BAYS • LID ONLY SLIDES SHUT WHEN ALL 43 PIECES ARE HOME

NEXT: CONNECTED LAB ▼ PAGE 07

THEN READ BOTTOM ROW (PAGE 07 → PAGE 08)

# 07 — THE STACKCALC CONNECTED LAB

PAGE / 08

How Your Kits & RPN Calculator Stack Work Together

05 06 07 08

SKU: ALGB-01 • KEEP IN LID

SIDE B • ECOSYSTEM (BOT-LEFT)

### ONE CONNECTED SYSTEM FROM KINDERGARTEN TO CALCULUS

Every StackCalc box uses the same modular grid and unit heights—so pieces from different kits actually fit and work together.

## WHERE YOUR BOX FITS IN THE 9-KIT LAB

| | | |
|---|---|---|
| BASE-10 **Ten-Frames Lab** Early Counting & Regrouping | DOMI-01 **Math Domino Studio** Doubles & 12-Frame Part-Whole | FRAC-01 **Fraction Studio** Proportional Towers & 4-Way Faces |
| PRIME-01 **Prime Factor Studio** Multiply by Stacking Prime Towers | ALGB-01 ◂ YOUR BOX **Algebra Studio** Quadratic Area Arrays & Factoring | TIME-01 **Planetary Clock** 12:1 Geared Hands & Time Wedges |
| PROB-01 **Probability Dice** Fair Polyhedrals & 10-Sector Spinner | VOLV-01 **Unit Circle Volvelle** Spin-to-Read Exact Trig Radicals | SC-32-HW **SC-32 RPN Calculator** 4-Register Stack Scientific Engine |

### WHY THIS MATTERS: ZERO "THROW-AWAY" MATH

In most classrooms, every grade uses a totally unrelated plastic toy. In StackCalc, the 7mm unit block you count with in 1st grade is the exact same height as the 1/12th fraction block and the [2] prime block later on!

1. ALGB-01 BOX LID ICON

EQN

[↱][EQN] & [↱][POLY]
EMBOSSED ON BOX LID
Build First → Press Key!

2. SC-6 ELEMENTARY RPN

3. SC-32 SCIENTIFIC RPN

▲ BOX ICON → CALCULATOR KEY CONTINUITY:
Every physical box lid carries the exact icon of the calculator key that computes that lesson!

## HOW BLOCKS CONNECT TO THE CALCULATOR

### SHARED MODULAR FOOTPRINT & HEIGHTS

- Constant [+1] tiles (19.0 × 19.0 mm) match the universal StackCalc grid footprint.
- Recessed sockets swallow the bottom peg so only true heights add.
- All storage boxes nest neatly side-by-side on any desk or shelf.

### WHY WE CALL IT "STACKCALC" (THE RPN SECRET)

- Bridges physical factoring to the SC-32 calculator: test any root x by evaluating on the RPN stack.
- Placing a block on the table = pressing [ENTER] to push onto the stack.
- Snapping two towers together = pressing [+] or [×] with zero parentheses!
- Mirrors the 4 stack levels on the SC-32 calculator: X, Y, Z, and T.

### FREE ACTIVITY DECKS & INTERACTIVE SIMULATORS

Download extra printable puzzle cards, teacher lesson guides, and interactive browser simulations anytime—no login needed.

Explore All 9 Kits & Free Guides at stackcalc.io

LIVE 4-REGISTER RPN STACK

T: −3 (Top Memory)
Z: 0 (3rd Level)
Y: −3 (Root 1) (1st Block)
X: −2 (Root 2) (Active Display)

HOW BLOCKS MAP TO STACK KEYS

1. PLACE 1ST PIECE ON STAGE:
Type number + [ENTER] → lifts to Y
2. ADD / STACK 2ND PIECE:
Type 2nd number in X, then [+]/[×]
3. PRESS BOX ICON KEY:
Press [↱][EQN] & [↱][POLY]
✓ Zero Parentheses • Always Visible!

EQN

BOX ICON = [↱][EQN] & [↱][POLY] • VERIFIED LESSON SEQUENCES:
• Lesson 1: 4 [ENTER] 2 [+] 4 [ENTER] 3 [+] [×] → X = 42
• Lesson 2: [↱] [POLY] 1 [A] 0 [B] 4 [+/−] [C] 0 [D] [ → X = 2

MODULAR 9-KIT CONTINUUM • FROM EARLY COUNTING TO AP CALCULUS

NEXT: PART ID TABLE → PAGE 08

# 08 — KEEP THIS CARD! PART IDs & VIDEO

PAGE / 08

Complete 43-Piece Reference & 60-Second 3D Video Guide

05 06 07 08

SKU: ALGB-01 • KEEP IN LID

SIDE B • PART REFERENCE (BOT-RIGHT)

## PART ID REFERENCE TABLE (43 TOTAL PIECES)

### ✓ KEEP THIS CARD INSIDE YOUR BOX LID!

Backpacks eat blocks. Dogs chew dice. Vacuum cleaners take no prisoners. Keep this card in your lid so you always have the exact Part IDs below for single-piece refills!

| PART NAME | QTY | COLOR / FINISH | PART ID |
|---|---|---|---|
| Quadratic Tiles (+x²) | 2x | Matte Blue | ALGB-ALGE-27b |
| Quadratic Tiles (−x²) | 2x | Muted Red | ALGB-ALGE-34c |
| Linear Rod Tiles (+x) | 6x | Matte Blue | ALGB-ALGE-2d2 |
| Linear Rod Tiles (−x) | 6x | Muted Red | ALGB-ALGE-2d2 |
| Constant Units (+1) | 12x | Matte Blue | ALGB-ALGE-76c |
| Constant Units (−1) | 12x | Muted Red | ALGB-ALGE-76c |
| Factoring Corner L-Rails | 2x | White PLA | ALGB-ALGE-ac0 |
| 7U Storage Box Body | 1x | Gray / White | ALGB-ALGE-49c |

### NEED A REPLACEMENT PIECE? NO NEED TO BUY A WHOLE NEW BOX.

Enter any Part ID above at stackcalc.io/parts to order individual replacement pieces.

COLOR & PIECE IDENTIFICATION KEY — MATCH ANY LOOSE PART

| Matte Blue | Muted Red | White PLA | Gray |
|---|---|---|---|
| Positive Tiles (+x², +x, +1 | Negative Tiles (−x², −x, −1 | Corner L-Rails (2x) | 7U Storage Case (1x) |

UNIVERSAL STACKCALC HEIGHT PITCH (1U = 7.00 MM • 12U = 84.00 MM ONE WHOLE)

Measure any loose piece in 7mm units (U) against this scale bar to identify it:

0U (0mm) 1U (7mm) 2U (14mm) 3U (21mm) 4U (28mm) 6U (42mm) 9U (63mm) 12U (84mm)

✓ DISTRICT PO & SAFETY: 10-Pack SKU ALGB-01-CLASS10 • CPSIA / ASTM F963 Non-Toxic Washable PLA/ABS

## WATCH THE 60-SECOND 3D VIDEO GUIDE

POINT CAMERA TO WATCH DEMO
stackcalc.io/products/algebra-boxes/
See animated 3D examples & pack-up walkthrough

### QUESTIONS, CLASSROOM SETS & SUPPORT

- Email us anytime: support@stackcalc.io
- School & District POs: school-orders@stackcalc.io

**Designed in Mountain View, CA • Non-Toxic & Washable**

COMPANION MANUAL CHAPTERS, STANDARDS & KEYSTROKES

BOOK CHAPTERS: SC-32 Manual Ch. 8 (Polynomial Solvers) • SC-6 Guide Ch. 8
STANDARDS: CCSS 8.EE.C.7, HSA-SSE.A.2, HSA-REI.B.4 • TEKS A.7B, A.8A (Factoring)
WATERLINE SPEC: 4-Bay Tile Deck + Dual 123mm L-Rail Frame

ALL 3 VERIFIED CALCULATOR LESSON SEQUENCES (SC32):
Ex 1: 4 [ENTER] 2 [+] 4 [ENTER] 3 [+] [×] → X=42
Ex 2: [↱] [POLY] 1 [A] 0 [B] 4 [+/−] [C] 0 [D] [EXE → X=2
Ex 3: [↱] [POLY] 1 [A] 5 [B] 6 [C] 0 [D] [EXEC] [RC → X=−2

EQN

BOX ICON KEY: [↱][EQN] & [↱][POLY]
✓ Cross-Referenced to Companion Book & Calculator Engine
Fold back to [01] & keep this reference card inside your box lid!

LOST A PIECE? GRAB SINGLE REPLACEMENTS ANYTIME AT stackcalc.io/parts

✓ KEEP THIS GUIDE IN BOX LID!

# 01 CLASSROOM PLANETARY CLOCK & INTERVAL SYSTEM

PAGE / 08

Real 12:1 Planetary Gears + Pie-Slice Time Wedges

01 02 03 04

SKU: TIME-01 • KEEP IN LID

SIDE A • COVER (TOP-LEFT)

**WHAT IS THIS BOX FOR? A CLOCK THAT ACTUALLY TELLS THE TRUTH.**

Most toy clocks have loose hands that lie to you. Ours has real planetary gears inside: spin the minute ring 360° and the hour hand moves 1 hour!

## REAL 12:1 PLANETARY GEARS INSIDE

Why is reading an analog clock tricky at 2:45? Because the hour hand is almost on the 3! Our geared clock links both hands mechanically:

- **1 Full Turn of Minute Ring (360°) = 30° Hour Hand Turn (1 Hour)**
- **Always In Sync: You can never put the hour hand in the wrong place!**
- Drop-In Pie Wedges: 15-minute sectors show quarter-hours as fractions.

See continuous gear motion and fraction slices at the exact same time!

### ANALOG-TO-DIGITAL PRACTICE LID

The box lid has digital clock slots on top! Snap in the minute blocks (:00, :15, :30, :45) and AM/PM tiles to match your analog dial.

### WHAT'S INSIDE YOUR BOX (19 PIECES)

- 1× Geared Planetary Clock Core (12:1 Ratio) + Hour & Minute Hands
- 4× Quarter-Hour Pie Wedges (:15, :30, :45, :60) + 12× Digital & AM/PM Tiles
- Keep this folded guide inside your box lid for examples & part IDs!

★ YOUR FIRST 60 SECONDS WITH THIS BOX — START HERE!

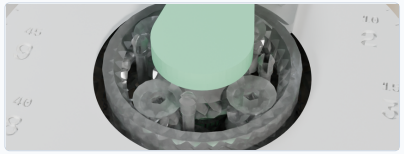
1. **SPIN THE MINUTE RING ONE FULL TURN:** Turn the Green minute hand 360° from 12 back to 12–watch the Red hour hand advance 1 hour (30°)!
2. **DROP PIE WEDGES ONTO THE FACE:** Lay two 15-minute wedges from 12 to 6 to cover half the circle (30 minutes = 1/2 hour).
3. **PRESS THE BOX ICON KEY [🕒] / [g][▸HMS]:** Type 3.30 [f][▸HR] 0.45 [f][▸HR] [+] [g][▸HMS] on your calculator to bridge across 4:00 to 4:15!

**BOX ICON = CALCULATOR [🕒] CLOCK MODE & [G][▸HMS] TIME KEYS**

The button icon embossed on your box lid (left) is the exact calculator key ([🕒] / [g][▸HMS] / [f][▸HR])! Build every lesson with your blocks, then press the key.

## 12:1 PLANETARY GEARS ↔ BOX [🕒] & [g][▸HMS] ICON

STEP A: HANDS-ON WITH EPICYCLIC GEARS — 1:1 CAD SCENE



Minute Turn: 360° = 60 Minutes | Hour Advance: 30° = 1 Hour | Box Icon Key: [🕒] & [g][▸HMS]

▲ **Spin the Green minute hand and watch the internal planet gears** drive the Red hour hand at a true 12:1 ratio!

STEP B: TRY IT ON YOUR CALCULATOR — BOX ICON: [🕒] / [g][▸HMS] /

FACTOR TREE: 60 MINUTES = 2 × 2 × 3 × 5

60 MIN (360° TURN) → 60 [g][FACT] | ÷ 4 = 15m Wedge → 15m (1/4) | 15m (1/4) | 15m (1/4) | 15m (1/4)

60 = 2²×3×5 • 60 ÷ 4 = 15 min

PRESS BOX ICON & RPM KEYS (SCG):

1. FACTOR 60: 6 0 g FACT 2²×3×5
2. 1/4 WEDGE: 4 ÷ = 15 min

✓ DISPLAY (X) = 15

KEYSTROKES: 60 [g] [FACT] 4 [÷]

Factor 60 minutes (2²×3×5) and divide by 4 quadrants to get each 15-minute wedge!

WELCOME TO HANDS-ON MATH • KEEP THIS GUIDE IN YOUR BOX LID

STEP 1: UNFOLD RIGHT → PAGE 02

▲ FOLD 1: FOLD BOTTOM HALF BEHIND TOP ▲

▼ UNFOLD STEP 2: OPEN DOWN FOR PAGES 03 & 04 ▼

# 02 HOW IT WORKS: THE CORE IDEA

PAGE / 08

12:1 Geared Hand Link & Modular-12 Clock Arithmetic

01 02 03 04

SKU: TIME-01 • KEEP IN LID

SIDE A • 1ST UNFOLD (TOP-RIGHT)

## HOW PLANETARY GEARS LINK ANGLES TO TIME

Inside your clock is an epicyclic ("planetary") gear train that divides every rotation of the outer minute ring by 12:

**Gear Ratio = ω_minute / ω_hour = 12 : 1**

The angle of the hour hand θ_hour is always locked to 1/12th of the angle swept by the minute hand θ_minute:

**θ_hour = (1 / 12) × θ_minute**

### INSIDE THE PLANETARY GEARBOX

- Sun Gear: 12-tooth center driver | Planet Gears: 18-tooth orbiting idlers
- Ring Gear: 48-tooth outer ring with smooth finger-turn grip
- Continuous Mesh: 1 full turn (360°) advances the hour hand by 30°
- Built to Last: Gears are permanently meshed so they never slip out of time

### TIME IS JUST FRACTIONS OF A CIRCLE

Why do we call 15 minutes "a quarter past" and 30 minutes "half past"? Drop the colored quadrant wedges onto the clock face and you'll see: 15 minutes is a 90° quarter-circle (1/4), and 30 minutes is a 180° half-circle (1/2)!

### 3 CLASSIC MATH TRAPS THIS KIT MAKES IMPOSSIBLE — VISUAL PROOF

✗ **TRAP 1 – Misreading 2:45 as '3:45':**
✓ **FIX:** Because the Red hour hand is geared to move gradually toward 3, learners see WHY it's almost 3:00.

✗ **TRAP 2 – Adding time like base-10 (1:45 + :30 ≠ 1:75):**
✓ **FIX:** Putting two :15 wedges past 1:45 (or pressing [g][▸HMS]) visibly crosses 12:00 into 2:15!

✗ **TRAP 3 – Not getting 'quarter past' or 'quarter to':**
✓ **FIX:** Each 15-minute wedge is literally 1/4 (one quarter) of the circular clock face.

CALCULATOR CHECK: 60 [g] [FACT] 4 [÷] = X = 15

## TIME TO ANGLE REFERENCE TABLE

| INTERVAL | MINUTE ARC | HOUR ARC | FRACTION |
|---|---|---|---|
| 60 Min (1 Hr) | 360° (Full) | 30.0° | 1 / 1 (Whole) |
| 45 Min (3/4) | 270° | 22.5° | 3 / 4 Circle |
| 30 Min (1/2) | 180° | 15.0° | 1 / 2 Circle |
| 15 Min (1/4) | 90° | 7.5° | 1 / 4 Circle |
| 5 Min (1/12) | 30° | 2.5° | 1 / 12 Circle |
| 1 Minute | 6° | 0.5° | 1 / 60 Circle |

### WHY YOUR BRAIN LOVES THIS

- No Batteries Needed: Powered 100% by your fingers and real gears.
- Zero Loose Screws: Monolithic gear assembly can't come apart in a desk.
- No Base-60 Headaches: Count elapsed time by dropping in 15-min wedges.
- Glow-in-the-Dark Hands: High-contrast hands are easy to read anywhere.
- Easy Care: Wipe clean or rinse in cool soapy water (never oil the gears).

1:1 CAD REFERENCE VIEW — BOX ICON & FORMULA QUICK-REFERENCE

1:1 BLOCK CAD ANATOMY

SOCKET (1.85mm) ▲ 1.85mm | 10 = 7.00 mm VISIBLE DATUM | 7mm | BASE PLUG (1.85mm) ▼ 1.85mm

✓ FLUSH PLUG-AND-SOCKET JOINT
Socket swallows plug completely:
**Only 7.00mm true heights add!**

**Box Button Icon Key:** [🕒] Clock Mode & [g][▸HMS] / [f][▸HR]

**Planetary Gear Law:** θ_hour = θ_minute ÷ 12

**1 Hour Advance:** 360° minute turn = 30° hour turn

**15-Min Quadrant Wedge:** 1/4 Hour = 90° = π/2 radians

**Add HH.MM Times:** T₃ [f][▸HR] T₂ [f][▸HR] [+] [g][▸HMS]

REAL PLANETARY GEARS • THE HANDS CAN NEVER FALL OUT OF SYNC

STEP 2: UNFOLD DOWN ▼ PAGES 03–04

# 03 EXAMPLE 1: HANDS-ON WALKTHROUGH

PAGE / 08

Spin the Minute Ring: Watch the Hour Hand Follow Automatically

01 02 03 04

SKU: TIME-01 • KEEP IN LID

SIDE A • 2ND UNFOLD (BOT-LEFT)

## 1 EXPLORING CONTINUOUS EPICYCLIC MOTION

1. **ROTATE THE MINUTE RING BY HAND:**
Grip the outer knurled ring and rotate it clockwise.
Watch the hour hand move automatically through internal gears!

2. **TEST 1 FULL REVOLUTION (60 MINUTES):**
- Start both hands at 12:00.
- Turn minute ring 360° until minute hand lands back on 12.
- **Observe hour hand: It has advanced exactly to 1:00 (30°)!**

3. **DIGITAL LID PRACTICE:**
Drop hour block [ 1 ] and interval block [ :00 ] into the 3x2 lid to bridge analog clock hands with digital displays (1:00 PM).

### WHY PLANETARY GEARING MATTERS FOR LEARNING

Toy teaching clocks often have loose hands that students move independently, teaching them to place the hour hand incorrectly (e.g. putting the hour hand on 2 at 2:50). The planetary gear train enforces realistic continuous movement!

STEP B: TRY EXAMPLE #1 ON YOUR CALCULATOR — [🕒] / [g][▸HMS] /

**STEP 1 (SPIN TO 3:30):**
Align both physical hands at 12:00, then spin the Green minute hand 3.5 turns to 3:30.

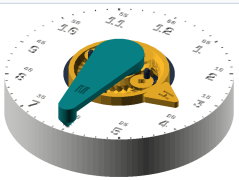
**STEP 2 (CONVERT WITH [f][▸HR]):**
Press 3.30 [f] [▸HR] → converts 3 hours 30 minutes into 3.5 decimal hours in Register X.

**STEP 3 (MULTIPLY BY 30°/HR):**
Press 30 [×] → X = 105 (proving the hour hand sits at 105°, halfway between 3 and 4)!

LIVE 4-REGISTER RPM STACK READOUT (AFTER KEYSTROKES):
**T: 0 | Z: 0 | Y: 0 | X: 105 → DISPLAY (X) = 105**

## LESSON 1: 12:1 PLANETARY CLOCK AT 3:30 ↔ [f][▸HR] KEY

STEP A: HANDS-ON WITH GEARED CLOCK HANDS — 1:1 CAD SCENE



Minute Hand: :30 = 180° (Half Turn) | Hour Hand: 3.5h = 105° (Halfway 3→4) | Gear Ratio: 12 : 1 Exact

▲ **When the minute hand spins 180° down to :30, the internal** planetary gears automatically advance the hour hand halfway to 4!

STEP B: TRY IT ON YOUR CALCULATOR — BOX ICON: [🕒] / [g][▸HMS] /

12:1 GEAR RATIO TREE AT 3:30

3:30 → 3.5 HOURS

3.30 [f][▸HR] = 3.5 | × 30° = 105° Hr | 3 Full Turns | + 180° (:30) | = 105° Hr

3.30 [f][▸HR] 30 [×] = 105°

PRESS BOX ICON & RPM KEYS (SCG):

1. TO DEC HR: 3.30 f ▸HR = 3.5 hr
2. HOUR ANGLE: 30 × = 105°

✓ DISPLAY (X) = 105

KEYSTROKES: 3.30 [f] [▸HR] 30 [×]

Spin the hands to 3:30, then convert 3.30 to 3.5 decimal hours with [f][▸HR] and multiply by 30°/hr!

✓ 4-REGISTER STACK: T: 0 | Z: 0 | Y: 0 | X: 105 → X = 105

LINKED 12:1 ROTATION • 360° MINUTE TURN = 30° HOUR ADVANCE

CONTINUE RIGHT → PAGE 04

# 04 EXAMPLE 2: CORE MATHEMATICAL PRINCIPLE

PAGE / 08

Elapsed Time as Pie Wedges: Three 15-Minute Slices = 45 Minutes

01 02 03 04

SKU: TIME-01 • KEEP IN LID

SIDE A • 2ND UNFOLD (BOT-RIGHT)

## 2 ELAPSED TIME AS ANGULAR ARC ADDITION

**THE PROBLEM: Find elapsed time from 2:15 PM to 3:45 PM.**
Subtracting time with 60-minute bases confuses many students.
Instead, use physical angular arc rotation:

1. **SET CLOCK TO 2:15 PM:**
Hour hand is past 2; minute hand points to 3 (15 min).

2. **ADVANCE MINUTE RING TO 3:45 PM:**
- Rotate 1 full circle (360°) to reach 3:15 PM → 60 Minutes.
- Rotate another half circle (180°) from 3:15 to 3:45 → 30 Minutes.

**Total Angular Turn = 360° + 180° = 540°**

3. **DROP IN QUADRANT ARCS TO CONFIRM:**
Drop in 15-minute quadrant wedges: 4 wedges (60m) + 2 wedges (30m)
**→ Total Elapsed Time = 6 wedges × 15 min = 90 Minutes (1 hr 30 min)!**

### ✓ NO BASE-60 ARITHMETIC CONFUSION

**Elapsed Time = Arc Swept by Minute Ring**

**540° / 360° = 1.5 Hours = 90 Minutes**

Students literally see and feel the elapsed duration as an accumulation of physical quadrant wedges without having to carry or borrow minutes.

### ★ 2-PLAYER GAME: 'TIME TRAVELER RELAY' (2 MINS)

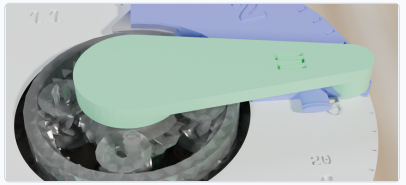
Player A sets a starting time on the geared clock (e.g., 3:30) and places 1 to 3 pie wedges beside the box. Player B predicts the ending time, lays the wedges onto the dial to verify, **and checks the answer on the calculator with [f][▸HR] and [g][▸HMS]!**

### ⚙ WHERE THIS SHOWS UP IN AUTOMOTIVE TRANSMISSIONS & ROBOTICS

Look inside the clear center window of your clock: that sun-planet-ring mechanism is an epicyclic planetary gearbox! The exact same gear geometry powers electric car drivetrains, **aircraft turbine reductions, and robotic arm joints.**

## LESSON 2: 15-MIN PIE WEDGES ↔ [g][▸HMS] TIME ADDITION

STEP A: HANDS-ON WITH PIE WEDGES ON DIAL — 1:1 CAD SCENE



1 Wedge = 15m = 1/4 Hr = 90° | 2 Wedges = 30m = 1/2 Hr = 180° | 3 Wedges = 45m = 3/4 Hr = 270°

▲ **Drop 15-minute pie wedges onto the clock face to see 3:30 + 0:45** = 4:15, then verify with [g][▸HMS]!

STEP B: TRY IT ON YOUR CALCULATOR — BOX ICON: [🕒] / [g][▸HMS] /

ELAPSED TIME TREE: 3:30 + 0:45 = 4:15

3:30 + 3×[15m] WEDGES → 3.5h + 0.75h = 4.25 | [g][▸HMS] = 4.15

+15m (3:45) | +15m (4:00) | +15m (4:15)

Bridges Across 12:00 Datum → 4:15!

PRESS BOX ICON & RPM KEYS (SCG):

1. ADD HOURS: 3.30 f ▸HR 0.45 f ▸HR +
2. TO H.MM: g ▸HMS = 4.15 (4:15)

✓ DISPLAY (X) = 4.15

KEYSTROKES: 3.30 [f] [▸HR] 0.45 [f] [▸HR] [+] [g] [▸HMS]

Lay three 15-minute wedges past 3:30 on the dial, then add in [f][▸HR] and convert back with [g][▸HMS]!

✓ 4-REGISTER STACK: T: 0 | Z: 0 | Y: 0 | X: 4.15 → X = 4.15

TIME AS FRACTIONS OF A CIRCLE • 15 MIN = 1/4 HOUR = 90°

STEP 3: FLIP SHEET OVER ↻ PAGE 05

# 05 — EXAMPLE 3 & DESK CHALLENGES

PAGE / 08 · Quarter-Hour Wedges & 60-Minute Factoring + 3 Desk Challenges

05 06 07 08 · SKU: TIME-01 • KEEP IN LID · SIDE B • FLIP SIDE (TOP-LEFT)

## 3 QUARTER-HOUR WEDGES & 60-MINUTE FACTORING

**THE GOAL: Show how 60 minutes factors into 4 clean 15-minute quadrant wedges!**

**1. INSERT 15-MIN WEDGE (:15):**
Drop the 90° colored quadrant wedge into the 12:00 to 3:00 dial sector.

**2. ADVANCE THE MINUTE HAND:**
Rotate the outer ring 90°—the minute hand traces the wedge edge cleanly to :15.

**3. SNAP DIGITAL BLOCKS IN LID:**
Seat [03], [:15], and [PM] into the lid sockets to link analog and digital notation.

**✓ 60 IS A HIGHLY COMPOSITE NUMBER: $2^2 \times 3 \times 5$**

**An hour has 60 minutes because 60 has 12 factors: 1, 2, 3, 4, 5, 6, 10, 12, 15, 20, 30, 60.**
Four 15-min wedges fill 360°: 60 / 4 = 15 min (90° each) with zero fractions!
Synchronized 12:1 planetary gears advance the hour hand 1/4 of an hour (7.5°).
Verify on calculator: 60 [g][FACT] 4 [÷] confirms 15 min per quarter-hour wedge!

**PRO TIP: CLOSE YOUR EYES & FEEL THE ANSWER**
Once you build an example, run your fingers across the pieces. Your hands remember physical shapes and heights way longer than memorized rules!

EXAMPLE 3: VISUAL DECOMPOSITION TREE

STEP B: PRESS BOX ICON & RPN KEYS (SC6)




FULL KEYSTROKES: 60 [g] [FACT] 4 [÷]
✓ 4-REGISTER STACK: T: 0 | Z: 0 | Y: 0 | X: 15 → X = 15

## 3 DESK CHALLENGES — TRY THESE NOW!

**WARM-UP #1** — **Set the clock to 4:30. What angle is between 12 and the minute hand?**
Step A: Build with blocks on your Lid Tray • Step B: Verify on SC-32 Calculator
✓ Build first, then flip booklet upside-down to check Answer #1 below!

**LEVEL-UP #2** — **How many 15-minute quadrant wedges fit between 1:00 and 2:30?**
Step A: Build with blocks on your Lid Tray • Step B: Verify on SC-32 Calculator
✓ Build first, then flip booklet upside-down to check Answer #2 below!

**BRAIN-MELTER #3** — **Spin the minute hand 2 full turns (720°). How far did the hour hand move?**
Step A: Build with blocks on your Lid Tray • Step B: Verify on SC-32 Calculator
✓ Build first, then flip booklet upside-down to check Answer #3 below!

SELF-CHECK ANSWER KEY (ROTATE BOOKLET 180° TO READ)
#3: 180° (1/2 circle) | #2: 6 wedges (90 minutes) | #1: 2 hours (60°)

**FOR PARENTS & TEACHERS (Grades K–4 (Time, Elapsed Intervals, Fractions & Angles))**

- CCSS.MATH.CONTENT.1.MD.B.3: Tell and write time in hours and half-hours.
- CCSS.MATH.CONTENT.3.MD.A.1: Tell and write time to the nearest minute & measure intervals.
- CCSS.MATH.CONTENT.4.MD.C.5: Recognize angles as geometric shapes formed by two rays.
- **CCSS.MATH.PRACTICE.MP2, MP4, MP5, MP7: Reason, Model, Use Tools & Structure.**

Next up on Page 06 → How to stack everything back in the box in 10 seconds!

**BOSS CHALLENGE #4** — **BLOCKS → BOX ICON KEY SPEED-RUN!**
1. Build Examples #1, #2, and #3 hands-on with your physical pieces.
2. Press the Box Button Icon ([ ] / [g][+HMS] / [f][+HR]) and keystrokes on your calculator:
• 60 [g] [FACT] 4 [÷] → X = 15

**CHECK OFF YOUR MASTERY MILESTONES:**
- ☐ 1. Demonstrated Lessons #1, #2 & #3 hands-on with the physical blocks
- ☐ 2. Solved all 3 Desk Challenges above (check upside-down key if stuck!)
- ☐ 3. Demonstrated each lesson on the calculator using the Box Button Icon
- ☐ 4. Packed all pieces back into the box in under 10 seconds (see Page 06 →)

TEST YOURSELF • ANSWERS INCLUDED SO YOU CAN CHECK RIGHT AWAY

NEXT: HOW TO PACK UP → PAGE 06

SIDE B (PAGES 05–08) • READ TOP ROW FIRST

# 06 — HOW TO STACK IT ALL BACK IN

PAGE / 08 · The 10-Second Self-Checking Pack-Up Puzzle

05 06 07 08 · SKU: TIME-01 • KEEP IN LID · SIDE B • PACK-UP GUIDE (TOP-RIGHT)

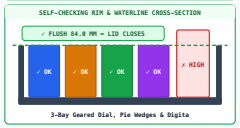
## TOP-DOWN BOX MAP: WHERE EVERY PIECE LIVES

Putting the blocks away is a 10-second puzzle. Match each piece to its labeled home bay below so everything nests flush and the lid snaps shut:

**✓ CUSTOM CIRCULAR CRADLE & WEDGE BAYS**
Drop the geared clock dial into the large circular cradle on the left.
Stack the 4 quarter-hour pie wedges into the middle bay, and snap the 6 digital minute blocks (:00 to :45) and AM/PM tags into the right-hand organizer slots.
One quick glance confirms all 19 pieces are present before you close the lid!

1. STORAGE CASE & COMPARTMENTS

2. WHY THE LID SELF-CHECKS EVERY PIECE




▲ SELF-INDEXING BAYS & EMBOSSED BOX BUTTON ICON:
Calibrated bays guide every piece home, and the lid icon links directly to your calculator.

## THE 10-SECOND PACK-UP ROUTINE

**HOW TO STACK EVERYTHING BACK IN:**

**1. SORT BY COLOR & NUMBER:**
Group matching blocks or tiles together on your desk first.

**2. STACK & DROP INTO LABELED BAYS:**
Follow the Top-Down Box Map on the left. If a tower looks too tall or too short, a block is in the wrong bay—swap it until everything levels!

**3. TUCK THIS CARD IN THE LID & SNAP SHUT:**
Fold this guide back to [01] and rest it right on top before closing.

**DIGITAL DISPLAY LID & GEAR CARE**
- The top of the lid has recessed sockets for building digital times (HH : MM AM/PM).
- Dry-running gears: never add oil or grease (it attracts dust). Just rinse in cool water if needed.
- Turn the minute ring to move the hands—never force the slow hour hand backward!

**SHORT A PIECE AT CLEAN-UP TIME?**
First check under notebooks and chair legs! If a piece truly vanished, **see PAGE 08 (bottom-right) for your complete Part ID Reference Table.**

**✓ TIME-01 4-POINT CLEAN-UP CHECKLIST**

| | |
|---|---|
| 1. CRADLE CLOCK CORE: | Seat the geared clock dial flat into the left circular bay. |
| 2. STACK 4 WEDGES: | Place all 4 quarter-hour pie wedges (Q1–Q4) into the center bay. |
| 3. SEAT DIGITAL TILES: | Snap the 6 minute tiles (:00..:45) & AM/PM blocks into the right slots. |
| 4. CLEAR DIGITAL LID: | Remove any tiles from the top lid sockets & snap the 3×2 lid shut! |

**MODULAR CLASSROOM & BOOKSHELF NESTING (42.0 MM PITCH)**
Every StackCalc box is sized in exact multiples of 42.0 mm (1×1, 3×1, 3×2).
All 9 kit boxes interlock side-by-side in a classroom cart or drawer with zero wasted space, and every embossed box icon matches its calculator operation key!

MOLDED CLOCK CRADLE •DEDICATED BAYS FOR ALL 19 PIECES

NEXT: CONNECTED LAB ▼ PAGE 07

THEN READ BOTTOM ROW (PAGE 07 → PAGE 08)

# 07 — THE STACKCALC CONNECTED LAB

PAGE / 08 · How Your Kits & RPN Calculator Stack Work Together

05 06 07 08 · SKU: TIME-01 • KEEP IN LID · SIDE B • ECOSYSTEM (BOT-LEFT)

**ONE CONNECTED SYSTEM FROM KINDERGARTEN TO CALCULUS**
Every StackCalc box uses the same modular grid and unit heights—so pieces from different kits actually fit and work together.

## WHERE YOUR BOX FITS IN THE 9-KIT LAB

| | | |
|---|---|---|
| BASE-10 **Ten-Frames Lab** Early Counting & Regrouping | DOMI-01 **Math Domino Studio** Doubles & 12-Frame Part-Whole | FRAC-01 **Fraction Studio** Proportional Towers & 4-Way Faces |
| PRIME-01 **Prime Factor Studio** Multiply by Stacking Prime Towers | ALGB-01 **Algebra Studio** Quadratic Area Arrays & Factoring | TIME-01 ◂ YOUR BOX **Planetary Clock** 12:1 Geared Hands & Time Wedges |
| PROB-01 **Probability Dice** Fair Polyhedrals & 10-Sector Spinner | VOLV-01 **Unit Circle Volvelle** Spin-to-Read Exact Trig Radicals | SC-32-HW **SC-32 RPN Calculator** 4-Register Stack Scientific Engine |

**WHY THIS MATTERS: ZERO "THROW-AWAY" MATH**
In most classrooms, every grade uses a totally unrelated plastic toy.
In StackCalc, the 7mm unit block you count with in 1st grade is the exact same height as the 1/12th fraction block and the [2] prime block later on!

1. TIME-01 BOX LID ICON

2. SC-6 ELEMENTARY RPN

3. SC-32 SCIENTIFIC RPN

▲ BOX ICON ↔ CALCULATOR KEY CONTINUITY:
Every physical box lid carries the exact icon of the calculator key that computes that lesson!

## HOW BLOCKS CONNECT TO THE CALCULATOR

**SHARED MODULAR FOOTPRINT & HEIGHTS**
- Time interval blocks (:00..:45) snap into the same 19.6mm sockets used across all StackCalc kits.
- Recessed sockets swallow the bottom peg so only true heights add.
- All storage boxes nest neatly side-by-side on any desk or shelf.

**WHY WE CALL IT "STACKCALC" (THE RPN SECRET)**
- Every hour mark is 30° (π/6 rad)—connecting early clock reading directly to the Unit Circle Volvelle!
- Placing a block on the table = pressing [ENTER] to push onto the stack.
- Snapping two towers together = pressing [+] or [×] with zero parentheses!
- Mirrors the 4 stack levels on the SC-32 calculator: X, Y, Z, and T.

**FREE ACTIVITY DECKS & INTERACTIVE SIMULATORS**
Download extra printable puzzle cards, teacher lesson guides, and interactive browser simulations anytime—no login needed.
Explore All 9 Kits & Free Guides at stackcalc.io

LIVE 4-REGISTER RPN STACK

HOW BLOCKS MAP TO STACK KEYS

1. PLACE 1ST PIECE ON STAGE:
Type number + [ENTER] → lifts to Y
2. ADD / STACK 2ND PIECE:
Type 2nd number in X, then [+]/[×]
3. PRESS BOX ICON KEY:
Press [ ] / [g][+HMS] / [f][
✓ Zero Parentheses • Always Visible!

BOX ICON = [ ] / [g][+HMS] / [f][÷HR] • VERIFIED LESSON SEQUENCES:
• Lesson 1: 3.30 [f] [÷HR] 30 [×] → X = 105
• Lesson 2: 3.30 [f] [÷HR] 0.45 [f] [÷HR] [+] [g] [÷MM → X = 4.15

MODULAR 9-KIT CONTINUUM • FROM EARLY COUNTING TO AP CALCULUS

NEXT: PART ID TABLE → PAGE 08

# 08 — KEEP THIS CARD! PART IDs & VIDEO

PAGE / 08 · Complete 19-Piece Reference & 60-Second 3D Video Guide

05 06 07 08 · SKU: TIME-01 • KEEP IN LID · SIDE B • PART REFERENCE (BOT-RIGHT)

## PART ID REFERENCE TABLE (19 TOTAL PIECES)

**✓ KEEP THIS CARD INSIDE YOUR BOX LID!**
Backpacks eat blocks. Dogs chew dice. Vacuum cleaners take no prisoners. Keep this card in your lid so you always have the exact Part IDs below for single-piece refills!

| PART NAME | QTY | COLOR / FINISH | PART ID |
|---|---|---|---|
| 12:1 Planetary Clock Dial | 1x | Blue / Gold | TIME-CLOC-db6 |
| 15-Min Pie Wedges (Q1–Q4) | 4x | Red / Blue | TIME-CLOC-db6 |
| Minute Blocks (:00 to :45) | 6x | Blue | TIME-CLOC-1f0 |
| AM / PM & Hour Blocks | 6x | Gold / Red | TIME-CLOC-aa0 |
| 3x2 Storage Case & Lid | 2x | Black | TIME-CLOC-db6 |

**NEED A REPLACEMENT PIECE? NO NEED TO BUY A WHOLE NEW BOX.**
Enter any Part ID above at stackcalc.io/parts to order individual replacement pieces.

COLOR & PIECE IDENTIFICATION KEY · MATCH ANY LOOSE PART

| Blue / Gold | Red / Blue | Green | Black |
|---|---|---|---|
| 12:1 Geared Clock Dial | 15-Min Pie Wedges | Hour & Minute Hands | 3x2 Storage Case & Lid |

**UNIVERSAL STACKCALC HEIGHT PITCH (1U = 7.00 MM • 12U = 84.00 MM ONE WHOLE)**
Measure any loose piece in 7mm units (U) against this scale bar to identify it:

✓ DISTRICT PO & SAFETY: 10-Pack SKU TIME-01-CLASS10 • CPSIA / ASTM F963 Non-Toxic Washable PLA/ABS

## WATCH THE 60-SECOND 3D VIDEO GUIDE

POINT CAMERA TO WATCH DEMO
stackcalc.io/products/classroom-clock/
See animated 3D examples & pack-up walkthrough

**QUESTIONS, CLASSROOM SETS & SUPPORT**
- Email us anytime: support@stackcalc.io
- School & District POs: school-orders@stackcalc.io

**Designed in Mountain View, CA • Non-Toxic & Washable**

COMPANION MANUAL CHAPTERS, STANDARDS & KEYSTROKES

**BOOK CHAPTERS:** SC-6 Guide Ch. 5 (Geared Time & Angles) • SC-32 Manual Ch. 4
**STANDARDS:** CCSS 1.MD.B.3, 2.MD.C.7, 3.MD.A.1, 4.MD.C.5 • TEKS 2.6G, 3.7C (Time & Angles)
**WATERLINE SPEC:** 3-Bay Geared Dial, Pie Wedges & Digital Tile Deck

**ALL 3 VERIFIED CALCULATOR LESSON SEQUENCES (SCG):**
Ex 1: 3.30 [f] [÷HR] 30 [×] → X=105
Ex 2: 3.30 [f] [÷HR] 0.45 [f] [÷HR] [+] [g] [÷MM] → X=4.15
Ex 3: 60 [g] [FACT] 4 [÷] → X=15

BOX ICON KEY: [ ] / [g][+HMS] / [f][÷HR]
✓ Cross-Referenced to Companion Book & Calculator Engine
Fold back to [01] & keep this reference card inside your box lid!

LOST A PIECE? GRAB SINGLE REPLACEMENTS ANYTIME AT stackcalc.io/parts

✓ KEEP THIS GUIDE IN BOX LID!

◂ PAGE 07 (ECOSYSTEM) → PAGE 08 (KEEP THIS CARD FOR PART IDs & QR) ▸

# 01 — TEN-FRAMES & PLACE VALUE ARITHMETIC LAB

PAGE / 08

Two-Color Counter Sets, Landmark '10' Blocks & Dual 5×2 Trays

01 02 03 04

SKU: BASE-10 • KEEP IN LID

SIDE A • COVER (TOP-LEFT)

**WHAT IS THIS BOX FOR? PUT AWAY YOUR FINGERS & SEE NUMBERS AT A GLANCE.**

Your brain can instantly recognize up to 5 objects without counting. This 2×5 Ten-Frame turns addition and "carrying the 1" into visual patterns!

## 5-ANCHOR SUBITIZING & TWO-COLOR COUNTERS

Recognizing a small quantity without counting one-by-one is called "subitizing." Our Ten-Frame organizes numbers into two rows of 5:

- **Two-Color Counters: Flip between Red (Side A) and Green (Side B)**
- **Tactile Finger Rings: Grooved surfaces you can feel without looking**
- **Stackable Plugs: 10 counters snap together into a solid 10-tower**

Once you see 8 as "a full row of 5 plus 3 more," finger-counting disappears!

**10 ONES = 1 SOLID TEN-ROD (70 MM)**

Stack 10 unit counters into a tower: it stands 70.0 mm tall — the exact same height as one solid purple '10' rod! That's place value in your hand.

**WHAT'S INSIDE YOUR BOX (56 PIECES)**

- 40 Counter Blocks (20× White Set A + 20× Tan Set B)
- 12× Landmark '10' Blocks (6× Set A + 6× Set B) + 2× 5×2 Trays + Box & Lid
- Keep this folded guide inside your box lid for examples & part IDs!

**★ YOUR FIRST 60 SECONDS WITH THIS BOX** — START HERE!

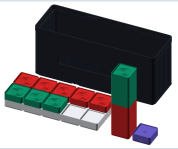
1. **FLIP THE TWO-COLOR COUNTERS:** Notice every counter is Red on one side and Green on the other—flip to show two addends!
2. **FILL ONE ROW OF 5 ON THE LID:** Drop 5 Red counters along the top row and 3 Green below: your eyes see 5 + 3 = 8 instantly!
3. **PRESS THE BOX ICON KEY [▦]:** Type 8 [ENTER] 2 [÷] 5 [÷] [▦] on your calculator—the 2×5 grid icon on your box is the [▦] key!

**BOX ICON = CALCULATOR [▦] TEN-FRAME & [PV] PLACE VALUE KEYS**

The button icon embossed on your box lid (left) is the exact calculator key ([▦] Ten-Frame & [PV])! Build every lesson with your blocks, then press the key.

## 5-ANCHOR SUBITIZING (5+3=8) ↔ BOX [▦] & [PV] ICONS

STEP A: HANDS-ON WITH 5-ANCHOR SUBITIZING — 1:1 CAD SCENE



| Top Row = 5 | 5 Red + 3 Green | 2 Empty Sockets |
|---|---|---|
| Instant Benchmark | See 8 at a Glance | 10 − 8 = 2 |

▲ **Because each 10-frame is arranged in two rows of 5, your eyes recognize 5 + 3 = 8 and 2 empty spots instantly!**

STEP B: TRY IT ON YOUR CALCULATOR — BOX ICON: [▦] Ten-Frame & [P

5-ANCHOR TREE: 8 = 5 + 3 (2 SHY OF 10)

10-FRAME CAPACITY = 10

Filled: 5+3 = 8 | 10 − 8 = 2 [PV]

Top Row: 5 Red | Bot Row: 3 Green | Empty = 2

5 [ENTER] 3 [+] 10 [xey] [−] [PV] → 2

PRESS BOX ICON & RPN KEYS (SCG):

1. 5-ANCHOR: 5 [ENTER] 3 [+] = 8

2. 10-COMP: 10 [xey] [−] [PV] = 2

✓ DISPLAY (X) = 2

**KEYSTROKES: 5 [ENTER] 3 [+] 10 [xey] [−] [PV]**

Build 5 + 3 = 8 on the lid tray, then subtract from 10 and press [PV] to verify the 2 empty sockets!

**WELCOME TO HANDS-ON MATH • KEEP THIS GUIDE IN YOUR BOX LID**

STEP 1: UNFOLD RIGHT → PAGE 02

▲ FOLD 1: FOLD BOTTOM HALF BEHIND TOP ▲

# 02 — HOW IT WORKS: THE CORE IDEA

PAGE / 08

5-Anchor Subitizing & Trading 10 Ones for 1 Ten-Block

01 02 03 04

SKU: BASE-10 • KEEP IN LID

SIDE A • 1ST UNFOLD (TOP-RIGHT)

## WHY WE GROUP IN TENS (PLACE VALUE)

Our entire number system is built on powers of 10:

**10 × 10^k = 10^(k+1)**

Whenever ten ones fill up a Ten-Frame, they trade up into a single unit of the next higher place value:

- **10 Unit Counters (10 × 1) = 1 Solid Ten-Rod (1 × 10)**
- **10 Ten-Rods (10 × 10) = 1 Hundred-Flat (1 × 100)**

Because the Ten-Frame physically holds at most 10 counters, "carrying" and "borrowing" stop being mysterious rules — you see the tray fill up!

**HARMONIZED 7.00 MM STACKING HEIGHT**

Each counter adds 7.00 mm of height (1U), matching our 1/12th Fraction block and Prime block [2]. A stack of 10 counters is 70.0 mm tall — matching the solid '10' rod right down to the millimeter.

**TACTILE GROOVES & EASY LIFT NOTCHES**

Every counter has concentric finger rings you can feel with your eyes closed, and every tray socket has side thumb-notches so small fingers can pop counters out effortlessly.

**3 CLASSIC MATH TRAPS THIS KIT MAKES IMPOSSIBLE** — VISUAL PROOF

✗ **TRAP 1 — Counting on fingers one-by-one:**
✓ **FIX:** The 2×5 grid trains your brain to see 8 as '5 and 3' or '2 shy of 10' in a split second.

✗ **TRAP 2 — Blindly 'carrying the 1' without meaning:**
✓ **FIX:** Trading 10 stacked counters (70mm) for 1 Purple '10' rod (70mm) proves what that '1' actually is!

✗ **TRAP 3 — Forgetting 10-complements (7 + 7 = 10):**
✓ **FIX:** Any partially filled 10-frame shows the complement as the empty sockets staring right back at you.

CALCULATOR CHECK: 5 [ENTER] 3 [+] 10 [xey] [−] [PV] → X = 2

## COMPONENT & DIMENSION GUIDE

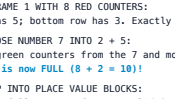
| COMPONENT | DIMENSIONS | VALUE | COLOR |
|---|---|---|---|
| Dual Counter | Ø18.0 × 7.00 mm | 1 Unit | Red / Green |
| 10-Stack Tower | Ø18.0 × 70.0 mm | 10 Units | Red / Green |
| Solid '10' Rod | 19.6 × 70.0 mm | 1 Ten | Purple |
| Operator Block | 19.6 × 19.6 mm | + − = | Gold |
| Ten-Frame Lid | 126 × 42 × 10 mm | 2x5 Tray | Matte Black |
| 3x1 Storage Bin | 126 × 42 × 49 mm | 4 Towers | Matte Black |

**WHY YOUR BRAIN LOVES THIS**

- Instant Visual Benchmarks: A full row is always 5; a full tray is always 10.
- Easy Finger-Lift Notches: Counters pop in and out without frustration.
- Stackable Into Towers: Snap 10 counters together to build a 10-rod.
- Fast Cleanup: 4 wells hold 10 counters each — if all 4 are level, none are lost!

1:1 CAD REFERENCE VIEW — BOX ICON & FORMULA QUICK-REFERENCE

1:1 BLOCK CAD ANATOMY




✓ FLUSH PLUG-AND-SOCKET JOINT
Socket swallows plug completely:
Only 7.00mm true heights add!

**Box Button Icon Keys:**
[▦] Ten-Frame & [PV] Place Value

**5-Anchor Subitizing:**
8 = 5 + 3 | 7 = 5 + 2 | 9 = 5 + 4

**10-Complement Law:**
n + (10 − n) = 10 (Filled + Empty = 10)

**Make-a-Ten Addition:**
8 + 7 = (8 + 2) + 5 = 10 + 5 = 15

**Tens & Ones Keystroke:**
N [ENTER] 10 [g][÷R] → X Tens, Y Ones

**BASE-10 PLACE VALUE • 10 UNIT COUNTERS = 1 LANDMARK '10' BLOCK**

STEP 2: UNFOLD DOWN ▼ PAGES 03–04

▼ UNFOLD STEP 2: OPEN DOWN FOR PAGES 03 & 04 ▼

# 03 — EXAMPLE 1: HANDS-ON WALKTHROUGH

PAGE / 08

Instant Recognition: See 5 + 3 = 8 Without Counting

01 02 03 04

SKU: BASE-10 • KEEP IN LID

SIDE A • 2ND UNFOLD (BOT-LEFT)

## 1 SUBITIZING 8 & BRIDGING TEN (8 + 7 = 15)

1. **MODEL 5-ANCHOR SUBITIZING (NUMBER 8):**
   • Place 5 Red counters in the top row and 3 Red in the bottom row.
   → Eyes see 5 + 3 = 8 and 2 empty sockets (8 + 2 = 10)!
2. **BRIDGE ACROSS 10 TO ADD 8 + 7:**
   • Start with 8 Red counters in the frame and 7 Green counters outside.
   • Slide 2 Green counters into the 2 empty sockets to complete 10!
3. **READ THE MAKE-A-TEN SUM AT A GLANCE:**
   1 full Ten-Frame (10) + 5 Green counters remaining outside:
   → 8 + 7 = (8 + 2) + 5 = 10 + 5 = 15!

**BENCHMARK 5 AND 10 MENTAL ANCHORS**

Cognitive research proves that students who master 5 and 10 benchmarks calculate 3x faster than those relying on counting by ones. The physical ten-frame hard-wires these geometric benchmarks into working memory.

STEP B: TRY EXAMPLE #1 ON YOUR CALCULATOR — [▦] Ten-Frame & [P

**STEP 1 (FILL TO 10):**
Slide 2 Green counters into the left 10-frame on the lid and press 8 [ENTER] 2 [+] (= 10).

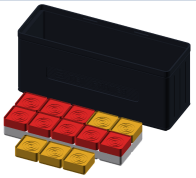
**STEP 2 (ADD REMAINING 5):**
Press 5 [+] → X = 15 (matching 1 full 10-frame + 5 Green counters in the right frame).

**STEP 3 (PRESS BOX ICON [▦]):**
Press [▦] on your SC-6 calculator to display 15 as a live visual Ten-Frame on screen!

LIVE 4-REGISTER RPN STACK READOUT (AFTER KEYSTROKES):
T: 0 | Z: 0 | Y: 0 | X: 15 → DISPLAY (X) = 15

## LESSON 1: MAKE-A-TEN (8+7 = 10+5) ↔ [▦] TEN-FRAME KEY

STEP A: HANDS-ON WITH COUNTERS ON LID — 1:1 CAD SCENE



| Left 10-Frame | Right 10-Frame | Total Sum |
|---|---|---|
| 8 Red + 2 Green = 10 | 5 Green Remaining | 10 + 5 = 15 |

▲ **Slide 2 Green counters across to finish filling the left 10-frame: 8 + 7 transforms into 10 + 5 = 15!**

STEP B: TRY IT ON YOUR CALCULATOR — BOX ICON: [▦] Ten-Frame & [P

MAKE-A-TEN NUMBER BOND: 8 + 7 = 15

8 + 7 = 15 [▦]

8 + 2 = 10 (Full) | 7 Splits: 2 + 5

8 Red | +2 Green | +5 Green

8 [ENTER] 2 [÷] 5 [÷] [▦] → 15

PRESS BOX ICON & RPN KEYS (SCG):

1. MAKE 10: 8 [ENTER] 2 [+] = 10

2. BOX ICON KEY: 5 [+] [▦] = 15

✓ DISPLAY (X) = 15

**KEYSTROKES: 8 [ENTER] 2 [÷] 5 [÷] [▦]**

Slide 2 counters on the lid to make 10, then press the Box Icon key [▦] on your SC-6 calculator!

✓ 4-REGISTER STACK: T: 0 | Z: 0 | Y: 0 | X: 15 → X = 15

**SUBITIZING SUPERPOWER • SEE NUMBERS IN CHUNKS OF 5 AND 10**

CONTINUE RIGHT → PAGE 04

# 04 — EXAMPLE 2: CORE MATHEMATICAL PRINCIPLE

PAGE / 08

Make-a-Ten Bridge: Why 8 + 7 Regroups Into 10 + 5 = 15

01 02 03 04

SKU: BASE-10 • KEEP IN LID

SIDE A • 2ND UNFOLD (BOT-RIGHT)

## 2 MAKING TEN REGROUPING: 8 + 7 = 15

**THE PROBLEM: Add 8 + 7 without counting on fingers.**

1. FILL FRAME 1 WITH 8 RED COUNTERS:
   Top row has 5; bottom row has 3. Exactly 2 spaces are empty.
2. DECOMPOSE NUMBER 7 INTO 2 + 5:
   • Take 2 green counters from the 7 and move them to Frame 1.
   → Frame 1 is now FULL (8 + 2 = 10)!
3. REGROUP INTO PLACE VALUE BLOCKS:
   • Swap the full Frame 1 for one solid '10' miniblock rod.
   • The remaining 5 green counters sit in Frame 2.
   → Total: 1 Ten + 5 Ones = 15!

**✓ THE "MAKING TEN" BRIDGE THEOREM**

8 + 7 = (8 + 2) + 5 = 10 + 5 = 15

**Carrying in columns is simply filling a physical frame.**

Students develop permanent mental addition agility by seeing the physical conservation of counters across two frames.

**★ 2-PLAYER GAME: 'FLASH TEN-FRAME SNAP!' (2 MINS)**

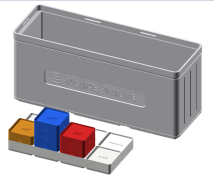
Player A secretly arranges counters on the lid's ten-frame behind a book, lifts the book for just 1 second, and covers it again. Player B must call out both the number of counters AND **how many empty spots are needed to make 10, then check with [▦]!**

**⚙ WHERE THIS SHOWS UP IN COMPUTER MEMORY & DIGITAL LOGIC**

Grouping individual units into fixed-size frames (10 in decimal, 8 bits = 1 byte in binary) is the foundation of positional number systems! When a frame fills up, it overflows ('carries') **one solid bundle into the next register column.**

## LESSON 2: BUNDLING 10 ONES INTO 1 TEN-ROD ↔ [g][÷R] KEY

STEP A: HANDS-ON WITH TEN-RODS & ONES — 1:1 CAD SCENE



| 10 Unit Counters | 1 Purple '10' Rod | Place Value Trade |
|---|---|---|
| 10 × 7mm = 70.0 mm | 70.0 mm Solid Tower | 15 + 10 = 1 R 5 |

▲ **Ten stacked unit counters (10 × 7.0 mm) stand at the exact same 70.0 mm height as one solid Purple '10' rod!**

STEP B: TRY IT ON YOUR CALCULATOR — BOX ICON: [▦] Ten-Frame & [P

PLACE VALUE REGROUPING: 15 = 1 TEN + 5 ONES

15 + 10 = 1 R 5

X = 1 Purple '10' | Y = 5 Unit Ones

10×7mm=70mm | 5×7mm=35mm | [g][÷R]

15 [ENTER] 10 [g][÷R] → X=1, Y=5

PRESS BOX ICON & RPN KEYS (SCG):

1. DIV BY 10: 15 [ENTER] 10 [g] [÷R]

2. TENS/ONES: X=1 Ten | Y=5 Ones

✓ DISPLAY (X) = 1

**KEYSTROKES: 15 [ENTER] 10 [g] [÷R]**

Dividing 15 by 10 with [g][÷R] puts Quotient = 1 Ten-Rod in X and Remainder = 5 Ones in Y!

✓ 4-REGISTER STACK: T: 0 | Z: 0 | Y: 5 | X: 1 → X = 1

**THE MAKE-A-TEN BRIDGE • HOW MENTAL ARITHMETIC ACTUALLY WORKS**

STEP 3: FLIP SHEET OVER ↻ PAGE 05

# 05 — EXAMPLE 3 & DESK CHALLENGES

PAGE / 08

5-Anchor Subitizing: 8 = 5 + 3 & 10-Complement + 3 Challenges

05 06 07 08

SKU: BASE-10 • KEEP IN LID

SIDE B • FLIP SIDE (TOP-LEFT)

## 3 5-ANCHOR SUBITIZING: SEE 8 AS 5 + 3 & 2 SHY OF 10

**THE GOAL: Subitize 8 in 0.1 seconds and spot its 10-complement instantly!**

**1. FILL TOP ROW (5 COUNTERS):**

Place 5 Set A counters across the top row to anchor the benchmark 5.

**2. ADD 3 ON BOTTOM ROW:**

Place 3 Set A counters on the bottom row: 5 on top + 3 below = 8 counters total.

**3. SPOT 2 EMPTY HOLES:**

Notice 2 empty spots remaining in the frame: 8 is exactly 2 shy of 10!

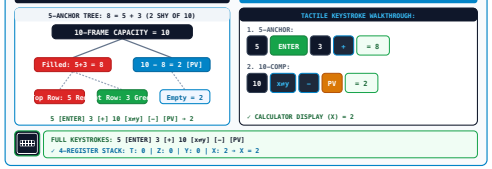
**✓ WHY 5-ANCHORS SUPERCHARGE MENTAL ARITHMETIC**

**The human eye can only subitize up to 4 items at once without counting 1-by-1.**
Grouping in 5s lets you recognize 8 as (5 + 3) instantly—no counting needed!
Because a full frame is 10, the two empty holes immediately show that 10 − 8 = 2.
Verify on calculator: 5 [ENTER] 3 [+] (→ 8) | 10 [x≷y] [−] [PV] (→ 2) confirms both!

**PRO TIP: CLOSE YOUR EYES & FEEL THE ANSWER**

Once you build an example, run your fingers across the pieces. Your hands remember physical shapes and heights way longer than memorized rules!

**EXAMPLE 3: VISUAL DECOMPOSITION TREE**




**STEP B: PRESS BOX ICON & RPN KEYS (SC6)**




FULL KEYSTROKES: 5 [ENTER] 3 [+] 10 [x≷y] [−] [PV]
✓ 4-REGISTER STACK: T: 0 | Z: 0 | Y: 0 | X: 2 → X = 2

## 3 DESK CHALLENGES — TRY THESE NOW!

**WARM-UP #1** — **Place 6 counters on the frame. How many empty spots to make 10?**
Step A: Build with blocks on your Lid Tray • Step B: Verify on SC-32 Calculator
✓ Build first, then flip booklet upside-down to check Answer #1 below!

**LEVEL-UP #2** — **Use two 10-frames to solve 9 + 6 by moving 1 counter across!**
Step A: Build with blocks on your Lid Tray • Step B: Verify on SC-32 Calculator
✓ Build first, then flip booklet upside-down to check Answer #2 below!

**BRAIN-MELTER #3** — **Build 37 using Landmark '10' blocks and unit counters. How many pieces total?**
Step A: Build with blocks on your Lid Tray • Step B: Verify on SC-32 Calculator
✓ Build first, then flip booklet upside-down to check Answer #3 below!

SELF-CHECK ANSWER KEY (ROTATE BOOKLET 180° TO READ)

#3: Three '10' blocks + 7 Counters

#1: 4 spots (6 + 4 = 10) | #2: (9 + 1) + 5 = 10 + 5 = 15

**FOR PARENTS & TEACHERS (Grades K–3 (Counting, Subitizing, Place Value & Regrouping))**

- CCSS.MATH.CONTENT.K.OA.A.4: For any number from 1 to 9, find the number that makes 10.
- CCSS.MATH.CONTENT.1.OA.C.6: Add and subtract within 20 using the making-ten strategy.
- CCSS.MATH.CONTENT.1.NBT.B.2: Understand that the two digits of a two-digit number represent tens and ones.
- **CCSS.MATH.PRACTICE.MP2, MP4, MP5, MP7: Reason, Model, Use Tools & Structure.**

Next up on Page 06 → How to stack everything back in the box in 10 seconds!

**BOSS CHALLENGE #4** — **BLOCKS → BOX ICON KEY SPEED-RUN!**

1. Build Examples #1, #2, and #3 hands-on with your physical pieces.
2. Press the Box Button Icon ([▦] Ten-Frame & [PV]) and keystrokes on your calculator:
   • **5 [ENTER] 3 [+] 10 [x≷y] [−] [PV] → X = 2**

**CHECK OFF YOUR MASTERY MILESTONES:**

- ☐ 1. Demonstrated Lessons #1, #2 & #3 hands-on with the physical blocks
- ☐ 2. Solved all 3 Desk Challenges above (check upside-down key if stuck!)
- ☐ 3. Demonstrated each lesson on the calculator using the Box Button Icon
- ☐ **4. Packed all pieces back into the box in under 10 seconds (see Page 06 →)**

TEST YOURSELF • ANSWERS INCLUDED SO YOU CAN CHECK RIGHT AWAY

NEXT: HOW TO PACK UP → PAGE 06

SIDE B (PAGES 05–08) • READ TOP ROW FIRST

# 06 — HOW TO STACK IT ALL BACK IN

PAGE / 08

The 10-Second Self-Checking Pack-Up Puzzle

05 06 07 08

SKU: BASE-10 • KEEP IN LID

SIDE B • PACK-UP GUIDE (TOP-RIGHT)

## TOP-DOWN BOX MAP: WHERE EVERY PIECE LIVES

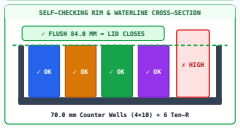
Putting the blocks away is a 10-second puzzle. Match each piece to its labeled home bay below so everything nests flush and the lid snaps shut:

**✓ 12-COLUMN STORAGE BOX (40 COUNTERS + 12 LANDMARK '10'S)**

The 7U box holds 40 Counter Blocks (20× Set A + 20× Set B) and 12× Landmark '10' Blocks (6× Set A + 6× Set B) across 12 internal columns, topped by the two 5×2 Ten-Frame Trays. When all 52 blocks and 2 trays are seated inside, everything packs flush so you can spot a missing counter from across the room!

**1. STORAGE CASE & COMPARTMENTS**

**2. WHY THE LID SELF-CHECKS EVERY PIECE**




▲ **SELF-INDEXING BAYS & EMBOSSED BOX BUTTON ICON:**
Calibrated bays guide every piece home, and the lid icon links directly to your calculator.

## THE 10-SECOND PACK-UP ROUTINE

**HOW TO STACK EVERYTHING BACK IN:**

1. **SORT BY COLOR & NUMBER:**
Group matching blocks or tiles together on your desk first.
2. **STACK & DROP INTO LABELED BAYS:**
Follow the Top-Down Box Map on the left. If a tower looks too tall or too short, a block is in the wrong bay—swap it until everything levels!
3. **TUCK THIS CARD IN THE LID & SNAP SHUT:**
Fold this guide back to [01] and rest it right on top before closing.

**DUAL 5×2 TEN-FRAME TRAYS & SNAPFINITY LID**

- The two removable 5×2 Ten-Frame Trays (20 spots total) sit side-by-side for hands-on addition.
- Use contrasting Set A (White) and Set B (Tan) counters to visualize addends and make-a-ten splits!
- Easy care: rinse in lukewarm soapy water and air dry.

**SHORT A PIECE AT CLEAN-UP TIME?**

First check under notebooks and chair legs! If a piece truly vanished, **see PAGE 08 (bottom-right) for your complete Part ID Reference Table.**

**✓ BASE-10 4-POINT CLEAN-UP CHECKLIST**

| | |
|---|---|
| 1. FOUR 70MM WELLS: | Stack 10 counters in each of the 4 round wells (all level at 70.0 mm). |
| 2. SEAT 6 TEN-RODS: | Place all 6 Purple '10' rods and 6 Gold sign blocks into the side bay. |
| 3. CLEAR 10-FRAME LID: | Remove any counters still seated in the 20 top lid sockets. |
| 4. FOLD CARD & SNAP: | Lay this folded guide inside and snap the 10-Frame Lid shut! |

**MODULAR CLASSROOM & BOOKSHELF NESTING (42.0 MM PITCH)**

Every StackCalc box is sized in exact multiples of 42.0 mm (1×1, 3×1, 3×2). All 9 kit boxes interlock side-by-side in a classroom cart or drawer with zero wasted space, and every embossed box icon matches its calculator operation key!

12-COLUMN STORAGE BOX • EQUAL STACKS PROVE NOTHING IS LOST

NEXT: CONNECTED LAB ▾ PAGE 07

THEN READ BOTTOM ROW (PAGE 07 → PAGE 08)

# 07 — THE STACKCALC CONNECTED LAB

PAGE / 08

How Your Kits & RPN Calculator Stack Work Together

05 06 07 08

SKU: BASE-10 • KEEP IN LID

SIDE B • ECOSYSTEM (BOT-LEFT)

**ONE CONNECTED SYSTEM FROM KINDERGARTEN TO CALCULUS**

Every StackCalc box uses the same modular grid and unit heights—so pieces from different kits actually fit and work together.

## WHERE YOUR BOX FITS IN THE 9-KIT LAB

| | | |
|---|---|---|
| **BASE-10** ◂ YOUR BOX<br>Ten-Frames Lab<br>Early Counting & Regrouping | **DOMI-01**<br>Math Domino Studio<br>Doubles & 12-Frame Part-Whole | **FRAC-01**<br>Fraction Studio<br>Proportional Towers & 4-Way Faces |
| **PRIME-01**<br>Prime Factor Studio<br>Multiply by Stacking Prime Towers | **ALGB-01**<br>Algebra Studio<br>Quadratic Area Arrays & Factoring | **TIME-01**<br>Planetary Clock<br>12:1 Geared Hands & Time Wedges |
| **PROB-01**<br>Probability Dice<br>Fair Polyhedrals & 10-Sector Spinner | **VOLV-01**<br>Unit Circle Volvelle<br>Spin-to-Read Exact Trig Radicals | **SC-32-HW**<br>SC-32 RPN Calculator<br>4-Register Stack Scientific Engine |

**WHY THIS MATTERS: ZERO "THROW-AWAY" MATH**

In most classrooms, every grade uses a totally unrelated plastic toy. In StackCalc, the 7mm unit block you count with in 1st grade is the exact same height as the 1/12th fraction block and the [2] prime block later on!

**1. BASE-10 BOX LID ICON**

**2. SC-6 ELEMENTARY RPN**

**3. SC-32 SCIENTIFIC RPN**

▲ **BOX ICON ↔ CALCULATOR KEY CONTINUITY:**
Every physical box lid carries the exact icon of the calculator key that computes that lesson!

## HOW BLOCKS CONNECT TO THE CALCULATOR

**SHARED MODULAR FOOTPRINT & HEIGHTS**

- Counter and Landmark '10' blocks share the exact 19.6mm socket pitch as Prime and Fraction blocks.
- Recessed sockets swallow the bottom peg so only true heights add.
- All storage boxes nest neatly side-by-side on any desk or shelf.

**WHY WE CALL IT "STACKCALC" (THE RPN SECRET)**

- Bundling 10 ones into a Landmark '10' block builds the intuition behind every base-10 calculator register.
- Placing a block on the table = pressing [ENTER] to push onto the stack.
- Snapping two towers together = pressing [+] or [×] with zero parentheses!
- Mirrors the 4 stack levels on the SC-32 calculator: X, Y, Z, and T.

**FREE ACTIVITY DECKS & INTERACTIVE SIMULATORS**

Download extra printable puzzle cards, teacher lesson guides, and interactive browser simulations anytime—no login needed.

Explore All 9 Kits & Free Guides at stackcalc.io

**LIVE 4-REGISTER RPN STACK**

**HOW BLOCKS MAP TO STACK KEYS**

1. **PLACE 1ST PIECE ON STAGE:**
Type number + [ENTER] → lifts to Y
2. **ADD / STACK 2ND PIECE:**
Type 2nd number in X, then [+]/[×]
3. **PRESS BOX ICON KEY:**
Press [▦] Ten-Frame & [PV]
✓ **Zero Parentheses • Always Visible!**

BOX ICON = [▦] Ten-Frame & [PV] • VERIFIED LESSON SEQUENCES:
• Lesson 1: 8 [ENTER] 2 [+] 5 [+] [▦] → X = 15
• Lesson 2: 15 [ENTER] 10 [g] [+R] → X = 1

MODULAR 9-KIT CONTINUUM • FROM EARLY COUNTING TO AP CALCULUS

NEXT: PART ID TABLE → PAGE 08

# 08 — KEEP THIS CARD! PART IDs & VIDEO

PAGE / 08

Complete 56-Piece Reference & 60-Second 3D Video Guide

05 06 07 08

SKU: BASE-10 • KEEP IN LID

SIDE B • PART REFERENCE (BOT-RIGHT)

## PART ID REFERENCE TABLE (56 TOTAL PIECES)

**✓ KEEP THIS CARD INSIDE YOUR BOX LID!**

Backpacks eat blocks. Dogs chew dice. Vacuum cleaners take no prisoners. Keep this card in your lid so you always have the exact Part IDs below for single-piece refills!

| PART NAME | QTY | COLOR / FINISH | PART ID |
|---|---|---|---|
| Counter Blocks — Set A | 20x | White PLA | BASE-CNTR-01A |
| Landmark '10' Blocks — Set | 6x | White PLA | BASE-LM10-01A |
| Counter Blocks — Set B | 20x | Tan PLA | BASE-CNTR-01B |
| Landmark '10' Blocks — Set | 6x | Tan PLA | BASE-LM10-01B |
| 5×2 Ten-Frame Trays | 2x | Matte Blue | BASE-TRAY-5X2 |
| 7U Storage Box & Lid | 2x | Gray / White | BASE-BOX-07U |

**NEED A REPLACEMENT PIECE? NO NEED TO BUY A WHOLE NEW BOX.**

Enter any Part ID above at stackcalc.io/parts to order individual replacement pieces.

**COLOR & PIECE IDENTIFICATION KEY** — MATCH ANY LOOSE PART

| White PLA | Tan PLA | Matte Blue | Gray |
|---|---|---|---|
| Set A Counters (20x) & '10 | Set B Counters (20x) & '10 | 5×2 Ten-Frame Trays (2x) | 7U Storage Box & Lid (2x) |

**UNIVERSAL STACKCALC HEIGHT PITCH (1U = 7.00 MM • 12U = 84.00 MM ONE WHOLE)**

Measure any loose piece in 7mm units (U) against this scale bar to identify it:

0U (0mm) 1U (7mm) 2U (14mm) 3U (21mm) 4U (28mm) 6U (42mm) 9U (63mm) 12U (84mm)

✓ DISTRICT PO & SAFETY: 10-Pack SKU BASE-10-CLASS10 • CPSIA / ASTM F963 Non-Toxic Washable PLA/ABS

## WATCH THE 60-SECOND 3D VIDEO GUIDE

**POINT CAMERA TO WATCH DEMO**

stackcalc.io/products/ten-frames/

See animated 3D examples & pack-up walkthrough

**QUESTIONS, CLASSROOM SETS & SUPPORT**

- Email us anytime: support@stackcalc.io
- School & District POs: school-orders@stackcalc.io

**Designed in Mountain View, CA • Non-Toxic & Washable**

**COMPANION MANUAL CHAPTERS, STANDARDS & KEYSTROKES**

**BOOK CHAPTERS:** SC-6 Guide Ch. 1 (Ten-Frames & Place Value) • SC-32 Manual Ch. 1
**STANDARDS:** CCSS K.OA.A.4, 1.OA.C.6, 1.NBT.B.2 • TEKS K.2I, 1.3D (Subitizing & Base-10)
**WATERLINE SPEC:** 70.0 mm Counter Wells (4×10) + 6 Ten-Rod Bays

**ALL 3 VERIFIED CALCULATOR LESSON SEQUENCES (SC6):**
Ex 1: 8 [ENTER] 2 [+] 5 [+] [▦] → X=15
Ex 2: 15 [ENTER] 10 [g] [+R] → X=1
Ex 3: 5 [ENTER] 3 [+] 10 [x≷y] [−] [PV] → X=2

**BOX ICON KEY: [▦] Ten-Frame & [PV]**
✓ Cross-Referenced to Companion Book & Calculator Engine
Fold back to [01] & keep this reference card inside your box lid!

LOST A PIECE? GRAB SINGLE REPLACEMENTS ANYTIME AT stackcalc.io/parts

✓ KEEP THIS GUIDE IN BOX LID!

# 01 PAGE / 08 — MATH DOMINO STUDIO & 12-FRAME REGROUPING LAB

28 Double-Six Dominoes, 2 Wildcards, 12-Frame Tray & Doubles Lid

01 02 03 04

SKU: DOMI-01 • KEEP IN LID

SIDE A • COVER (TOP-LEFT)

**WHAT IS THIS BOX FOR? STOP COUNTING ON YOUR FINGERS.**

Fingers are great for holding sandwiches, not for adding 9 + 5. These 28 dominoes and 12-frame tray train your eyes to see numbers in chunks.

## WELCOME TO SUBITIZING & 12-FRAME PLAY

Why count one-by-one when your brain can recognize patterns instantly?

- **28 Classic Dominoes: Every pair from [0|0] to [6|6] (168 dots total!)**
- **Doubles Practice Lid: Snap matching doubles (1+1 to 6+6) right on top**
- **The 12-Frame Tray: A raised bridge at 10 shows how numbers spill over**
- Secret Underside: Flip any tile over to spot the StackCalc register icon

Try this right now: Grab any two tiles and race to call out their total!

### THE "MAKE-A-TEN" BRIDGE

A raised tactile ridge between the 10th and 11th spots physically shows how 9 + 3 naturally regroups into 10 + 2 = 12 without counting by ones.

### WHAT'S IN YOUR BOX (32 PIECES)

- 28 Double-Six Domino Tiles + 2 Bonus Mystery Blank Tiles
- 1x 12-Frame Regrouping Tray with 10-Bridge Tactile Ridge
- 1x 3x1 Storage Case + 1x Snap-Fit Doubles Practice Lid

★ YOUR FIRST 60 SECONDS WITH THIS BOX — START HERE!

1. **FIND ALL 6 DOUBLES & SNAP ONTO THE LID:**
   Hunt for [1|1], [2|2], [3|3], [4|4], [5|5], and [6|6] and seat them into the Doubles Lid!
2. **SPIN ANY DOMINO 180° ON YOUR DESK:**
   Take [2|5] (= 7) and rotate it half a turn: it becomes [5|2] (= 7)—proving a + b = b + a!
3. **DOUBLE IN 2 TAPS WITH [ENTER][+]:**
   Type 5 [ENTER] [+] 1 [+] [g][COUNT] on your calculator to solve [5|6] = 11!

**BOX ICON = CALCULATOR [G][COUNT] SUBITIZING & [G][A:B] RATIO KEYS**

The button icon embossed on your box lid (left) is the exact calculator key ([g][COUNT] & [g][a:b])! Build every lesson with your blocks, then press the key.

## NEAR-DOUBLES ([5|6] = 11) ↔ BOX [g][COUNT] ICON

STEP A: HANDS-ON WITH DOUBLES & NEAR-DOUBLES — 1:1 CAD SCENE

| Double [5\|5] | Near-Double [5\|6] | Box Icon Key |
|---|---|---|
| 5 + 5 = 10 | 10 + 1 = 11 | [g][COUNT] Subitize |

▲ Hold [5|6] beside [5|5] on the Doubles Lid: left halves match (5=5), right half has 1 extra dot (10+1=11)!

STEP B: TRY IT ON YOUR CALCULATOR — BOX ICON: [g][COUNT] & [g][a

NEAR-DOUBLE TREE: [5|6] = [5|5] + 1 = 11

DOMINO [5|6] = 11

Double [5|5] = 10 | + 1 Extra Pip

[ENTER] [+] | 1 [+] | [g][COUNT]=11

5 [ENTER] [+] 1 [+] [g][COUNT] → 11

PRESS BOX ICON & RPN KEYS (SC6):

1. DOUBLE 5: 5 ENTER × = 10
2. NEAR-DBL: 1 + g COUNT = 11

✓ DISPLAY (X) = 11

KEYSTROKES: 5 [ENTER] [+] 1 [+] [g] [COUNT]
In RPN, pressing [ENTER]|+| doubles 5 into 10 in two taps—then press 1 [+] [g][COUNT] for 11!

WELCOME TO HANDS-ON MATH • KEEP THIS GUIDE IN YOUR BOX LID

STEP 1: UNFOLD RIGHT → PAGE 02

▲ FOLD 1: FOLD BOTTOM HALF BEHIND TOP ▲

▼ UNFOLD STEP 2: OPEN DOWN FOR PAGES 03 & 04 ▼

# 02 PAGE / 08 — HOW IT WORKS: THE CORE IDEA

Triangular Number Set: T(7) = 28 Tiles (168 Dots Total)

01 02 03 04

SKU: DOMI-01 • KEEP IN LID

SIDE A • 1ST UNFOLD (TOP-RIGHT)

## TRIANGULAR NUMBERS & COMBINATORICS

Why are there exactly 28 tiles in a double-six domino box?
Because pairing numbers from 0 to n forms a triangular number:

**T(n+1) = (n + 1)(n + 2) / 2**

For double-six dominoes (n = 6):

**T(7) = (7 × 8) / 2 = 28 Unique Tiles**

And if you count every single dot across all 28 tiles:

**Total Dots = 8 × (0 + 1 + 2 + 3 + 4 + 5 + 6) = 8 × 21 = 168 Dots**

Every number from 0 to 6 appears on exactly 8 half-tiles!

### THE SECRET BELL CURVE OF DOMINO SUMS

Tile sums run from 0 [0|0] to 12 [6|6]. Notice that 6 is the most common sum of all (appearing 4 times: [0|6], [1|5], [2|4], [3|3]). Stacking tiles by their sum builds a real-life probability histogram right on your table!

### WHY DOUBLES UNLOCK NEAR-DOUBLES

Once your brain knows 6+6=12 by heart, tricky neighbors like 6+7 become effortless: just think "6+6 is 12, plus 1 more is 13!" Matching the 6 doubles into the lid sockets builds those mental anchor points automatically.

### 3 CLASSIC MATH TRAPS THIS KIT MAKES IMPOSSIBLE — VISUAL PROOF

- ✗ **TRAP 1 — Counting every dot one-by-one:**
  ✓ FIX: Standard dice/domino pip patterns let your visual cortex recognize 5 + 4 = 9 in 0.2 seconds!
- ✗ **TRAP 2 — Struggling with '7 + 8' or '5 + 6':**
  ✓ FIX: Hold [5|6] next to [5|5]: you immediately see it is just a Double Plus One (10 + 1 = 11).
- ✗ **TRAP 3 — Thinking 4 + 3 and 3 + 4 are two facts:**
  ✓ FIX: Rotating one physical domino tile 180° proves both facts are the exact same object!

CALCULATOR CHECK: 5 [ENTER] [+] 1 [+] [g] [COUNT] → X = 11

## DOMINO & TRAY QUICK REFERENCE

| PIECE | SIZE | ROLE | COLOR |
|---|---|---|---|
| Domino Tile | 39.2 × 19.6 × 7 mm | [a \| b] | White |
| Blank Tile | 39.2 × 19.6 × 7 mm | Wildcard | White |
| 12-Frame Tray | 126 × 42 × 10 mm | 10 + 2 | Blue |
| Doubles Lid | 126 × 42 × 10 mm | 1+1..6+6 | Black |
| 3x1 Storage Case | 126 × 42 × 49 mm | 3x1 Case | Black |

### BUILT FOR HANDS-ON TABLETOP PLAY

- Deeply recessed high-contrast dots you can count with your eyes closed.
- Sized to match the universal StackCalc grid (each tile = 2 miniblocks).
- Flip any tile over to use it as a mystery number or RPN stack token.
- Snug snap-on lid keeps all 30 tiles quiet inside a backpack.
- Easy to clean: wipe with a damp cloth or rinse in lukewarm soapy water.

1:1 CAD REFERENCE VIEW — 1:1 BLOCK CAD ANATOMY

✓ FLUSH PLUG-AND-SOCKET JOINT
Socket swallows plug completely: Only 7.00mm true heights add!

BOX ICON & FORMULA QUICK-REFERENCE

- **Box Button Icon Key:** [g][COUNT] Subitizing & [g][a:b] Ratio
- **Triangular Number Set:** T(7) = 7 × 8 ÷ 2 = 28 Unique Tiles
- **Doubles Fact Anchors:** [n|n] = 2n ↔ n [ENTER] [+]
- **Near-Doubles Shortcut:** [n | n+1] = 2n + 1 (Double + 1)
- **Proportional Pip Ratio:** [.] a [.] b [g][a:b] [g][SIMP]

COMBINATORICS MADE TACTILE • EVERY PAIR FROM [0|0] TO [6|6]

STEP 2: UNFOLD DOWN ▼ PAGES 03–04

# 03 PAGE / 08 — EXAMPLE 1: HANDS-ON WALKTHROUGH

Doubles Fact Matching on the Lid: [1|1] Through [6|6]

01 02 03 04

SKU: DOMI-01 • KEEP IN LID

SIDE A • 2ND UNFOLD (BOT-LEFT)

## 1 DOUBLES FACTS MATCHING ON LID

**1. POP THE DOUBLES PRACTICE SNAP LID:**
The top surface of your storage box lid features 6 pairs of recessed sockets labeled [ 1+1 ] through [ 6+6 ].

**2. HUNT FOR THE 6 DOUBLE TILES:**

- Find [1|1] → Snap into socket 1+1 (Sum = 2)
- Find [2|2] → Snap into socket 2+2 (Sum = 4)
- Find [3|3] → Snap into socket 3+3 (Sum = 6)
- Find [4|4] → Snap into socket 4+4 (Sum = 8)
- Find [5|5] → Snap into socket 5+5 (Sum = 10)
- Find [6|6] → Snap into socket 6+6 (Sum = 12)

**3. PRACTICE RAPID MENTAL DOUBLING:**
Tap each double in sequence. State the sum out loud without counting dots!

### WHY DOUBLES ARE THE KEY TO ARITHMETIC FLUENCY

Once a student knows 6 + 6 = 12 automatically, difficult near-doubles like 6 + 7 become trivial mental calculations: (6 + 6) + 1 = 13. The doubles lid physically locks these key anchors into place.

STEP B: TRY EXAMPLE #1 ON YOUR CALCULATOR — [g][COUNT] & [g][a

STEP 1 (COMPUTE 7×8):
Press 7 [ENTER] 8 [×] → X = 56 (if every ordered pair [a|b] and [b|a] were separate).

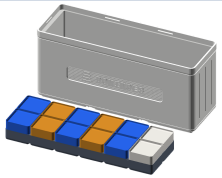
STEP 2 (DIVIDE BY 2):
Press 2 [÷] → X = 28 unique domino tiles (since rotating [a|b] by 180° gives [b|a]!).

STEP 3 (PRESS [g][COUNT]):
Press [g] [COUNT] to inspect the subitized counter pattern on your SC-6 display!

LIVE 4-REGISTER RPN STACK READOUT (AFTER KEYSTROKES):
T: 0 | Z: 0 | Y: 0 | X: 28 → DISPLAY (X) = 28

## LESSON 1: 28 DOUBLE-SIX TILES ↔ [g][COUNT] KEYSTROKES

STEP A: HANDS-ON WITH DOMINOES ON LID — 1:1 CAD SCENE



| 28 Unique Tiles | 6 Doubles Tiles | Tile Footprint |
|---|---|---|
| [0\|0] Through [6\|6] | Sums: 2, 4, 6, 8, 10, 1 | 39.2 × 19.6 mm (2U×1U) |

▲ Every possible pair from [0|0] to [6|6] appears once—snap the 6 Doubles tiles into the lid sockets to anchor mental math!

STEP B: TRY IT ON YOUR CALCULATOR — BOX ICON: [g][COUNT] & [g][a

TRIANGULAR COMBINATORICS: T(7) = 28 TILES

28 TILES (168 DOTS)

7 × 8 ÷ 2 = 28 | 6 Doubles [n|n]

[1|1]=2 | [3|3]=6 | [5|5]=10 | [6|6]=12

7 [ENTER] 8 [×] 2 [÷] [g][COUNT] → 28

PRESS BOX ICON & RPN KEYS (SC6):

1. T(7) FORMULA: 7 ENTER 8 × 2 ÷
2. BOX ICON KEY: g COUNT = 28 Tiles

✓ DISPLAY (X) = 28

KEYSTROKES: 7 [ENTER] 8 [×] 2 [÷] [g] [COUNT]
Compute the triangular number T(7) = 7×8÷2 = 28 tiles and press [g][COUNT] to view counters!
✓ 4-REGISTER STACK: T: 0 | Z: 0 | Y: 0 | X: 28 → X = 28

ADDITION DOUBLES ANCHORS • SNAP DOUBLES DIRECTLY ONTO THE LID

CONTINUE RIGHT → PAGE 04

# 04 PAGE / 08 — EXAMPLE 2: CORE MATHEMATICAL PRINCIPLE

Bridging Through 10 on the 12-Frame: (9 + 1) + 4 = 14

01 02 03 04

SKU: DOMI-01 • KEEP IN LID

SIDE A • 2ND UNFOLD (BOT-RIGHT)

## 2 BRIDGING THROUGH 10 ON THE 12-FRAME

**THE PROBLEM: Add domino [5|4] (9) and domino [3|2] (5).**

1. SEAT DOMINO [5|4] INTO THE 12-FRAME TRAY:
Its 9 pips (5 + 4 = 9) fill the first 9 sockets of the tray.
Notice: Exactly 1 socket remains before the red 10-Bridge line!

2. REGROUP FROM DOMINO [3|2] (5 PIPS):

- Move 1 pip across to reach the 10-Bridge line (9 + 1 = 10!).
- The remaining 4 pips cross the red ridge into the overflow sockets.

→ (9 + 1) + 4 = 10 + 4 = 14

### ✓ ASSOCIATIVE PROPERTY OF ADDITION

(A + B) + C = A + (B + C)

**9 + 5 = 9 + (1 + 4) = (9 + 1) + 4 = 10 + 4 = 14**

The tactile ridge physically forces the decomposition of 5 into 1 + 4. Students develop permanent mental arithmetic bridges instead of **counting by ones on their fingers!**

★ 2-PLAYER GAME: 'SUMS OF 7 SPEED HUNT' (2 MINS)
Spread all 28 dominoes face-up on the table. On 'GO!', both players race to grab every tile whose left and right pips add up to 7 ([0|7] doesn't exist—only [1|6], [2|5], and [3|4]!). **Then verify the total pip sum on your RPN calculator!**

⊙ WHERE THIS SHOWS UP IN COMBINATORICS & NETWORK GRAPH THEORY
Why are there 28 tiles in a double-six set? Because choosing 2 numbers from {0..6} with replacement is the triangular number T(7) = 7×8/2 = 28! Computer scientists use this exact **formula to count handshakes and network connections between servers.**

## LESSON 2: BRIDGING 10 ([5|4]+[3|2]=14) ↔ RPN STACK

STEP A: HANDS-ON WITH 12-FRAME TRAY — 1:1 CAD SCENE

| Domino [5\|4] = 9 | Domino [3\|2] = 5 | Total: 9 + 5 |
|---|---|---|
| Needs +1 for 10 | Splits into 1 + 4 | 10 + 4 = 14 |

▲ Place dominoes into the 12-Frame Tray to decompose part-whole sums and see why 9 + 5 regroups into 10 + 4 = 14!

STEP B: TRY IT ON YOUR CALCULATOR — BOX ICON: [g][COUNT] & [g][a

PART-WHOLE DECOMPOSITION: [5|4] + [3|2] = 14

[5|4] = 9 | [3|2] = 5

5 [ENTER] 4 [+] 3 [ENTER] 2 [+] [+] → 14

PRESS BOX ICON & RPN KEYS (SC6):

1. TILE [5|4]: 5 ENTER 4 + = 9
2. + TILE [3|2]: 3 ENTER 2 + + = 14

✓ DISPLAY (X) = 14

KEYSTROKES: 5 [ENTER] 4 [+] 3 [ENTER] 2 [+] [+]
Sum [5|4]=9 into Y and [3|2]=5 into X, then press [+] to combine both dominoes into 14!
✓ 4-REGISTER STACK: T: 0 | Z: 0 | Y: 0 | X: 14 → X = 14

PART-WHOLE DECOMPOSITION • SEE WHY 9 + 5 BECOMES 10 + 4

STEP 3: FLIP SHEET OVER ↻ PAGE 05

# 05 — EXAMPLE 3 & DESK CHALLENGES

PAGE / 08

Near-Doubles & Sum Histogram + 3 Desk Challenges

05 · 06 · 07 · 08

SKU: DOMI-01 • KEEP IN LID

SIDE B • FLIP SIDE (TOP-LEFT)

## 3 NEAR-DOUBLES: USE [5|5] TO SOLVE [5|6] IN 1 SECOND

**THE GOAL: Solve 5 + 6 without counting dots—just compare against a double!**

**1. SNAP [5|5] INTO THE LID:**
Seat domino [5|5] into its 5+5 = 10 socket on the lid.

**2. PLACE [5|6] RIGHT BESIDE IT:**
Hold domino [5|6] next to [5|5] and compare the two halves.

**3. SPOT THE +1 DIFFERENCE:**
Left halves match (5 = 5), right half has 1 extra dot: 10 + 1 = 11!

✓ WHY DOUBLES UNLOCK EVERY NEIGHBORING FACT

**Kids who memorize doubles (4+4=8, 5+5=10, 6+6=12) get 12 more facts for free!**
Whenever two numbers are neighbors (like 6 + 7), it's just a Double Plus One.
Instead of counting 7 fingers past 6, your eyes see [6|6] + 1 = 13 immediately.
Use the 2 blank wildcard tiles with a dry-erase marker to invent your own games!

PRO TIP: CLOSE YOUR EYES & FEEL THE ANSWER

Once you build an example, run your fingers across the pieces. Your hands
remember physical shapes and heights way longer than memorized rules!

EXAMPLE 3: VISUAL DECOMPOSITION TREE

NEAR-DOUBLE TREE: [5|6] = [5|5] + 1 = 11

DOMINO [5|6] = 11

Double [5|5] = 10 · + 1 Extra Pip

5 [ENTER] [+] · 1 [+] · [g][COUNT]=11

5 [ENTER] [+] 1 [+] [g][COUNT] → 11

STEP B: PRESS BOX ICON & RPN KEYS (SC6)

TACTILE KEYSTROKE WALKTHROUGH:

1. DOUBLE 5: 5 ENTER + = 10

2. NEAR-DBL: 1 + g COUNT = 11

✓ CALCULATOR DISPLAY (X) → 11

Σ+ FULL KEYSTROKES: 5 [ENTER] [+] 1 [+] [g] [COUNT]

✓ 4-REGISTER STACK: T: 0 | Z: 0 | Y: 0 | X: 11 → X = 11

## 3 DESK CHALLENGES — TRY THESE NOW!

WARM-UP #1 — Match all 6 doubles ([1

Step A: Build with blocks on your Lid Tray • Step B: Verify on SC-32 Calculator

✓ Build first, then flip booklet upside-down to check Answer #1 below!

LEVEL-UP #2 — Find all 4 dominoes in the box whose dots add up to 6!

Step A: Build with blocks on your Lid Tray • Step B: Verify on SC-32 Calculator

✓ Build first, then flip booklet upside-down to check Answer #2 below!

BRAIN-MELTER #3 — Place [6

Step A: Build with blocks on your Lid Tray • Step B: Verify on SC-32 Calculator

✓ Build first, then flip booklet upside-down to check Answer #3 below!

SELF-CHECK ANSWER KEY (ROTATE BOOKLET 180° TO READ)

#1: [1] to [6|6]) on the Tile! | Sums are 2, 4, 6, 8, 10, 12 (All evens!) | #2: [0|6], [1|5], [2|4], and [3|3] #3: 2] (8) and [2|2] (5) on the 12-Frame. What crosses the 10-bridge? | 10 + 3 = 13

FOR PARENTS & TEACHERS (Grades K–4 (Subitizing, Addition Doubles & Mental Math))

- CCSS.MATH.CONTENT.1.OA.C.6: Add and subtract within 20 using doubles and making ten.
- CCSS.MATH.CONTENT.2.OA.B.2: Fluently add and subtract within 20 using mental strategies.
- TEKS 111.3.b.3.D: Apply basic fact strategies to add and subtract within 20.
- **CCSS.MATH.PRACTICE.MP2, MP4, MP5, MP7: Reason, Model, Use Tools & Structure.**

Next up on Page 06 → How to stack everything back in the box in 10 seconds!

BOSS CHALLENGE #4 — BLOCKS → BOX ICON KEY SPEED-RUN!

1. Build Examples #1, #2, and #3 hands-on with your physical pieces.
2. Press the Box Button Icon ([g][COUNT] & [g][a:b]) and keystrokes on your calculator:
• 5 [ENTER] [+] 1 [+] [g] [COUNT] → X = 11

CHECK OFF YOUR MASTERY MILESTONES:

- □ 1. Demonstrated Lessons #1, #2 & #3 hands-on with the physical blocks
- □ 2. Solved all 3 Desk Challenges above (check upside-down key if stuck!)
- □ 3. Demonstrated each lesson on the calculator using the Box Button Icon
- □ 4. Packed all pieces back into the box in under 10 seconds (see Page 06 →)

TEST YOURSELF • ANSWERS INCLUDED SO YOU CAN CHECK RIGHT AWAY

NEXT: HOW TO PACK UP → PAGE 06

# 06 — HOW TO STACK IT ALL BACK IN

PAGE / 08

The 10-Second Self-Checking Pack-Up Puzzle

05 · 06 · 07 · 08

SKU: DOMI-01 • KEEP IN LID

SIDE B • PACK-UP GUIDE (TOP-RIGHT)

## TOP-DOWN BOX MAP: WHERE EVERY PIECE LIVES

Putting the blocks away is a 10-second puzzle. Match each piece to its
labeled home bay below so everything nests flush and the lid snaps shut:

Col 1 (6x) · Col 2 (6x) · Col 3 (6x) · Col 4 (6x) · Col 5 (6x)

✓ FIVE COLUMNS OF 6 TILES (30 TILES TOTAL)

The storage box holds 5 neat vertical stacks of 6 domino tiles each (5 × 6 = 30 tiles:
all 28 double-six dominoes plus the 2 blank wildcard tiles).
When all 30 tiles are stacked in the 5 columns, every stack sits flush with the
top rim—so a single missing domino is obvious before you snap the lid on!

1. STORAGE CASE & COMPARTMENTS

2. WHY THE LID SELF-CHECKS EVERY PIECE

SELF-CHECKING RIM & WATERLINE CROSS-SECTION

✓ FLUSH 84.0 MM = LID CLOSES

✓ OK · ✓ OK · ✓ OK · ✓ OK · ✗ HIGH

5 Columns × 6 Dominoes Flush Rim + 12-

▲ SELF-INDEXING BAYS & EMBOSSED BOX BUTTON ICON:
Calibrated bays guide every piece home, and the lid icon links directly to your calculator.

## THE 10-SECOND PACK-UP ROUTINE

HOW TO STACK EVERYTHING BACK IN:

**1. SORT BY COLOR & NUMBER:**
Group matching blocks or tiles together on your desk first.

**2. STACK & DROP INTO LABELED BAYS:**
Follow the Top-Down Box Map on the left. If a tower looks too tall
or too short, a block is in the wrong bay—swap it until everything levels!

**3. TUCK THIS CARD IN THE LID & SNAP SHUT:**
Fold this guide back to [01] and rest it right on top before closing.

DOUBLES PRACTICE LID & TRAYNESTING

- The Doubles Lid snaps directly onto the storage box and seats the 12-Frame Tray on top.
- Flip any domino tile over to see the debossed 4-level StackCalc RPN register icon.
- Easy care: wipe clean or rinse in lukewarm soapy water.

SHORT A PIECE AT CLEAN-UP TIME?

First check under notebooks and chair legs! If a piece truly vanished,
**see PAGE 08 (bottom-right) for your complete Part ID Reference Table.**

✓ DOMI-01 4-POINT CLEAN-UP CHECKLIST

1. FIVE 6-TILE STACKS: Stack 6 dominoes in each of the 5 columns (28 tiles + 2 wildcards = 30).
2. CHECK RIM LEVEL: Verify all 5 stacks sit completely flush with the box rim.
3. NEST 12-FRAME TRAY: Seat the Blue 6×2 12-Frame Tray and Doubles Practice Lid.
4. TUCK CARD & SNAP: Fold this booklet to Panel [01] and snap the lid closed!

MODULAR CLASSROOM & BOOKSHELF NESTING (42.0 MM PITCH)

Every StackCalc box is sized in exact multiples of 42.0 mm (1×1, 3×1, 3×2).
All 9 kit boxes interlock side-by-side in a classroom cart or drawer with zero
wasted space, and every embossed box icon matches its calculator operation key!

5 EQUAL COLUMNS OF 6 TILES • SPOT A MISSING DOMINO AT A GLANCE

NEXT: CONNECTED LAB ▾ PAGE 07

SIDE B (PAGES 05–08) • READ TOP ROW FIRST

THEN READ BOTTOM ROW (PAGE 07 → PAGE 08)

# 07 — THE STACKCALC CONNECTED LAB

PAGE / 08

How Your Kits & RPN Calculator Stack Work Together

05 · 06 · 07 · 08

SKU: DOMI-01 • KEEP IN LID

SIDE B • ECOSYSTEM (BOT-LEFT)

ONE CONNECTED SYSTEM FROM KINDERGARTEN TO CALCULUS

Every StackCalc box uses the same modular grid and unit heights—so pieces from different kits actually fit and work together.

## WHERE YOUR BOX FITS IN THE 9-KIT LAB

| | | |
|---|---|---|
| BASE-10 — Ten-Frames Lab — Early Counting & Regrouping | DOMI-01 ◂ YOUR BOX — Math Domino Studio — Doubles & 12-Frame Part-Whole | FRAC-01 — Fraction Studio — Proportional Towers & 4-Way Faces |
| PRIME-01 — Prime Factor Studio — Multiply by Stacking Prime Towers | ALGB-01 — Algebra Studio — Quadratic Area Arrays & Factoring | TIME-01 — Planetary Clock — 12:1 Geared Hands & Time Wedges |
| PROB-01 — Probability Dice — Fair Polyhedrals & 10-Sector Spinner | VOLV-01 — Unit Circle Volvelle — Spin-to-Read Exact Trig Radicals | SC-32-HW — SC-32 RPN Calculator — 4-Register Stack Scientific Engine |

WHY THIS MATTERS: ZERO "THROW-AWAY" MATH

In most classrooms, every grade uses a totally unrelated plastic toy.
In StackCalc, the 7mm unit block you count with in 1st grade is the exact
same height as the 1/12th fraction block and the [2] prime block later on!

1. DOMI-01 BOX LID ICON

Σ+

[g][COUNT] & [g][a:b]

EMBOSSED ON BOX LID

Build First → Press Key!

2. SC-6 ELEMENTARY RPN

3. SC-32 SCIENTIFIC RPN

▲ BOX ICON ↔ CALCULATOR KEY CONTINUITY:
Every physical box lid carries the exact icon of the calculator key that computes that lesson!

## HOW BLOCKS CONNECT TO THE CALCULATOR

SHARED MODULAR FOOTPRINT & HEIGHTS

- Each domino tile (39.2 × 19.6 mm) equals exactly two standard StackCalc unit squares.
- Recessed sockets swallow the bottom peg so only true heights add.
- All storage boxes nest neatly side-by-side on any desk or shelf.

WHY WE CALL IT "STACKCALC" (THE RPN SECRET)

- Flip any tile over to reveal the 4-register RPN stack icon—linking dot patterns to calculator registers.
- Placing a block on the table = pressing [ENTER] to push onto the stack.
- Snapping two towers together = pressing [+] or [×] with zero parentheses!
- Mirrors the 4 stack levels on the SC-32 calculator: X, Y, Z, and T.

FREE ACTIVITY DECKS & INTERACTIVE SIMULATORS

Download extra printable puzzle cards, teacher lesson guides,
and interactive browser simulations anytime—no login needed.

Explore All 9 Kits & Free Guides at stackcalc.io

LIVE 4-REGISTER RPN STACK

T: 0 (Top Memory)
Z: 0 (3rd Level)
Y: 0 (1st Block)
X: 28 (T{7} Tiles) (Active Display)

HOW BLOCKS MAP TO STACK KEYS

1. PLACE 1ST PIECE ON STAGE:
Type number + [ENTER] → lifts to Y
2. ADD / STACK 2ND PIECE:
Type 2nd number in X, then [+]/[×]
3. PRESS BOX ICON KEY:
Press [g][COUNT] & [g][a:b]
✓ Zero Parentheses • Always Visible!

Σ+ BOX ICON = [g][COUNT] & [g][a:b] • VERIFIED LESSON SEQUENCES:
• Lesson 1: 7 [ENTER] 8 [×] 2 [+] [g] [COUNT] → X = 28
• Lesson 2: 5 [ENTER] 4 [+] 3 [ENTER] 2 [×] [+] → X = 14

MODULAR 9-KIT CONTINUUM • FROM EARLY COUNTING TO AP CALCULUS

NEXT: PART ID TABLE → PAGE 08

# 08 — KEEP THIS CARD! PART IDs & VIDEO

PAGE / 08

Complete 32-Piece Reference & 60-Second 3D Video Guide

05 · 06 · 07 · 08

SKU: DOMI-01 • KEEP IN LID

SIDE B • PART REFERENCE (BOT-RIGHT)

## PART ID REFERENCE TABLE (32 TOTAL PIECES)

✓ KEEP THIS CARD INSIDE YOUR BOX LID!

Backpacks eat blocks. Dogs chew dice. Vacuum cleaners take no prisoners. Keep this card
in your lid so you always have the exact Part IDs below for single-piece refills!

| PART NAME | QTY | COLOR / FINISH | PART ID |
|---|---|---|---|
| Double-Six Domino Tiles | 28x | White | DOMI-TILE-01a |
| Blank Wildcard Dominoes | 2x | White | DOMI-BLNK-01a |
| 6x2 12-Frame Tray | 1x | Blue | DOMI-TRAY-d4e |
| Doubles Practice Snap Lid | 1x | Black | DOMI-LID-b7c |
| 3x1 Storage Case | 1x | Black | DOMI-BOX-a1b |

NEED A REPLACEMENT PIECE? NO NEED TO BUY A WHOLE NEW BOX.

Enter any Part ID above at stackcalc.io/parts to order individual replacement pieces.

COLOR & PIECE IDENTIFICATION KEY — MATCH ANY LOOSE PART

White — Domino Tiles (28x + 2x) · Blue — 12-Frame Regrouping Tray · Black — 3x1 Storage Case · Gray — Doubles Practice Lid

UNIVERSAL STACKCALC HEIGHT PITCH (1U = 7.00 MM • 12U = 84.00 MM ONE WHOLE)

Measure any loose piece in 7mm units (U) against this scale bar to identify it:

0U (0mm) 1U (7mm) 2U (14mm) 3U (21mm) 4U (28mm) 6U (42mm) 9U (63mm) 12U (84mm)

✓ DISTRICT PO & SAFETY: 10-Pack SKU DOMI-01-CLASS10 • CPSIA / ASTM F963 Non-Toxic Washable PLA/ABS

## WATCH THE 60-SECOND 3D VIDEO GUIDE

POINT CAMERA TO WATCH DEMO

stackcalc.io/products/math-dominoes/

See animated 3D examples & pack-up walkthrough

QUESTIONS, CLASSROOM SETS & SUPPORT

- Email us anytime: support@stackcalc.io
- School & District POs: school-orders@stackcalc.io

**Designed in Mountain View, CA • Non-Toxic & Washable**

COMPANION MANUAL CHAPTERS, STANDARDS & KEYSTROKES

BOOK CHAPTERS: SC-6 Guide Ch. 2 (Domino Pairs & Ratios) • SC-32 Manual Ch. 2
STANDARDS: CCSS K.OA.A.3, 1.OA.B.3, 2.OA.B.2, 6.RP.A.1 • TEKS 1.3C, 6.4B (Pairs & Ratios)
WATERLINE SPEC: 5 Columns × 6 Dominoes Flush Rim + 12-Frame Tray

ALL 3 VERIFIED CALCULATOR LESSON SEQUENCES (SCG):
Ex 1: 7 [ENTER] 8 [×] 2 [+] [g] [COUNT] → X=28
Ex 2: 5 [ENTER] 4 [+] 3 [ENTER] 2 [×] [+] → X=14
Ex 3: 5 [ENTER] [+] 1 [×] [g] [COUNT] → X=11

Σ+ BOX ICON KEY: [g][COUNT] & [g][a:b]
✓ Cross-Referenced to Companion Book & Calculator Engine
Fold back to [01] & keep this reference card inside your box lid!

LOST A PIECE? GRAB SINGLE REPLACEMENTS ANYTIME AT stackcalc.io/parts

✓ KEEP THIS GUIDE IN BOX LID!

# 01 PROBABILITY DICE & MODULAR SPINNER PACK

PAGE / 08

5 Symmetrical Platonic Dice + 10-Sector Decimal Spinner Lid

01 02 03 04

SKU: PROB-01 • KEEP IN LID

SIDE A • COVER (TOP-LEFT)

**WHAT IS THIS BOX FOR? OUTSMART YOUR "GUT FEELING."**

Human brains are terrible at guessing odds. Five Platonic solids and a 10-sector rotary spinner let you test probability right on your desk.

## WELCOME TO YOUR POCKET PROBABILITY LAB

Why guess the odds when you can roll and spin them for real?

- 5 Platonic Solids: 4, 6, 8, 12, and 20-sided symmetrical dice
- Free-Spinning Lid Rotor: Flick the pointer for an 8-second spin
- 10-Sector Decimal Dial: Turns fractions into 10% decimal slices
- Symmetrical Balance: Every face has an equal shot at landing up

Try this right now: Snap the red pointer onto the lid and give it a flick!

### THE LID THAT DOUBLES AS A SPINNER

The storage lid has a built-in 0–9 decimal dial. Snap the pointer into the center hub to run random trials, or pop it inside the case when you pack up.

### WHAT'S IN YOUR BOX (9 PIECES)

- 5 Polyhedral Dice (d4 Tetrahedron, d6 Cube, d8 Octahedron, d12, d20)
- 1x Balanced Spinner Pointer + 1x Smooth-Spin Center Hub Pin
- 1x 1x1 Pocket Storage Case + 1x 10-Sector Decimal Dial Lid

★ YOUR FIRST 60 SECONDS WITH THIS BOX — START HERE!

1. SNAP THE SPINNER POINTER ONTO THE LID:
Seat the hub pin and pointer into the 0–9 dial lid and give it a flick—each sector is 10%!
2. ROLL ALL 5 PLATONIC DICE AT ONCE:
Compare the 4-sided pyramid (d4 = 25% per face) up to the 20-sided icosahedron (d20 = 5% per face).
3. PRESS THE BOX ICON KEY [🎲 & [Σ+]:
Press 6 [🎲 on your SC-6 to roll digitally, or log physical rolls with [Σ+] and [g][x̄]!

BOX ICON = CALCULATOR [🎲 DICE SIM, [Σ+] & [G][x̄] STATS KEYS

The button icon embossed on your box lid (left) is the exact calculator key ([🎲, [Σ+] & [g][x̄])! Build every lesson with your blocks, then press the key.

## d20 PRIME ODDS (8/20 = 2/5) ↔ BOX [🎲] & [FDISP] KEYS

STEP A: HANDS-ON WITH d20 & POLYHEDRA — 1:1 CAD SCENE

d20 = 20 Faces — 8 Primes (2..19) | Exact Fraction — 8/20 = 2/5 = 40% | Box Icon Key — [🎲 & [Σ+] Stats

▲ Count the 8 prime numbers on the 20-sided icosahedron (2,3,5,7,11,13,17,19) and reduce 8/20 → 2/5 with [FDISP]!

STEP B: TRY IT ON YOUR CALCULATOR — BOX ICON: [🎲, [Σ+] & [g][x̄

PROBABILITY TREE: d20 PRIME ROLL ODDS

d20 = 20 EQUAL FACES

8 Primes (40%) | 12 Non-Primes (60%)

2,3,5,7 | 11,13,17,19 | [FDISP]=2/5

8 [ENTER] 20 [÷] [FDISP] → 2/5 (40%)

PRESS BOX ICON & RPN KEYS (SCG):

1. d20 PRIMES: 8 ENTER 20 ÷ = 0.4
2. REDUCE FRAC: FDISP = 2/5 (40%)

✓ DISPLAY (X) = 2/5

KEYSTROKES: 8 [ENTER] 20 [÷] [FDISP]

Divide 8 prime faces by 20 total faces on the d20 and press [FDISP] to see 2/5!

WELCOME TO HANDS-ON MATH • KEEP THIS GUIDE IN YOUR BOX LID

STEP 1: UNFOLD RIGHT → PAGE 02

▲ FOLD 1: FOLD BOTTOM HALF BEHIND TOP ▲

# 02 HOW IT WORKS: THE CORE IDEA

PAGE / 08

Platonic Symmetry & Fair Odds: P(Single Outcome) = 1 / Faces

01 02 03 04

SKU: PROB-01 • KEEP IN LID

SIDE A • 1ST UNFOLD (TOP-RIGHT)

## FAIR ODDS, EXPECTED VALUE & THE LONG RUN

For any symmetrical die or balanced spinner with F equal faces:

**P(single outcome = k) = 1 / F**

What is the average score you should expect over many rolls (1 to F)?

**Expected Value E[X] = (F + 1) / 2**

And how widely do the rolls spread out around that average?

**Variance Var(X) = (F² − 1) / 12**

Notice that for a 6-sided die, the average roll is 3.5 — a number that doesn't even exist on the die!

### WHY ONLY 5 PLATONIC DICE EXIST IN THE UNIVERSE

Over 2,300 years ago, Euclid proved there are only 5 3D shapes where every face is an identical regular polygon and every corner meets at the exact same angle: 4, 6, 8, 12, and 20 faces. You hold all 5 in this box!

### THE LAW OF LARGE NUMBERS

In 5 rolls, anything can happen. In 500 rolls, the average settles right onto E[X]. That's how casinos, insurance companies, and weather models turn wild short-term randomness into rock-solid long-term predictions.

3 CLASSIC MATH TRAPS THIS KIT MAKES IMPOSSIBLE — VISUAL PROOF

✗ TRAP 1 — The Gambler's Fallacy ('7 is due!'):
✓ FIX: Dice and spinners have zero memory—after spinning three 9s in a row, the next spin is still 10%.

✗ TRAP 2 — Thinking 2 and 7 are equally likely on 2d6:
✓ FIX: Only (1,1) makes 2, whereas 6 different pairs make 7 (16.7% vs 2.8%)!

✗ TRAP 3 — Expecting 10 spins to hit 0–9 once each:
✓ FIX: Real randomness clumps in small samples; convergence only emerges over dozens of trials.

CALCULATOR CHECK: 8 [ENTER] 20 [÷] [FDISP] → X = 2/5

## PLATONIC SOLIDS & SPINNER ODDS TABLE

| SOLID | FACES (F) | AVG ROLL | SINGLE HIT ODDS |
|---|---|---|---|
| d4 (Tetrahedron) | 4 Faces | 2.50 | 1/4 = 25.0% |
| d6 (Cube) | 6 Faces | 3.50 | 1/6 ≈ 16.7% |
| d8 (Octahedron) | 8 Faces | 4.50 | 1/8 = 12.5% |
| d12 (Dodecahedron) | 12 Faces | 6.50 | 1/12 ≈ 8.33% |
| d20 (Icosahedron) | 20 Faces | 10.50 | 1/20 = 5.00% |
| Spinner Dial | 10 Sectors | 4.50 (0–9) | 1/10 = 10.0% |

### BUILT FOR FAIR & FAST EXPERIMENTS

- Rounded tumble edges ensure every die rolls smoothly across the desk.
- Low-friction center hub pin lets the pointer spin freely for 8+ seconds.
- Deeply recessed high-contrast numbers won't rub off during classroom play.
- Pocket-sized 1x1 case snaps into any modular StackCalc desk tray.

1:1 CAD REFERENCE VIEW — BOX ICON & FORMULA QUICK-REFERENCE

1:1 BLOCK CAD ANATOMY

SOCKET (1.85mm) ▲ 1.85mm

1U = 7.00 mm VISIBLE DATUM

7mm

BASE PLUG (1.85mm) ▼ 1.85mm

✓ FLUSH PLUG-AND-SOCKET JOINT

Socket swallows plug completely: Only 7.00mm true heights add!

Box Button Icon Keys: [🎲 Dice Roll, [Σ+] & [g][x̄] Mean

Single Fair Die Law: P(face) = 1 / F (Sum of all faces = 1.0)

Opposite Faces Rule: Opposite faces on fair d(F) sum to F + 1

10-Sector Spinner: 1 sector = 1/10 = 0.10 = 10% = 36°

Independent Events: P(A and B) = P(A) × P(B)

PLATONIC GEOMETRY • 5 FAIR SOLIDS + 10-SECTOR DECIMAL SPINNER

STEP 2: UNFOLD DOWN ▼ PAGES 03–04

▼ UNFOLD STEP 2: OPEN DOWN FOR PAGES 03 & 04 ▼

# 03 EXAMPLE 1: HANDS-ON WALKTHROUGH

PAGE / 08

Run 20 Spinner Trials on the 0–9 Decimal Dial Lid

01 02 03 04

SKU: PROB-01 • KEEP IN LID

SIDE A • 2ND UNFOLD (BOT-LEFT)

## 1 UNIFORM ODDS: 20 SPINNER TRIALS

**1. SNAP ROTARY NEEDLE ONTO DIAL LID:**
Seat the pointer arm over the pivot pin on the 10-sector dial lid.
Flick it lightly: verify that it spins freely for at least 8 seconds.

**2. CONDUCT 20 TRIAL SPINS:**
- Spin the needle 20 times in sequence.
- Tally each result into sectors 0 through 9.

Theoretical Expected Count = 20 × (1/10) = 2 hits per sector.

**3. OBSERVE EMPIRICAL VARIANCE:**
Some sectors will have 3 or 4 hits; others may have 0 or 1.
→ This is genuine random variance, not equipment bias!

### THE LAW OF LARGE NUMBERS IN PRACTICE

Combine tallies across 5 lab partner groups (100 total spins): watch the empirical frequencies level out toward exactly 10% each! Students tangibly experience why large sample sizes produce certainty.

STEP B: TRY EXAMPLE #1 ON YOUR CALCULATOR — [🎲, [Σ+] & [g][x̄

STEP 1 (ENTER d6 ODDS):
Press 1 [ENTER] 6 [÷] → pushes P(d6 face) = 0.1667 into Register X.

STEP 2 (MULTIPLY SPINNER):
Press 1 [ENTER] 10 [÷] [×] → multiplies by P(Spinner digit) = 0.10 to get 0.0167.

STEP 3 (EXACT FRACTION):
Press [FDISP] → displays exact compound probability X = 1/60!

LIVE 4-REGISTER RPN STACK READOUT (AFTER KEYSTROKES):
T: 0 | Z: 0 | Y: 0 | X: 1/60 → DISPLAY (X) = 1/60

## LESSON 1: 5 PLATONIC DICE & SPINNER ↔ [FDISP] ODDS

STEP A: HANDS-ON WITH DICE & SPINNER — 1:1 CAD SCENE

5 Fair Polyhedra — d4, d6, d8, d12, d20 | Decimal Spinner Lid — 10 Sectors (0–9 = 10% e | Compound Odds — 1/6 × 1/10 = 1/60

▲ Every Platonic die gives exact fractional odds (1/N), while the 10-sector lid spinner gives exact decimal odds (0.10 = 10%)!

STEP B: TRY IT ON YOUR CALCULATOR — BOX ICON: [🎲, [Σ+] & [g][x̄

INDEPENDENT EVENTS TREE: d6 × SPINNER

6 × 10 = 60 OUTCOMES

d6: P(A) = 1/6 | Spinner: P(B) = 1/1

[ENT] 6 [÷] | [ENT] 10 [÷] | [FDISP]=1/6

P(A and B) = 1/6 × 1/10 = 1/60

PRESS BOX ICON & RPN KEYS (SCG):

1. P(A) & P(B): 1 ENTER 6 ÷ 1 ENTER 10 ÷
2. MULTIPLY: × FDISP = 1/60

✓ DISPLAY (X) = 1/60

KEYSTROKES: 1 [ENTER] 6 [÷] 1 [ENTER] 10 [÷] [×] [FDISP]

Roll the d6 (1/6) and spin the 0–9 lid (1/10), then multiply on the calculator and press [FDISP] for 1/60!

✓ 4-REGISTER STACK: T: 0 | Z: 0 | Y: 0 | X: 1/60 → X = 1/60

THEORETICAL VS REAL LIFE • WATCH THE AVERAGES SETTLE OVER TIME

CONTINUE RIGHT → PAGE 04

# 04 EXAMPLE 2: CORE MATHEMATICAL PRINCIPLE

PAGE / 08

Two-Dice Sum Bell Curve: Why 7 Appears 6 Times More Than 2

01 02 03 04

SKU: PROB-01 • KEEP IN LID

SIDE A • 2ND UNFOLD (BOT-RIGHT)

## 2 CENTRAL LIMIT THEOREM: TWO d6 DICE SUM

**THE EXPERIMENT: Roll one d12 die vs. rolling two d6 dice.**

1. SINGLE d12 DIE (UNIFORM DISTRIBUTION):
Every number from 1 to 12 has exactly 1 face (P = 1/12 ≈ 8.33%).
The histogram is completely flat across all numbers.

2. SUM OF TWO d6 DICE (COMPOUND DISTRIBUTION):
There are 6 × 6 = 36 equally likely microstates (a, b):
- Sum 2: Only 1 way: (1, 1) → P = 1/36 (2.8%)
- Sum 7: Exactly 6 ways: (1,6), (2,5), (3,4), (4,3), (5,2), (6,1)

→ P(Sum = 7) = 6/36 = 1/6 (16.7%) — 6x more likely than rolling a 2!

### ✓ BIRTH OF THE BELL CURVE (CENTRAL LIMIT THEOREM)

Adding independent random variables creates a bell curve!

**Uniform + Uniform = Triangular Distribution (Mode = 7)**

Students literally watch a smooth bell curve emerge out of discrete square dice rolls. This is the foundational intuition for all **modern statistics and Gaussian normal distributions!**

★ 2-PLAYER GAME: 'PLATONIC HIGH-LOW SHOWDOWN' (2 MINS)

Player A picks the d12 + Spinner (0–9). Player B picks the d20. Before rolling, both players calculate who has the higher expected average (d12 avg 6.5 + Spinner avg 4.5 = 11.0 vs d20 **avg 10.5!), then log 10 rounds into [Σ+] and press [g][x̄]!**

⊙ WHERE THIS SHOWS UP IN QUANTUM PHYSICS, AI & MONTE CARLO SIMS

Why are there only 5 regular polyhedra (d4, d6, d8, d12, d20) in the universe? Euclid proved in 300 BC that no other flat regular polygons can close into a 3D corner! Today, engineers use **random sampling ('Monte Carlo simulation') to predict weather and train AI models.**

## LESSON 2: 2d6 EMPIRICAL TRIALS ↔ [Σ+] & [g][x̄] MEAN

STEP A: HANDS-ON WITH 2d6 BELL CURVE — 1:1 CAD SCENE

Sum = 2 or 12 — 1/36 = 2.8% (Rare) | Sum = 7 (Peak) — 6/36 = 1/6 = 16.7% | Log Rolls in [Σ+] — [g][x̄] Reads Mean

▲ Six different pairs ((1,6), (2,5), (3,4), (4,3), (5,2), (6,1)) add to 7—log your rolls with [Σ+] and press [g][x̄]!

STEP B: TRY IT ON YOUR CALCULATOR — BOX ICON: [🎲, [Σ+] & [g][x̄

2d6 SAMPLE MEAN TREE: ROLLS 7, 4, 10

3 TRIALS IN [Σ+]

Σx = 7+4+10 = 21 | n = 3 Rolls

7 [Σ+] | 4 [Σ+] | 10 [Σ+] [g][x̄]=7

[g][CLE] 7 [Σ+] 4 [Σ+] 10 [Σ+] [g][x̄] → 7

PRESS BOX ICON & RPN KEYS (SCG):

1. LOG ROLLS: g CLE 7 Σ+ 4 Σ+ 10 Σ+
2. READ MEAN: g x̄ = 7 (Σ: n=3)

✓ DISPLAY (X) = 7

KEYSTROKES: [g] [CLE] 7 [Σ+] 4 [Σ+] 10 [Σ+] [g] [x̄]

Log your physical dice or spinner trials into [Σ+], then press [g][x̄] to see the sample mean converge!

✓ 4-REGISTER STACK: T: 0 | Z: 3 | Y: 0 | X: 7 → X = 7

COMPOUND PROBABILITY • WHY TWO DICE CREATE A TRIANGULAR BELL CURVE

STEP 3: FLIP SHEET OVER ↻ PAGE 05

# 05 — EXAMPLE 3 & DESK CHALLENGES

PAGE / 08

Comparing Die Odds vs. Spinner Odds + 3 Desk Challenges

05 06 07 08

SKU: PROB-01 • KEEP IN LID

SIDE B • FLIP SIDE (TOP-LEFT)

## ③ HEAD-TO-HEAD: ROLL THE D20 VS. SPIN THE 0–9 DIAL

**THE GOAL: Compare a 20-sided die against the 10-sector decimal spinner!**

**1. ROLL THE 20-SIDED DIE (d20):**

Each face from 1 to 20 has a 1/20 = 5% chance of landing on top.

**2. GROUP INTO EVEN VS. ODD:**

How many faces are > 18? Only two (19 and 20): 2/20 = 1/10 = 10%.

**3. TEST ON THE 0–9 SPINNER:**

Spinning a '9' on the 10-sector lid also has a 1/10 = 10% chance—identical odds!

**✓ TRANSLATING FRACTIONS INTO DECIMALS BY SPINNING**

**Dice naturally teach fractions: 1/4, 1/6, 1/8, 1/12, and 1/20.**

The 10-sector spinner naturally teaches decimals and percents: 1 sector = 0.10 = 10%.
Using both together shows that a 1-in-4 roll on the d4 (25%) is halfway between 2 sectors (20%) and 3 sectors (30%) on the decimal spinner!

**PRO TIP: CLOSE YOUR EYES & FEEL THE ANSWER**

Once you build an example, run your fingers across the pieces. Your hands remember physical shapes and heights way longer than memorized rules!

**EXAMPLE 3: VISUAL DECOMPOSITION TREE**

PROBABILITY TREE: d20 PRIME ROLL ODDS

d20 = 20 EQUAL FACES

8 Primes (40%) | 12 Non-Primes (60%)

2,3,5,7 | 11,13,17,19 | [FDISP]=2/5

8 [ENTER] 20 [÷] [FDISP] → 2/5 (40%)

**STEP B: PRESS BOX ICON & RPN KEYS (SCG)**

TACTILE KEYSTROKE WALKTHROUGH:

1. d20 PRIMES: 8 ENTER 20 ÷ = 0.4
2. REDUCE FRAC: FDISP = 2/5 (40%)

✓ CALCULATOR DISPLAY (X) = 2/5

FULL KEYSTROKES: 8 [ENTER] 20 [÷] [FDISP]

✓ 4-REGISTER STACK: T: 0 | Z: 0 | Y: 0 | X: 2/5 → X = 2/5

## 3 DESK CHALLENGES — TRY THESE NOW!

**WARM-UP #1** — **Spin the 0–9 dial 30 times. Expected hits per number = 3. Which won?**

Step A: Build with blocks on your Lid Tray • Step B: Verify on SC-32 Calculator

✓ Build first, then flip booklet upside-down to check Answer #1 below!

**LEVEL-UP #2** — **Roll the d12 and d20 together. Which die is more likely to roll a 1?**

Step A: Build with blocks on your Lid Tray • Step B: Verify on SC-32 Calculator

✓ Build first, then flip booklet upside-down to check Answer #2 below!

**BRAIN-MELTER #3** — **Roll the d20: what is the exact probability of rolling a prime number?**

Step A: Build with blocks on your Lid Tray • Step B: Verify on SC-32 Calculator

✓ Build first, then flip booklet upside-down to check Answer #3 below!

SELF-CHECK ANSWER KEY (ROTATE BOOKLET 180° TO READ)

#3: 8 primes / 20 = 40%!

#2: Roll random(1,20) | #2: d12 (1/12 = 8.3% vs 1/20 = 5%)

#1: Roll random(0,9) | #1: 3 hits = 10% (Expected)

**FOR PARENTS & TEACHERS (Grades 5–10 (Probability, Statistics & Data Science))**

- CCSS.MATH.CONTENT.7.SP.C.5: Understand that probability is a number between 0 and 1.
- CCSS.MATH.CONTENT.7.SP.C.6: Approximate probability by collecting data on chance processes.
- CCSS.MATH.CONTENT.7.SP.C.8: Find probabilities of compound events using tables and lists.
- **CCSS.MATH.PRACTICE.MP2, MP4, MP5, MP7: Reason, Model, Use Tools & Structure.**

Next up on Page 06 → How to stack everything back in the box in 10 seconds!

**BOSS CHALLENGE #4** — **BLOCKS → BOX ICON KEY SPEED-RUN!**

1. Build Examples #1, #2, and #3 hands-on with your physical pieces.
2. Press the Box Button Icon ([▣], [Σ+] & [g][x]) and keystrokes on your calculator:

• 8 [ENTER] 20 [÷] [FDISP] → X = 2/5

**CHECK OFF YOUR MASTERY MILESTONES:**

- ☐ 1. Demonstrated Lessons #1, #2 & #3 hands-on with the physical blocks
- ☐ 2. Solved all 3 Desk Challenges above (check upside-down key if stuck!)
- ☐ 3. Demonstrated each lesson on the calculator using the Box Button Icon
- ☐ 4. Packed all pieces back into the box in under 10 seconds (see Page 06 →)

TEST YOURSELF • ANSWERS INCLUDED SO YOU CAN CHECK RIGHT AWAY

NEXT: HOW TO PACK UP → PAGE 06

SIDE B (PAGES 05–08) • READ TOP ROW FIRST

# 06 — HOW TO STACK IT ALL BACK IN

PAGE / 08

The 10-Second Self-Checking Pack-Up Puzzle

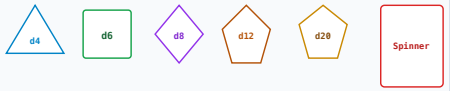
05 06 07 08

SKU: PROB-01 • KEEP IN LID

SIDE B • PACK-UP GUIDE (TOP-RIGHT)

## TOP-DOWN BOX MAP: WHERE EVERY PIECE LIVES

Putting the blocks away is a 10-second puzzle. Match each piece to its labeled home bay below so everything nests flush and the lid snaps shut:




**✓ CUSTOM CAVITIES FOR ALL 5 PLATONIC SOLIDS**

Each of the 5 polyhedral dice (d4, d6, d8, d12, d20) drops into its own shaped pocket inside the 1x1 storage case, alongside the spinner pointer and pin. Before snapping the dial lid shut, a 2-second look confirms all 5 dice are in place and none rolled under the couch!

**1. STORAGE CASE & COMPARTMENTS**

**2. WHY THE LID SELF-CHECKS EVERY PIECE**

SELF-CHECKING RIM & WATERLINE CROSS-SECTION

✓ FLUSH 84.0 MM = LID CLOSES

✓ OK | ✓ OK | ✓ OK | ✓ OK | ✗ HIGH

5 Shaped Polyhedral Pockets + 10-Secto

▲ SELF-INDEXING BAYS & EMBOSSED BOX BUTTON ICON:
Calibrated bays guide every piece home, and the lid icon links directly to your calculator.

## THE 10-SECOND PACK-UP ROUTINE

**HOW TO STACK EVERYTHING BACK IN:**

1. **SORT BY COLOR & NUMBER:**
Group matching blocks or tiles together on your desk first.
2. **STACK & DROP INTO LABELED BAYS:**
Follow the Top-Down Box Map on the left. If a tower looks too tall or too short, a block is in the wrong bay—swap it until everything levels!
3. **TUCK THIS CARD IN THE LID & SNAP SHUT:**
Fold this guide back to [01] and rest it right on top before closing.

**10-SECTOR SPINNER LID & POINTER STORAGE**

- Snap the pointer onto the lid's center hub to spin, or tuck it inside the box for travel.
- Pocket-sized 1x1 case docks neatly beside your SC-32 calculator or larger kit boxes.
- Keep away from toddlers under 3 (polyhedral dice are small parts).

**SHORT A PIECE AT CLEAN-UP TIME?**

First check under notebooks and chair legs! If a piece truly vanished, **see PAGE 08 (bottom-right) for your complete Part ID Reference Table.**

**✓ PROB-01 4-POINT CLEAN-UP CHECKLIST**

1. SEAT 5 POLYHEDRA: Drop d4, d6, d8, d12, and d20 into their shaped pockets inside the 1×1 case.
2. TUCK POINTER & PIN: Store the Blue spinner pointer and Clear hub pin in the side bay.
3. FOLD CARD TO [01]: Fold this reference booklet to Panel [01] for compact storage.
4. SNAP SPINNER LID: Click the 10-sector decimal dial lid onto the 1×1 case!

**MODULAR CLASSROOM & BOOKSHELF NESTING (42.0 MM PITCH)**

Every StackCalc box is sized in exact multiples of 42.0 mm (1×1, 3×1, 3×2). All 9 kit boxes interlock side-by-side in a classroom cart or drawer with zero wasted space, and every embossed box icon matches its calculator operation key!

SHAPED DIE CAVITIES • EVERY POLYHEDRON HAS ITS OWN HOME SOCKET

NEXT: CONNECTED LAB ▼ PAGE 07

THEN READ BOTTOM ROW (PAGE 07 → PAGE 08)

# 07 — THE STACKCALC CONNECTED LAB

PAGE / 08

How Your Kits & RPN Calculator Stack Work Together

05 06 07 08

SKU: PROB-01 • KEEP IN LID

SIDE B • ECOSYSTEM (BOT-LEFT)

**ONE CONNECTED SYSTEM FROM KINDERGARTEN TO CALCULUS**

Every StackCalc box uses the same modular grid and unit heights—so pieces from different kits actually fit and work together.

## WHERE YOUR BOX FITS IN THE 9-KIT LAB

| | | |
|---|---|---|
| BASE-10 **Ten-Frames Lab** Early Counting & Regrouping | DOMI-01 **Math Domino Studio** Doubles & 12-Frame Part-Whole | FRAC-01 **Fraction Studio** Proportional Towers & 4-Way Faces |
| PRIME-01 **Prime Factor Studio** Multiply by Stacking Prime Towers | ALGB-01 **Algebra Studio** Quadratic Area Arrays & Factoring | TIME-01 **Planetary Clock** 12:1 Geared Hands & Time Wedges |
| PROB-01 ◂ YOUR BOX **Probability Dice** Fair Polyhedrals & 10-Sector Spinner | VOLV-01 **Unit Circle Volvelle** Spin-to-Read Exact Trig Radicals | SC-32-HW **SC-32 RPN Calculator** 4-Register Stack Scientific Engine |

**WHY THIS MATTERS: ZERO "THROW-AWAY" MATH**

In most classrooms, every grade uses a totally unrelated plastic toy. In StackCalc, the 7mm unit block you count with in 1st grade is the exact same height as the 1/12th fraction block and the [2] prime block later on!

1. PROB-01 BOX LID ICON

[▣], [Σ+] & [g][x]
EMBOSSED ON BOX LID
Build First → Press Key!

2. SC-6 ELEMENTARY RPN

3. SC-32 SCIENTIFIC RPN

▲ BOX ICON → CALCULATOR KEY CONTINUITY:
Every physical box lid carries the exact icon of the calculator key that computes that lesson!

## HOW BLOCKS CONNECT TO THE CALCULATOR

**SHARED MODULAR FOOTPRINT & HEIGHTS**

- The compact 1x1 case shares the exact modular grid footprint of the rest of the StackCalc lab.
- Recessed sockets swallow the bottom peg so only true heights add.
- All storage boxes nest neatly side-by-side on any desk or shelf.

**WHY WE CALL IT "STACKCALC" (THE RPN SECRET)**

- Enter your experimental dice tallies into the SC-32 calculator to compute live means and sums!
- Placing a block on the table = pressing [ENTER] to push onto the stack.
- Snapping two towers together = pressing [+] or [×] with zero parentheses!
- Mirrors the 4 stack levels on the SC-32 calculator: X, Y, Z, and T.

**FREE ACTIVITY DECKS & INTERACTIVE SIMULATORS**

Download extra printable puzzle cards, teacher lesson guides, and interactive browser simulations anytime—no login needed.

Explore All 9 Kits & Free Guides at stackcalc.io

LIVE 4-REGISTER RPN STACK

T: 0 (Top Memory)
Z: 0 (3rd Level)
Y: 0 (1st Block)
X: 1/60 (P(A)×P(B)) (Active Display)

HOW BLOCKS MAP TO STACK KEYS

1. PLACE 1ST PIECE ON STAGE: Type number + [ENTER] → lifts to Y
2. ADD / STACK 2ND PIECE: Type 2nd number in X, then [+]/[×]
3. PRESS BOX ICON KEY: Press [▣], [Σ+] & [g][x]
✓ Zero Parentheses • Always Visible!

BOX ICON = [▣], [Σ+] & [g][x] • VERIFIED LESSON SEQUENCES:
• Lesson 1: 1 [ENTER] 6 [÷] 1 [ENTER] 10 [÷] [×] [FDIS → X = 1/60
• Lesson 2: [g] [CLI] 7 [Σ+] 4 [Σ+] 10 [Σ+] [g] [x] → X = 7

MODULAR 9-KIT CONTINUUM • FROM EARLY COUNTING TO AP CALCULUS

NEXT: PART ID TABLE → PAGE 08

# 08 — KEEP THIS CARD! PART IDs & VIDEO

PAGE / 08

Complete 9-Piece Reference & 60-Second 3D Video Guide

05 06 07 08

SKU: PROB-01 • KEEP IN LID

SIDE B • PART REFERENCE (BOT-RIGHT)

## PART ID REFERENCE TABLE (9 TOTAL PIECES)

**✓ KEEP THIS CARD INSIDE YOUR BOX LID!**

Backpacks eat blocks. Dogs chew dice. Vacuum cleaners take no prisoners. Keep this card in your lid so you always have the exact Part IDs below for single-piece refills!

| PART NAME | QTY | COLOR / FINISH | PART ID |
|---|---|---|---|
| Tetrahedron Die (d4) | 1x | Black / Gold | PROB-DICE-04a |
| 6-Sided Cube Die (d6) | 1x | Black / Gold | PROB-DICE-04a |
| Octahedron Die (d8) | 1x | Black / Gold | PROB-DICE-04a |
| Dodecahedron Die (d12) | 1x | Black / Gold | PROB-DICE-04a |
| Icosahedron Die (d20) | 1x | Black / Gold | PROB-DICE-04a |
| Balanced Spinner Pointer | 1x | Blue | PROB-SPIN-7e5 |
| Smooth-Spin Hub Pin | 1x | Clear | PROB-SPIN-7e5 |
| 1x1 Case & Dial Lid | 2x | Black | PROB-DICE-04a |

**NEED A REPLACEMENT PIECE? NO NEED TO BUY A WHOLE NEW BOX.**

Enter any Part ID above at stackcalc.io/parts to order individual replacement pieces.

COLOR & PIECE IDENTIFICATION KEY — MATCH ANY LOOSE PART

Black / Gold — Platonic Dice (d4..d20)
Blue — Balanced Spinner Pointer
Clear — Smooth-Spin Hub Pin
Black — 1x1 Case & Dial Lid

UNIVERSAL STACKCALC HEIGHT PITCH (1U = 7.00 MM • 12U = 84.00 MM ONE WHOLE)

Measure any loose piece in 7mm units (U) against this scale bar to identify it:

0U (0mm) 1U (7mm) 2U (14mm) 3U (21mm) 4U (28mm) 6U (42mm) 9U (63mm) 12U (84mm)

✓ DISTRICT PO & SAFETY: 10-Pack SKU PROB-01-CLASS10 • CPSIA / ASTM F963 Non-Toxic Washable PLA/ABS

## WATCH THE 60-SECOND 3D VIDEO GUIDE

POINT CAMERA TO WATCH DEMO
stackcalc.io/products/dice-and-spinner/
See animated 3D examples & pack-up walkthrough

**QUESTIONS, CLASSROOM SETS & SUPPORT**

- Email us anytime: support@stackcalc.io
- School & District POs: school-orders@stackcalc.io

**Designed in Mountain View, CA • Non-Toxic & Washable**

COMPANION MANUAL CHAPTERS, STANDARDS & KEYSTROKES

BOOK CHAPTERS: SC-6 Guide Ch. 6 (Probability & Stats) • SC-32 Manual Ch. 9
STANDARDS: CCSS 6.SP.A.1, 7.SP.C.5, 7.SP.C.8 • TEKS 7.6E, 7.6I (Sample Space & Mean)
WATERLINE SPEC: 5 Shaped Polyhedral Pockets + 10-Sector Spinner Lid

ALL 3 VERIFIED CALCULATOR LESSON SEQUENCES (SCG):
Ex 1: 1 [ENTER] 6 [÷] 1 [ENTER] 10 [÷] [×] [FDISP] → X=1/60
Ex 2: [g] [CLI] 7 [Σ+] 4 [Σ+] 10 [Σ+] [g] [x] → X=7
Ex 3: 8 [ENTER] 20 [÷] [FDISP] → X=2/5

BOX ICON KEY: [▣], [Σ+] & [g][x]
✓ Cross-Referenced to Companion Book & Calculator Engine
Fold back to [01] & keep this reference card inside your box lid!

LOST A PIECE? GRAB SINGLE REPLACEMENTS ANYTIME AT stackcalc.io/parts

✓ KEEP THIS GUIDE IN BOX LID!

# 01 — UNIT CIRCLE VOLVELLE & TRIGONOMETRIC CALCULATOR

PAGE / 08

Spin-to-Read Rotating Wheel Chart for All 16 Special Trig Angles

01 02 03 04

SKU: VOLV-01 • KEEP IN LID

SIDE A • COVER (TOP-LEFT)

**WHAT IS THIS BOX FOR? NEVER MEMORIZE A TRIG TABLE AGAIN.**

Sines and cosines aren't random decimals to cram before a test—they're just the X and Y shadows of a spinning wheel of radius 1.

## WELCOME TO THE POCKET TRIG WHEEL

Instead of memorizing 16 separate angles, spin one dial:

- Degrees & Radians Together: See 30° and π/6 lined up side-by-side
- Exact Radical Windows: Read (cos θ, sin θ) as (√3/2, 1/2) instantly
- Rotating Radius Arm: Sweeps a unit radius (r = 1.00) around all 4 quadrants
- Automatic Sign Check: Shows where X and Y turn negative in Q2, Q3, and Q4

Try this right now: Turn the pointer to 45° (π/4) and read the window!

### WHY IS IT CALLED A "VOLVELLE"?

Medieval astronomers used rotating paper and brass wheel charts—called volvelles—to compute star positions without pen and paper. Yours computes trig!

### WHAT'S IN YOUR BOX (5 PIECES)

- 1x Base Stator Disc (0°–360° Special Angles & X/Y Coordinate Grid)
- 1x Rotating Radian Dial (Gold) + 1x Radial Cursor Arm
- 1x Snap-Fit Center Hub Pin + 1x Protective Travel Slipcase

★ YOUR FIRST 60 SECONDS WITH THIS BOX — START HERE!

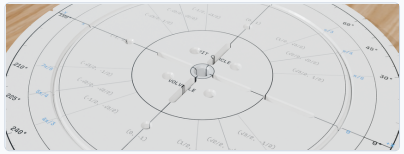
1. **DIAL THE CURSOR ARM TO 30° (π/6):** Look through the cursor window: read Degrees (30°), Radians (π/6), and Exact Coordinates (√3/2, 1/2)!
2. **COMPARE 30° AND 60° (π/3):** Spin to 60°: notice how √3/2 and 1/2 simply swap places because the triangle flips across 45°!
3. **PROJECT WITH [r][•y,x] ON SC-32:** Type 30 [ENTER] 1 [r][•y,x] on your SC-32 to get X = 0.8660 (√3/2) and Y = 0.5000 (1/2) simultaneously!

**DISC ICON = CALCULATOR [COS], [SIN] & [r][•Y,X] PHASOR KEYS**

The button icon embossed on your box lid (left) is the exact calculator key ([DEG•RAD] & [r][•y,x])! Build every lesson with your blocks, then press the key.

## EXACT RADICALS (√3/2) ↔ [COS] & [√x] VERIFICATION

STEP A: HANDS-ON WITH THE VOLVELLE DISC — 1:1 CAD SCENE



cos(30°) = √3/2 = 0.8660 on SC-32 | cos(45°) = √2/2 = 0.7071 on SC-32 | sin(30°) = 1/2 = 0.5000 on SC-32

▲ Read the exact symbolic radical √3/2 in your Volvelle window, then verify 3 [√x] 2 [÷] = 30 [COS] on SC-32!

STEP B: TRY IT ON YOUR CALCULATOR — BOX ICON: [DEG•RAD] & [r][•y

RADICAL TREE: θ = 30° (π/6) RIGHT TRIANGLE

θ = 30° (π/6 RAD)

Volvelle: √3/2 | SC-32: 30 [COS]

3 [√x] 2 [÷] | Y=0.8660 | X=0.8660

3 [√x] 2 [÷] 30 [COS] → Y=X=0.8660

PRESS BOX ICON & RPN KEYS (SC32):

1. RADICAL √3/2: 3 [√x] 2 [÷] → 0.8660

2. COMPARE COS: 30 [COS] → Y=X=0.8660

✓ DISPLAY (X) = 0.866

KEYSTROKES: 3 [√x] 2 [÷] 30 [COS]

Evaluate √3/2 from the Volvelle window with 3 [√x] 2 [÷], then press 30 [COS] to see Y and X match!

WELCOME TO HANDS-ON MATH • KEEP THIS GUIDE IN YOUR BOX LID

STEP 1: UNFOLD RIGHT → PAGE 02

▲ FOLD 1: FOLD BOTTOM HALF BEHIND TOP ▲

▼ UNFOLD STEP 2: OPEN DOWN FOR PAGES 03 & 04 ▼

# 02 — HOW IT WORKS: THE CORE IDEA

PAGE / 08

Unit Circle Coordinates: (cos θ, sin θ) on a Circle of Radius 1

01 02 03 04

SKU: VOLV-01 • KEEP IN LID

SIDE A • 1ST UNFOLD (TOP-RIGHT)

## HOW THE UNIT CIRCLE CONNECTS GEOMETRY & TRIG

A unit circle is simply a circle of radius 1 centered at (0, 0): $x^2 + y^2 = 1$.
When you turn the pointer arm to any angle θ from the positive x-axis:

**Horizontal Shadow x = cos(θ) | Vertical Height y = sin(θ)**

Because the pointer arm always has length 1, the right triangle it forms with the x-axis automatically obeys Pythagoras:

$\cos^2(\theta) + \sin^2(\theta) = 1^2 = 1$

And the slope (steepness) of the pointer arm is the tangent:

$\tan(\theta) = \text{Rise} / \text{Run} = \sin(\theta) / \cos(\theta)$

### ALL 16 SPECIAL ANGLES AT YOUR FINGERTIPS

Every multiple of 30° (π/6) and 45° (π/4) around the circle comes from just two classic right triangles: the 30–60–90 triangle and the 45–45–90 square diagonal. Once you know Quadrant I, the other three quadrants are mirror images!

### NEVER MISS A MINUS SIGN AGAIN

Left of center? Cosine (x) is negative. Below center? Sine (y) is negative. Looking at the physical pointer position makes plus and minus signs obvious at a glance, without relying on fragile mnemonics.

### 3 CLASSIC MATH TRAPS THIS KIT MAKES IMPOSSIBLE — VISUAL PROOF

✗ TRAP 1 — Confusing which is sin and which is cos:
✓ FIX: Alphabetical order! C comes before S, just like X comes before Y: (cos θ, sin θ) = (x, y).

✗ TRAP 2 — Mixing up √3/2 and 1/2 at 30° vs 60°:
✓ FIX: Look at the physical wheel: at 30°, the point is wide (x = √3/2 ≈ 0.87) and low (y = 1/2 = 0.50)!

✗ TRAP 3 — Reversing θ and r in [r][•y,x]:
✓ FIX: On SC-32, push angle θ into Y first, type radius r in X, then press [r][•y,x]: 30 [ENTER] 1 [r][•y,x]!

CALCULATOR CHECK: 3 [√x] 2 [÷] 30 [COS] → X = 0.866

## SPECIAL ANGLES & EXACT RADICAL VALUES

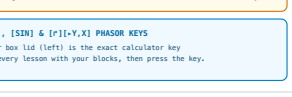
| DEGREES | RADIANS | COS (X) | SIN (Y) | TAN |
|---|---|---|---|---|
| 0° | 0 | 1 | 0 | 0 |
| 30° | π / 6 | √3 / 2 | 1 / 2 | √3 / 3 |
| 45° | π / 4 | √2 / 2 | √2 / 2 | 1 |
| 60° | π / 3 | 1 / 2 | √3 / 2 | √3 |
| 90° | π / 2 | 0 | 1 | undef |
| 120° | 2π / 3 | −1 / 2 | √3 / 2 | −√3 |
| 180° | π | −1 | 0 | 0 |
| 270° | 3π / 2 | 0 | −1 | undef |

### DESIGNED FOR DESK & BACKPACK

- Compact 110 mm outer disc slips easily into a binder pouch or jacket pocket.
- Smooth snap-fit center hub holds your chosen angle steady while you read.
- Deeply embossed angle ticks and radical fractions never fade or rub off.
- Pairs directly with the SC-32 calculator's [SIN], [COS], and [TAN] keys.

1:1 CAD REFERENCE VIEW — BOX ICON & FORMULA QUICK-REFERENCE

1:1 BLOCK CAD ANATOMY




✓ FLUSH PLUG-AND-SOCKET JOINT
Socket swallows plug completely: Only 7.00mm true heights add!

Box / Disc Icon Keys: [COS], [SIN], [r]|[•RAD] & [r][•y,x]

Coordinate Law: (x, y) = (cos θ, sin θ) | tan θ = sin θ / cos θ

Pythagorean Identity: $\cos^2(\theta) + \sin^2(\theta) = 1^2 = 1$ (for ALL θ)

30° (π/6) ↔ 60° (π/3): 30°: (√3/2, 1/2) | 60°: (1/2, √3/2)

SC-32 Polar → Rect: θ [ENTER] r [r] [•y,x] → X=r·cosθ, Y=r·sinθ

GEOMETRY MEETS TRIGONOMETRY • COSINE IS X, SINE IS Y

STEP 2: UNFOLD DOWN ▼ PAGES 03–04

# 03 — EXAMPLE 1: HANDS-ON WALKTHROUGH

PAGE / 08

Dial In 45° (π/4): Read Exact Radicals (√2/2, √2/2) in the Window

01 02 03 04

SKU: VOLV-01 • KEEP IN LID

SIDE A • 2ND UNFOLD (BOT-LEFT)

## 1 ALIGNING SPECIAL ANGLES & RADICALS

**1. ROTATE INNER ROTOR TO 45° (π/4):**
Grip the textured rotor rim and rotate until the cursor line aligns exactly with the 45° (π/4) graduation tick.

**2. READ EXACT VALUES THROUGH THE CURSOR WINDOW:**

- Radial Position: Exact unit circle boundary (r = 1.00)
- Horizontal Cosine (x): cos(45°) = √2 / 2 ≈ 0.7071
- Vertical Sine (y): sin(45°) = √2 / 2 ≈ 0.7071
- Tangent Slope: tan(45°) = 1.000 (Equal legs: x = y)

**3. TEST OTHER SPECIAL ANGLES:**
Rotate to 30° (π/6): cos = √3/2, sin = 1/2. Rotate to 60° (π/3): cos = 1/2, sin = √3/2.

### ELIMINATE PENCIL & PAPER GUESSWORK

Students frequently swap √3/2 and 1/2 on exams. Looking at the volvelle makes it visually undeniable: at 30°, the horizontal base is much longer than the height, so cos(30°) must be √3/2 (0.866) and sin(30°) must be 1/2 (0.5)!

STEP B: TRY EXAMPLE #1 ON YOUR CALCULATOR — [DEG•RAD] & [r][•y

STEP 1 (DIAL 45° / π/4): Align the Volvelle cursor arm with 45° → window reads (√2/2, √2/2) and tan(45°) = 1.

STEP 2 (SQUARE COS 45°): On your SC-32 calculator, press 45 [COS] [^] [x²] → X = 0.5000.

STEP 3 (ADD SIN² 45°): Press 45 [SIN] [^] [x²] [+] → 0.5 + 0.5 = 1 (proving $\cos^2\theta + \sin^2\theta = 1$)!

LIVE 4-REGISTER RPN STACK READOUT (AFTER KEYSTROKES):
T: 0 | Z: 0 | Y: 0 | X: 1 → DISPLAY (X) = 1

## LESSON 1: DIAL 45° (π/4) ON VOLVELLE ↔ PYTHAGOREAN KEYST

STEP A: HANDS-ON WITH THE VOLVELLE DISC — 1:1 CAD SCENE

16 Special Angles Degrees (°) & Radians ( | Exact Radicals (cos θ, sin θ) in Windo | Radius R = 1 cos²θ + sin²θ = 1

▲ Dial the cursor arm to 45° (π/4): the window frames (√2/2, √2/2), and squaring + adding on SC-32 gives 1!

STEP B: TRY IT ON YOUR CALCULATOR — BOX ICON: [DEG•RAD] & [r][•y

PYTHAGOREAN TREE: θ = 45° (π/4 RAD)

cos²(45°) + sin²(45°) = 1

45 [COS] [^] [x²] | 45 [SIN] [^] [x²]

.7071² = 0.5 | .7071² = 0.5 | [+] → 1

45 [COS] [^] [x²] 45 [SIN] [^] [x²] [+] → 1

PRESS BOX ICON & RPN KEYS (SC32):

1. COS²(45°): 45 [COS] [^] [x²] → 0.5

2. + SIN²(45°): 45 [SIN] [^] [x²] [+] → 1

✓ DISPLAY (X) = 1

KEYSTROKES: 45 [COS] [^] [x²] 45 [SIN] [^] [x²] [+]

Dial 45° on the Volvelle, then square cos(45°) and sin(45°) on SC-32 (no redundant ENTER before [^][x²])!

✓ 4-REGISTER STACK: T: 0 | Z: 0 | Y: 0 | X: 1 → X = 1

EXACT RADICAL FRACTIONS • NEVER CONFUSE √3/2 AND 1/2 AGAIN

CONTINUE RIGHT → PAGE 04

# 04 — EXAMPLE 2: CORE MATHEMATICAL PRINCIPLE

PAGE / 08

Pythagorean Identity: Why cos²(θ) + sin²(θ) Always Equals 1

01 02 03 04

SKU: VOLV-01 • KEEP IN LID

SIDE A • 2ND UNFOLD (BOT-RIGHT)

## 2 PYTHAGOREAN IDENTITY: cos²θ + sin²θ = 1

**THE BIG DISCOVERY: Why does cos²θ + sin²θ ALWAYS equal 1?**

1. TRACE LEGS AT ANY ANGLE θ:
- The horizontal base of the triangle is length |cos θ|.
- The vertical height of the triangle is length |sin θ|.
- The hypotenuse is the radial arm connecting origin to (cos θ, sin θ).

2. APPLY THE PYTHAGOREAN THEOREM:
Base² + Height² = Hypotenuse²
$(\cos\theta)^2 + (\sin\theta)^2 = (1.00)^2 = 1$

3. ROTATE PAST 90° INTO QUADRANT II (e.g. 120°):
cos(120°) = −1/2, sin(120°) = √3/2.
$(-1/2)^2 + (\sqrt{3}/2)^2 = 1/4 + 3/4 = 4/4 = 1.00$! (Signs square away!)

### ✓ UNIT RADIUS CONSERVATION THEOREM

The identity is not a separate formula to memorize.
**It is literally the Pythagorean theorem on the circle!**
No matter how fast or far you spin the volvelle rotor, the tip of the radial arm always remains exactly 1 unit from the origin.
**Pythagoras and trigonometry are one single unified truth.**

★ 2-PLAYER GAME: 'QUADRANT RADICAL SPRINT' (2 MINS)

Player A calls out a radian angle (like 5π/6 or 7π/4). Player B has 5 seconds to predict the plus/minus signs of (cos, sin) and the reference triangle (30°, 45°, or 60°), spins the **Volvelle cursor arm to verify, and checks with 0 [ENTER] 1 [r][•y,x] on SC-32!**

⊙ WHERE THIS SHOWS UP IN AUDIO SYNTHESIS, AC POWER & GAME ENGINES

Why are 'volvelles' shaped like rotating discs? Astronomers used paper and brass volvelles for centuries before electronic computers! Every 3D video game rotation matrix, musical synthesizer **oscillator, and AC electrical grid wave runs on (cos wt, sin wt).**

## LESSON 2: 30° VOLVELLE WINDOW ↔ [r][•y,x] POLAR-TO-RECT

STEP A: HANDS-ON WITH 30° (π/6) PROJECTION — 1:1 CAD SCENE

30° = π/6 rad (√3/2, 1/2) | X = cos 30° 0.8660 on SC-32 | Y = sin 30° 0.5000 on SC-32

▲ Dial 30° on the Volvelle, then press 30 [ENTER] 1 [r][•y,x] on SC-32 to project X=0.8660 and Y=0.5000!

STEP B: TRY IT ON YOUR CALCULATOR — BOX ICON: [DEG•RAD] & [r][•y

POLAR-TO-RECT PROJECTION: θ=30°, r=1

30° [ENTER] 1 [r][•y,x]

X = r·cos(30°) | Y = r·sin(30°)

X = 0.8660 | (√3/2) | Y = 0.5000

30 [ENTER] 1 [r][•y,x] → X=0.8660, Y=0.5

PRESS BOX ICON & RPN KEYS (SC32):

1. θ IN Y, r IN X: 30 [ENTER] 1 [r] [•y,x]

2. READ X & Y: X=0.8660 | Y=0.5000

✓ DISPLAY (X) = 0.866

KEYSTROKES: 30 [ENTER] 1 [r] [•y,x]

Enter angle θ = 30° into Y and radius r = 1 into X, then press [r][•y,x] to project both coordinates at once!

✓ 4-REGISTER STACK: T: 0 | Z: 0 | Y: 0.5 | X: 0.8660254038 → X = 0.866

PYTHAGORAS ON A CIRCLE • HYPOTENUSE IS ALWAYS EQUAL TO 1

STEP 3: FLIP SHEET OVER ↻ PAGE 05

◂ PAGE 03 (LEFT) → PAGE 04 (RIGHT) • THEN FLIP SHEET OVER FOR PAGES 05–08 ▸

# 05 — EXAMPLE 3 & DESK CHALLENGES

PAGE / 08

Reference Angles Across All 4 Quadrants + 3 Desk Challenges

05 06 07 08

SKU: VOLV-01 • KEEP IN LID

SIDE B • FLIP SIDE (TOP-LEFT)

## 3 SYMMETRY ACROSS QUADRANTS: 30°, 150°, 210°, AND 330°

**THE GOAL: See why four different angles share the exact same (√3/2, 1/2) numbers!**

**1. START AT 30° (π/6) IN QUADRANT I:**

Read the window: cos(30°) = +√3/2 (long horizontal), sin(30°) = +1/2 (short height).

**2. SPIN TO 150° (5π/6) IN QUADRANT II:**

It's the exact same 30° triangle flipped left! Now x is negative: (−√3/2, +1/2).

**3. CHECK 210° (QIII) AND 330° (QIV):**

At 210° both are negative (−√3/2, −1/2); at 330° only y is negative (+√3/2, −1/2).

### ✓ HOW 3 TRIANGLES UNLOCK ALL 16 SPECIAL ANGLES

**You don't need to memorize 16 separate coordinate pairs around the circle.**
Every angle is just a 30°, 45°, or 60° right triangle reflected across the X and Y axes!
Look at the pointer arm first: is it left of center (cos < 0)? Below center (sin < 0)?
Verify any angle on your SC-32 calculator: press [3][0][COS] to see 0.8660 (√3/2)!

**PRO TIP: CLOSE YOUR EYES & FEEL THE ANSWER**

Once you build an example, run your fingers across the pieces. Your hands remember physical shapes and heights way longer than memorized rules!

EXAMPLE 3: VISUAL DECOMPOSITION TREE

RADICAL TREE: θ = 30° (π/6 RAD) RIGHT TRIANGLE

θ = 30° (π/6 RAD)

Volvelle: √3/2 | SC-32: 30 [COS]

3 [√x] 2 [÷] | Y=0.8660 | X=0.8660

3 [√x] 2 [÷] 30 [COS] → Y=X=0.8660

STEP 8: PRESS BOX ICON & RPN KEYS (SC32)

TACTILE KEYSTROKE WALKTHROUGH:

1. RADICAL √3/2:

3 √x 2 ÷ 0.8660

2. COMPARE COS:

30 COS Y=X=0.8660

✓ CALCULATOR DISPLAY (X) = 0.866

FULL KEYSTROKES: 3 [√x] 2 [÷] 30 [COS]

✓ 4-REGISTER STACK: T: 0 | Z: 0 | Y: 0.866025403 | X: 0.866025403 → X = 0.866

## 3 DESK CHALLENGES — TRY THESE NOW!

**WARM-UP #1** — **Turn to 150° (5π/6). What are the exact values of cos θ and sin θ?**

Step A: Build with blocks on your Lid Tray • Step B: Verify on SC-32 Calculator

✓ Build first, then flip booklet upside-down to check Answer #1 below!

**LEVEL-UP #2** — **Turn to 225° (5π/4) in Quadrant III. Why is tan(225°) positive +1?**

Step A: Build with blocks on your Lid Tray • Step B: Verify on SC-32 Calculator

✓ Build first, then flip booklet upside-down to check Answer #2 below!

**BRAIN-MELTER #3** — **At which two angles around the circle is tan(θ) undefined?**

Step A: Build with blocks on your Lid Tray • Step B: Verify on SC-32 Calculator

✓ Build first, then flip booklet upside-down to check Answer #3 below!

SELF-CHECK ANSWER KEY (ROTATE BOOKLET 180° TO READ)

θ = π RADIAN '(Z/NE) ,ΔEZ PWR (Z/π) .86 :tΣ

it+ = (Z/Z←) ÷ (Z/Z←) !Σ# | Z/T+ = NTS 'Z/Z← = 9O3 :tΣ#

**FOR PARENTS & TEACHERS (Grades 9–12 & College (Geometry, Pre-Calculus & Calculus))**

- CCSS.MATH.CONTENT.HSF.TF.A.2: Explain how the unit circle extends trig functions to all real numbers.
- CCSS.MATH.CONTENT.HSF.TF.A.3: Use special triangles to determine geometrically the values of sin, cos, tan.
- CCSS.MATH.CONTENT.HSF.TF.C.8: Prove the Pythagorean identity $\sin^2(\theta) + \cos^2(\theta) = 1$.
- **CCSS.MATH.PRACTICE.MP2, MP4, MP5, MP7: Reason, Model, Use Tools & Structure.**

Next up on Page 06 → How to stack everything back in the box in 10 seconds!

**BOSS CHALLENGE #4** — **BLOCKS → BOX ICON KEY SPEED-RUN!**

1. Build Examples #1, #2, and #3 hands-on with your physical pieces.
2. Press the Box Button Icon ([DEG→RAD] & [r][+,y,x]) and keystrokes on your calculator:

• 3 [√x] 2 [÷] 30 [COS] → X = 0.866

**CHECK OFF YOUR MASTERY MILESTONES:**

- ☐ 1. Demonstrated Lessons #1, #2 & #3 hands-on with the physical blocks
- ☐ 2. Solved all 3 Desk Challenges above (check upside-down key if stuck!)
- ☐ 3. Demonstrated each lesson on the calculator using the Box Button Icon
- ☐ 4. Packed all pieces back into the box in under 10 seconds (see Page 06 →)

TEST YOURSELF • ANSWERS INCLUDED SO YOU CAN CHECK RIGHT AWAY

NEXT: HOW TO PACK UP → PAGE 06

# 06 — HOW TO STACK IT ALL BACK IN

PAGE / 08

The 10-Second Self-Checking Pack-Up Puzzle

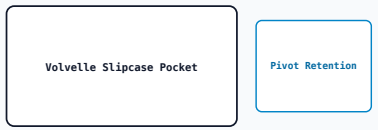
05 06 07 08

SKU: VOLV-01 • KEEP IN LID

SIDE B • PACK-UP GUIDE (TOP-RIGHT)

## TOP-DOWN BOX MAP: WHERE EVERY PIECE LIVES

Putting the blocks away is a 10-second puzzle. Match each piece to its labeled home bay below so everything nests flush and the lid snaps shut:

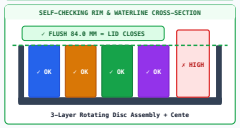



### ✓ SLIDES ASSEMBLED INTO THE PROTECTIVE SLIPCASE

You never need to take the volvelle apart after assembly—the base disc, gold radian rotor, cursor arm, and center pin stay snapped together as one pocket instrument. When you finish your problem set, align the cursor arm with 0° and slide the entire wheel straight into its protective travel slipcase!

1. STORAGE CASE & COMPARTMENTS

2. WHY THE LID SELF-CHECKS EVERY PIECE




▲ SELF-INDEXING BAYS & EMBOSSED BOX BUTTON ICON:
Calibrated bays guide every piece home, and the lid icon links directly to your calculator.

## THE 10-SECOND PACK-UP ROUTINE

**HOW TO STACK EVERYTHING BACK IN:**

1. **SORT BY COLOR & NUMBER:**
Group matching blocks or tiles together on your desk first.
2. **STACK & DROP INTO LABELED BAYS:**
Follow the Top-Down Box Map on the left. If a tower looks too tall or too short, a block is in the wrong bay—swap it until everything levels!
3. **TUCK THIS CARD IN THE LID & SNAP SHUT:**
Fold this guide back to [01] and rest it right on top before closing.

### HUB PIN CARE & BINDER STORAGE

- The snap-fit center pin holds your chosen angle steady without loosening over time.
- If dust ever gets under the rotor, rinse the wheel in cool water and pat dry.
- Tuck this folded reference guide right inside the slipcase pocket alongside the wheel!

**SHORT A PIECE AT CLEAN-UP TIME?**

First check under notebooks and chair legs! If a piece truly vanished, **see PAGE 08 (bottom-right) for your complete Part ID Reference Table.**

**✓ VOLV-01 4-POINT CLEAN-UP CHECKLIST**

| | |
|---|---|
| 1. ALIGN CURSOR TO 0°: | Rotate the Gold cursor arm back to the 0° (2π) home position. |
| 2. CHECK PIVOT CAP: | Ensure the center retention pin and cap are seated flush. |
| 3. FOLD CARD TO [01]: | Fold this reference booklet to Panel [01] alongside the disc. |
| 4. SLIDE INTO SLEEVE: | Slip the Volvelle disc and folded guide into the protective sleeve! |

**MODULAR CLASSROOM & BOOKSHELF NESTING (42.0 MM PITCH)**

Every StackCalc box is sized in exact multiples of 42.0 mm (1×1, 3×1, 3×2). All 9 kit boxes interlock side-by-side in a classroom cart or drawer with zero wasted space, and every embossed box icon matches its calculator operation key!

SLIPCASE TRAVEL SLEEVE • KEEPS YOUR VOLVELLE SAFE IN ANY BINDER

NEXT: CONNECTED LAB ▼ PAGE 07

SIDE B (PAGES 05–08) • READ TOP ROW FIRST

THEN READ BOTTOM ROW (PAGE 07 → PAGE 08)

# 07 — THE STACKCALC CONNECTED LAB

PAGE / 08

How Your Kits & RPN Calculator Stack Work Together

05 06 07 08

SKU: VOLV-01 • KEEP IN LID

SIDE B • ECOSYSTEM (BOT-LEFT)

**ONE CONNECTED SYSTEM FROM KINDERGARTEN TO CALCULUS**

Every StackCalc box uses the same modular grid and unit heights—so pieces from different kits actually fit and work together.

## WHERE YOUR BOX FITS IN THE 9-KIT LAB

| | | |
|---|---|---|
| BASE-10 **Ten-Frames Lab** Early Counting & Regrouping | DOMI-01 **Math Domino Studio** Doubles & 12-Frame Part-Whole | FRAC-01 **Fraction Studio** Proportional Towers & 4-Way Faces |
| PRIME-01 **Prime Factor Studio** Multiply by Stacking Prime Towers | ALGB-01 **Algebra Studio** Quadratic Area Arrays & Factoring | TIME-01 **Planetary Clock** 12:1 Geared Hands & Time Wedges |
| PROB-01 **Probability Dice** Fair Polyhedrals & 10-Sector Spinner | VOLV-01 ◂ YOUR BOX **Unit Circle Volvelle** Spin-to-Read Exact Trig Radicals | SC-32-HW **SC-32 RPN Calculator** 4-Register Stack Scientific Engine |

**WHY THIS MATTERS: ZERO "THROW-AWAY" MATH**

In most classrooms, every grade uses a totally unrelated plastic toy. In StackCalc, the 7mm unit block you count with in 1st grade is the exact same height as the 1/12th fraction block and the [2] prime block later on!

1. VOLV-01 BOX LID ICON

[DEG-RAD] & [r][+,y,x]
EMBOSSED ON BOX LID
Build First → Press Key!

2. SC-6 ELEMENTARY RPN

3. SC-32 SCIENTIFIC RPN

▲ BOX ICON → CALCULATOR KEY CONTINUITY:
Every physical box lid carries the exact icon of the calculator key that computes that lesson!

## HOW BLOCKS CONNECT TO THE CALCULATOR

**SHARED MODULAR FOOTPRINT & HEIGHTS**

- Every 30° (π/6) step on the Volvelle matches one hour on the Classroom Planetary Clock!
- Recessed sockets swallow the bottom peg so only true heights add.
- All storage boxes nest neatly side-by-side on any desk or shelf.

**WHY WE CALL IT "STACKCALC" (THE RPN SECRET)**

- Pair with your SC-32 calculator: read exact radicals on the wheel, then compute decimals on the RPN stack.
- Placing a block on the table = pressing [ENTER] to push onto the stack.
- Snapping two towers together = pressing [+] or [×] with zero parentheses!
- Mirrors the 4 stack levels on the SC-32 calculator: X, Y, Z, and T.

**FREE ACTIVITY DECKS & INTERACTIVE SIMULATORS**

Download extra printable puzzle cards, teacher lesson guides, and interactive browser simulations anytime—no login needed.

Explore All 9 Kits & Free Guides at stackcalc.io

LIVE 4-REGISTER RPN STACK

T: 0 (Top Memory)
Z: 0 (3rd Level)
Y: 0 (1st Block)
X: 1 (cos²45°+sin²45°) (Active Display)

HOW BLOCKS MAP TO STACK KEYS

1. **PLACE 1ST PIECE ON STAGE:**
Type number + [ENTER] → lifts to Y
2. **ADD / STACK 2ND PIECE:**
Type 2nd number in X, then [+]/[×]
3. **PRESS BOX ICON KEY:**
Press [DEG→RAD] & [r][+,y,x]
✓ **Zero Parentheses • Always Visible!**

BOX ICON = [DEG→RAD] & [r][+,y,x] • VERIFIED LESSON SEQUENCES:
• Lesson 1: 45 [COS] [↰] [x²] 45 [SIN] [↰] [x²] [+] → X = 1
• Lesson 2: 30 [ENTER] 1 [r] [+,y,x] → X = 0.866

MODULAR 9-KIT CONTINUUM • FROM EARLY COUNTING TO AP CALCULUS

NEXT: PART ID TABLE → PAGE 08

# 08 — KEEP THIS CARD! PART IDs & VIDEO

PAGE / 08

Complete 5-Piece Reference & 60-Second 3D Video Guide

05 06 07 08

SKU: VOLV-01 • KEEP IN LID

SIDE B • PART REFERENCE (BOT-RIGHT)

## PART ID REFERENCE TABLE (5 TOTAL PIECES)

**✓ KEEP THIS CARD INSIDE YOUR BOX LID!**

Backpacks eat blocks. Dogs chew dice. Vacuum cleaners take no prisoners. Keep this card in your lid so you always have the exact Part IDs below for single-piece refills!

| PART NAME | QTY | COLOR / FINISH | PART ID |
|---|---|---|---|
| Base Stator Disc (0°–360°) | 1x | Black | VOLV-VOLV-a17 |
| Polar Radian Rotor Disc | 1x | Gold | VOLV-VOLV-a17 |
| Exact Radical Cursor Arm | 1x | Blue | VOLV-VOLV-a17 |
| Snap-Fit Center Hub Pin | 1x | Clear | VOLV-VOLV-a17 |
| Protective Travel Slipcase | 1x | Gray | VOLV-VOLV-a17 |

**NEED A REPLACEMENT PIECE? NO NEED TO BUY A WHOLE NEW BOX.**

Enter any Part ID above at stackcalc.io/parts to order individual replacement pieces.

COLOR & PIECE IDENTIFICATION KEY — MATCH ANY LOOSE PART

| Black | Gold | Blue | Gray |
|---|---|---|---|
| Base Stator Disc | Polar Radian Rotor | Radial Cursor Arm | Protective Slipcase |

UNIVERSAL STACKCALC HEIGHT PITCH (1U = 7.00 MM • 12U = 84.00 MM ONE WHOLE)

Measure any loose piece in 7mm units (U) against this scale bar to identify it:

0U (0mm) 1U (7mm) 2U (14mm) 3U (21mm) 4U (28mm) 6U (42mm) 9U (63mm) 12U (84mm)

✓ DISTRICT PO & SAFETY: 10-Pack SKU VOLV-01-CLASS10 • CPSIA / ASTM F963 Non-Toxic Washable PLA/ABS

## WATCH THE 60-SECOND 3D VIDEO GUIDE

POINT CAMERA TO WATCH DEMO

stackcalc.io/products/unit-circle-volvelle/

See animated 3D examples & pack-up walkthrough

**QUESTIONS, CLASSROOM SETS & SUPPORT**

- Email us anytime: support@stackcalc.io
- School & District POs: school-orders@stackcalc.io

**Designed in Mountain View, CA • Non-Toxic & Washable**

COMPANION MANUAL CHAPTERS, STANDARDS & KEYSTROKES

**BOOK CHAPTERS:** SC-32 Manual Ch. 5 (Trigonometry & Phasors) • Appendix C
**STANDARDS:** CCSS HSF-TF.A.1, HSF-TF.A.2, HSF-TF.C.8 • TEKS P.4A, P.4B (Unit Circle)
**WATERLINE SPEC:** 3-Layer Rotating Disc Assembly + Conte Retention Pin

ALL 3 VERIFIED CALCULATOR LESSON SEQUENCES (SC32):
Ex 1: 45 [COS] [↰] [x²] 45 [SIN] [↰] [x²] [+] → X=1
Ex 2: 30 [ENTER] 1 [r] [+,y,x] → X=0.866
Ex 3: 3 [√x] 2 [÷] 30 [COS] → X=0.866

BOX ICON KEY: [DEG→RAD] & [r][+,y,x]
✓ Cross-Referenced to Companion Book & Calculator Engine
Fold back to [01] & keep this reference card inside your box lid!

LOST A PIECE? GRAB SINGLE REPLACEMENTS ANYTIME AT stackcalc.io/parts

✓ KEEP THIS GUIDE IN BOX LID!